UNIVERSITY OF SKÖVDE

DOCTORAL DISSERTATION

HUMAN POSTURE AND MOTION PREDICTION FOR AUTOMOTIVE ERGONOMICS DESIGN

Enhancing Functionality and Accuracy in Digital Human Modelling Tools

ESTELA PÉREZ LUQUE

Informatics

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ABSTRACT

Product development (PD) increasingly relies on digital tools to support the process of exploring, generating, and evaluating product design proposals. Ergonomics plays a critical role in ensuring that product designs align with human capabilities and needs. Digital human modelling (DHM) tools can simulate human-product interactions and assess ergonomics virtually, before physical prototypes exist. In vehicle design, DHM tools are frequently applied in occupant packaging activities, supporting the design of vehicle interiors that accommodate a diverse user population. Still, although commonly used in industry, DHM tools have various limitations. One challenge is their limited ability to predict human postures and motions with sufficient accuracy. This inaccuracy is the result of current simulation procedures and the prediction models used. To compensate for this, DHM tool users often require significant manual adjustments to produce realistic postures, making the process time-consuming, subjective, and difficult to reproduce. Moreover, the simulation procedures themselves can be complex and inefficient, reducing their accessibility and usefulness in iterative design work. These limitations often lead to costly and time-consuming validation activities involving real users.

This thesis addresses these challenges by developing and evaluating methods and models to enhance the functionality and accuracy of posture and motion predictions in DHM tools. The main contributions are: (1) identifying current practices and challenges in industry when applying DHM tools for ergonomics in PD, (2) developing methods that increase the functionality of DHM tools through improved simulations methods, and (3) developing and evaluating posture and motion prediction models that support more reliable and efficient virtual ergonomics assessments. Collectively, the findings support a more proactive, systematic, and human-centred approach to ergonomics in PD processes.

SAMMANFATTNING

Produktutveckling förlitar sig alltmer på digitala verktyg för att stödja processen med att utforska, generera och utvärdera produktdesignförslag. Ergonomi spelar en avgörande roll för att säkerställa att produktdesignen är anpassad till människans förmågor och behov. Digitala verktyg för mänsklig modellering (digital human modelling - DHM) kan simulera interaktioner mellan människa och produkt och bedöma ergonomin virtuellt, innan det finns fysiska prototyper. Inom fordonsdesign används DHM-verktyg ofta i aktiviteter som rör förar- och passagerarergonomi, för att stödja utformningen av fordonsinteriörer som passar en mångfald av användare. Dock, även om DHM-verktyg ofta används i industrin, så finns begränsningar av olika slag. En begränsning är DHM-verktygens förmåga att förutsäga mänskliga kroppsställningar och -rörelser med tillräcklig noggrannhet. Denna brist på noggrannhet beror på de nuvarande simuleringsförfarandena och de prediktionsmodeller som används. För att kompensera för detta behöver användare av DHM-verktyg ofta göra betydande manuella justeringar för att åstadkomma realistiska kroppsställningar, vilket gör processen tidskrävande, subjektiv och svår att reproducera. Dessutom kan simuleringsförfarandena i sig vara komplexa och ineffektiva, vilket minskar deras tillgänglighet och användbarhet i iterativt designarbete. Dessa begränsningar leder ofta till kostsamma och tidskrävande valideringsaktiviteter som involverar verkliga användare.

Avhandlingen behandlar dessa utmaningar genom att utveckla och utvärdera metoder och modeller för att förbättra funktionaliteten och noggrannheten av prediktioner av kroppsställningar och -rörelser i DHM-verktyg. De viktigaste bidragen är: (1) identifiering av rådande praktik och utmaningar i industrin vid användning av DHM-verktyg för ergonomi i produktutveckling, (2) utveckling av metoder som ökar funktionaliteten hos DHM-verktyg genom förbättrade simuleringsmetoder, och (3) utveckling och utvärdering av prediktionsmodeller för kroppsställningar och -rörelser som stöder mer tillförlitliga och effektiva virtuella ergonomiska bedömningar. Sammantaget stöder resultaten ett mer proaktivt, systematiskt och människocentrerat förhållningssätt för beaktande av ergonomi i produktutvecklingsprocesser.

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I don't know what to say, I just want to cry.

That's the loop I have been stuck in for weeks. It's surprisingly hard to put into words the mix of emotions I carry as I reach the end of this chapter.

When I arrived in Skövde on August 23, 2017, I never imagined that I would stay this long and that this place would come to feel like home. I was just 23, had flawless skin, was full of curiosity, and had so much to learn. Today, I am a little more grown-up, I would like to think a bit wiser, and definitely have more stories, wrinkles, and a few white hairs to tell. Let's just say I have been… PhD-affected.

Doing a PhD is no piece of cake, and it does not get easier when you are far from your loved ones. It is a full-on rollercoaster ride, and the trick is all about learning how to stay steady through it. But I couldn't feel luckier to have done it in Sweden. It's been tough, yes, but also filled with lots of fun and growth. So here goes my attempt to thank those who have been with me throughout this journey.

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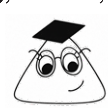

PUBLICATIONS

The appended publications in this dissertation are listed below. Estela Pérez Luque's contributions, as the main author of all appended publications, are specified for each publication according to the CRediT taxonomy (https://credit.niso.org). Co-authors include the author's supervision team: Prof. Dan Högberg, Dr. Erik Brolin, and Dr. Maurice Lamb. Additional collaborations during the thesis include Dr. Aitor Iriondo Pascual at the University of Skövde, Pernilla Nurbo, Technical Leader Ergonomics at Volvo Cars, and Prof. James Yang and Dr. Seunghun Lee at Texas Tech University.

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C. **Perez Luque, E.**, Iriondo Pascual, A., Högberg, D., Lamb, M. & Brolin, E. (2025). Simulation-based Multi-objective Optimization Combined with a DHM tool for Occupant Packaging Design. International Journal of Industrial Ergonomics, 105, 103690. https://doi.org/10.1016/j.ergon.2024.103690

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2. Fontinovo, E., **Perez Luque, E.**, Papetti, A., Högberg, D., Hanson, L., Truijen, S. & Scataglini, S. (2025). Comparison between Wearable Inertial Measurement System and 4D Stereophotogrammetry for Ergonomics Risk Assessment. Proceedings of the DHM2025 Conference, (accepted).
3. Lamb M., Brundin M., **Perez Luque E.** & Billing E. (2022) Eye-Tracking Beyond Peripersonal Space in Virtual Reality: Validation and Best Practices. Front. Virtual Real. 3:864653. https://doi.org/10.3389/frvir.2022.864653
4. Garcia Rivera, F., Högberg, D., Lamb, M. & **Perez Luque, E.** (2022). DHM Supported Assessment of the Effects of Using an Exoskeleton During Work. International Journal of Human Factors Modelling and Simulation, 7, 231–246. https://doi.org/10.1504/ijhfms.2021.10048920
5. Iriondo Pascual, A., Högberg, D., Syberfeldt, A., Brolin, E., **Perez Luque, E.**, Hanson, L. & Lämkull, D. (2022). Multi-objective Optimization of Ergonomics and Productivity by Using an Optimization Framework. Proceedings of the 21st Congress of the International Ergonomics Association (IEA 2021): Volume V: Methods & Approaches, 374–378. https://doi.org/10.1007/978-3-030-74614-8_46
6. Iriondo Pascual, A., Högberg, D., Lämkull, D., **Perez Luque, E.**, Syberfeldt, A. & Hanson, L. (2021). Optimization of Productivity and Worker Well-Being by Using a Multi-Objective Optimization Framework, IISE Transactions on Occupational Ergonomics and Human Factors, 9:3-4, 143-153. https://doi.org/10.1080/24725838.2021.1997834
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10. Igelmo, V., Syberfeldt, A., Högberg, D., García Rivera, F. & **Perez Luque, E.** (2020). Aiding Observational Ergonomic Evaluation Methods Using MOCAP Systems Supported by AI-Based Posture Recognition. DHM2020 : Proceedings of the 6th International Digital Human Modeling Symposium, August 31 – September 2, 2020, 419–429. https://doi.org/10.3233/ATDE200050

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INTRODUCTION

1 INTRODUCTION

This chapter describes the background of the research area as well as the aim, research questions, and objectives of the thesis.

1.1 BACKGROUND

Successful product development involves complex design processes that aim to meet customer needs while complying with industry standards and governmental requirements. Product development (PD) is the set of activities required to create a clear and complete definition of a product (Mattson and Sorensen, 2020). Activities in a PD process typically require several areas and aspects, including mechanical engineering, electrical engineering, manufacturing engineering, industrial design, ergonomics, and user experience design. Identifying customers' needs and how a product will affect the end-user is a core activity in PD and an essential part of product design (Ulrich et al., 2020). Engineers and designers in charge of considering, designing, and testing human-product interactions typically perform human-product interaction studies at relatively late stages of the PD process, often due to time constraints, limited resources, lack of detailed product information available early in development, or absence of appropriate tools and methods (Miehling et al., 2013; Sinha et al., 2024). This tendency persists even though there are clear benefits to proactively addressing human-related aspects at earlier stages (Bernard et al., 2020; Schröppel et al., 2021).

One of the fields concerned with human-product interaction during PD is ergonomics. Ergonomics (or its synonym human factors) is the discipline concerned with understanding human interactions with other system elements to optimize human well-being and overall system performance (IEA, 2025). Therefore, ergonomics designers are responsible for identifying and ensuring that human interaction needs are fulfilled during PD (Hartmann and Brennecke, 2008). To do so, ergonomics designers can use various tools to assess ergonomics aspects during the PD process, including physical prototypes, user tests and interviews, and human-product interaction simulations using digital human modelling (DHM) tools.

Consider a person getting into a new car. They place their hands on the steering wheel and their feet on the pedals. They may adjust the seat for comfort or to improve visibility. Further adjustments, such as modifying the seat height or backrest angle, may be made to better reach the climate controls or touchscreen interface. The position of the body, his/her posture, is shaped by the vehicle's design, the intended use of the vehicle, and the person's unique anthropometry and preferences. Another individual, taller, for example, would likely adopt a different posture and make different adjustments to the same seat and controls. Hence, ergonomics designers must consider individuals' characteristics and preferences early in the design process to ensure that products, whether vehicle cabins or another complex system, accommodate diverse user needs. One way to do this is by using digital simulation tools, e.g. DHM tools, that predict how people with different body characteristics might interact with a variety of product designs.

Most product design work is performed using computer-aided design (CAD) tools, which are used to create and modify digital models of products. These are often complemented by computer-aided engineering (CAE) tools, which support simulation and analysis tasks such as finite element modelling and computational fluid dynamics, enabling engineers to further evaluate product performance virtually during the development process. DHM tools are an extension of CAD tools that incorporate human models into the digital design process for considering human interactions with the designed products, providing early insights to designers (Chaffin, 2005; Ahmed et al., 2021). DHM tools enable testing virtual humans, ensuring that product designs fit the users' requirements in terms of health, comfort, safety, and ergonomics (Scataglini and Paul, 2019). Despite their potential, DHM tools can be expensive, time-consuming, difficult to use, and subjective, limiting the potential insights within the design process (Paul and Wischniewski, 2012; Ranger et al., 2018; Lämkull and Zdrodowski, 2020). While effective for general predictions of possible human motions, DHM simulation accuracy (i.e., similarity to observed motions) is quite low and often fails to reflect realistic behavioural motion variability even when anthropometric diversity is considered (Wolf et al., 2020).

DHM tools in ergonomic design have been used across various design contexts, where ergonomics designers' goals, workflows, and tools differ significantly (Zhu et al., 2019; Hanson et al., 2021). Consequently, it can be difficult to identify the source(s) of ergonomics designers' challenges, especially related to creating consistent and accurate DHM simulations. While DHM tools promise opportunity for more successful ergonomic outcomes in the PD process, it remains unclear which improvements to DHM tools and development processes might enhance ergonomics designers' contribution to the design process during the early stages of PD.

In the automotive industry, simulation software and DHM tools are widely used to analyse and improve vehicle occupant packaging, which refers to the arrangement of vehicle interior space to accommodate drivers and passengers safely and comfortably (Gkikas, 2016). While DHM tools are intended to support the consideration of human aspects early in the design process, their current limitations often force designers to rely heavily on their own technical expertise to improve the obtained simulation results (Lämkull and Zdrodowski, 2020). The improvement of simulation results often involves manual adjustments to get more accurate or realistic driving simulations, which introduce biases and an unspecified level of subjectivity to the results (Chaffin, 2007; Jun et al., 2019). Those manual adjustments defined by DHM users typically consist of constraints that may relate to the geometric features of the vehicle or the digital model of the human itself, but constraints can also be based on assumptions about human behaviour. The manual adjustment processes can be difficult, time-consuming, and subjective, easily causing non-repeatable simulation results. Today, DHM tools still have limitations in considering human diversity and behaviour, and lack reliable methods to simulate, analyse, and evaluate vehicle ergonomics and user accommodation (Bhise, 2016; Brolin et al., 2020). Often, DHM tools predictions are "good enough" but do not necessarily reflect actual human movements, requiring constant evaluation from DHM tools users (Lämkull and Zdrodowski, 2020; Demirel, Ahmed, and Duffy, 2021). Further, predictions produced by DHM tools are hard to evaluate objectively due to the lack of standardized driver and passenger ergonomics assessment methods, defining what is acceptable and what is not, within DHM tools (Schmidt et al., 2023).

The development of methods and models for human motion prediction, as well as improvements in DHM functionality overall, would provide ergonomics designers with deeper insights into human interactions earlier in the PD process, reducing the need for extensive real-world validations. Consequently, reductions in time-consuming validations would lead to time and cost savings in the PD process and possibly improved product design solutions as well. The improved simulation results could also be used as input for other departments or roles involved in the vehicle design process, such as vehicle safety, which uses finite-element-based human body models (HBM) for virtual simulations of vehicle crash scenarios.

1.2 AIM AND RESEARCH QUESTIONS

The presented problems indicate that current approaches struggle to accurately and consistently simulate human-product interactions. To address these issues,

this thesis aims to advance DHM tools through the development and evaluation of methods and models that enhance the functionality and accuracy of human posture and motion simulations, to support decision-making when addressing ergonomics in PD processes.

Here, *functionality* refers to the ability of DHM tools to support relevant ergonomic tasks and simulate human-product interactions effectively, while *accuracy* describes how closely the simulation results match real human behaviour. The research is situated within the domain of virtual ergonomics and simulation, specifically focusing on the use of DHM tools in PD. The context is automotive ergonomics design, one of the most prominent application areas for DHM tools, and where automotive development and manufacturing is a major industry sector in Sweden. While the work originates in this domain, the developed methods and models are intended to be applicable beyond the automotive industry, potentially supporting other fields concerned with human-product interaction. This research is expected to benefit designers, engineers, ergonomists, and product developers, as well as the research community working in the areas of DHM, simulation-based design support tools, and human-product interaction.

The thesis aims to identify areas for further enhancement by addressing knowledge gaps in existing workflows and ergonomics design practices. To structure this work, the thesis aim is divided into three research questions (RQ1-3), each targeting specific challenges: state-of-the-practice DHM usage, simulation functionality, and human motion prediction modelling. The three specific research questions are oriented around at least one objective (O1-4).

RQ1 *DHM tools usage*. How are DHM tools currently used to address ergonomics in PD processes?

- O1 Characterize the use of DHM tools for addressing ergonomics during the PD process.

RQ2 *Simulation functionality*. How can simulation functionality of DHM tools be improved to support decision-making in PD processes?

- O2 Develop methods to improve the functionality of DHM simulations for human-product interaction in PD processes.

RQ3 *Prediction modelling*. How can human posture and motion prediction models be improved and evaluated for DHM tool implementation?

- O3 Develop and evaluate human posture prediction models to increase simulation accuracy in PD processes.
- O4 Develop and evaluate human motion prediction models to increase simulation accuracy in PD processes.

As a first step, RQ1 aims to distinguish how ergonomics designers in industry use DHM tools to address ergonomics in the PD process. Understanding ergonomics designers' objectives, current methods, and tools is essential for a holistic view of the PD process within the automotive industry. Further, the identification of their main challenges during PD clarifies research gaps and guides the development of the research. O1 supports RQ1.

Subsequently, RQ2 and RQ3 aim to investigate and develop potential solutions for improving the functionality and accuracy of ergonomic simulations in PD processes. The limited functionality and accuracy of DHM tools, which constitutes the central research problem of this thesis, may not necessarily be caused by the tools themselves. Instead, the issue may be attributed to missing or using inadequate methods and models for simulating human-product interaction in applications such as automotive ergonomics design. Existing methods for simulating human-product interaction and how they are currently used in ergonomic simulation tools are reviewed to clarify needs and develop methods and models that enable more accurate human-product interaction studies in virtual PD processes.

RQ2 focuses on improving simulation functionality by analysing current procedures and practices for simulating seated driving tasks, aiming to identify how existing methods can be refined, extended, or new methods developed to better support decision-making in DHM applications. The objective is to describe methods as guidelines or procedures, i.e. a series of steps that help DHM users work systematically when digitally analysing human-product interactions. O2 supports RQ2.

Finally, RQ3 goes a step further into the human-product interaction modelling. RQ3 focuses on developing and evaluating posture and motion prediction models. Model improvements may involve the integration of new variables, constraints, or the combination of different modelling approaches. In this context, *models* refer to the mathematical representations of human biomechanical systems, cognitive systems, and designed interaction systems, e.g. car interiors. Additionally, evaluation of the models is also addressed to demonstrate their potential applicability to DHM tools. O3 and O4 supports RQ3.

1.3 LIMITATIONS

This thesis focuses on predicting human postures and motions from a kinematic perspective. Dynamic aspects such as inertia effects caused by masses being subject to accelerations are not considered, even though they influence motion and may be relevant for the occupant packaging context.

The research also excludes modelling of muscle deformation or elastic body responses, which falls outside the scope of this work. While such properties affect posture, comfort, and joint loading, their simulation would require detailed anatomical data and significantly more computational effort.

The developed methods are applied in the context of automotive occupant packaging design. While their applicability in other domains is anticipated, it is not evaluated in the current work.

Finally, this thesis builds on and aims to enhance existing DHM tools without the intention to develop new software. Software modifications were made only when necessary to facilitate implementation or testing.

FRAME OF REFERENCE

2 FRAME OF REFERENCE

This chapter provides the context and theoretical background for the research field of this thesis.

2.1 ERGONOMICS IN PRODUCT DEVELOPMENT

The significant movements within today's design world are technology-driven, sustainable, and human-centred design (Doherty, 2024). Each is characterised by distinct core values: technical innovation, environmental sustainability, and human-centricity, resulting in notable differences in products, systems, or services (Giacomin, 2014; Ulrich et al., 2020). In line with these trends, Industry 5.0 expands upon these design values by explicitly positioning human well-being, sustainability, and resilience as core objectives for future industrial and technological systems. Rather than viewing technology purely as a driver of efficiency, Industry 5.0 advocates for a human-centric approach where technological innovation is aligned with societal needs and values, promoting a symbiotic relationship between humans and digital systems (Bolis et al., 2025).

According to the International Organization for Standardization (ISO), human-centred design is an approach to systems design and development that aims to enhance usability by focusing on the users, their needs, and the context of use, applying knowledge and techniques from human factors, ergonomics, and usability engineering (ISO 9241-11, 2018). While alternative design approaches tend to focus on material qualities or an individual designer's creativity, a distinctive feature of human-centred design is that the people intended to use a product are the central focus of design activities, questions, and insights. Creativity is thus directed towards understanding and addressing users' needs, preferences, and experiences during product interaction. To achieve this, it is essential to understand the different needs of individuals when interacting with a product so that human diversity and variability are adequately considered. Diversity refers to differences between individuals, such as body proportions, age, strength, or cognitive abilities, while variability relates to differences in behaviour, such as changes in posture and motion during task performance.

In human-centred design, human needs are generally categorised into physical, cognitive, and emotional aspects (Wikberg-Nilsson et al., 2021). Physical needs relate to ensuring that the product or system operates safely, reliably, and effectively to support human physical interaction. The cognitive needs relate to users knowing how to interact with and use the product, e.g. the product is easy to understand. Emotional needs focus on creating meaningful solutions that enhance the user's experience, e.g. a product that the users like (Wikberg-Nilsson et al., 2021). Human-centred design processes require collaboration among different design professionals, including ergonomists/human factors experts, user experience and usability specialists, and industrial or engineering designers, in order to develop products satisfying human needs.

Wilson (2000) explains the two aspects that the discipline of ergonomics covers in more detail: (a) the theoretical understanding of all interactions in human-technology systems, and (b) the application of such understanding in design. Thus, the foundation of ergonomics is the understanding of purposive interaction between people and artefacts. This understanding goes beyond describing interactions between a person, task, equipment, and environment, to include an understanding of why behaviours unfold as they do and what factors can affect behaviours. To develop this understanding, ergonomics practitioners often embrace scientific methodologies and approaches. The second aspect, (b), involves improving human interactions and conditions by applying the understanding of human behaviour to design interactions in real settings, through the development of systems, products, and environments that are adapted to users' capabilities, limitations, and needs.

In essence, ergonomics focuses on optimizing the interaction of humans with products and environments, following the principle of adapting the product or environment to human needs and conditions. A product may be a workplace, a task, a tool, equipment, or a job (Wilson, 2000). Ergonomics is a broad field involving several areas since it promotes a holistic approach and considers different factors affecting human well-being. Ergonomics can be divided into three fields or domains of specialisation (IEA, 2025):

- *Physical ergonomics* is concerned with anthropometric, physiological, and biomechanical characteristics.
- *Cognitive ergonomics* is concerned with mental processes and cognitive abilities such as motor response, memory, reasoning, and perception.
- *Organizational ergonomics* is concerned with improving sociotechnical aspects of the system, including organizational structures, processes, and policies.

To support their work across these domains, ergonomics designers rely on a range of tools to assess and refine the interaction between users and systems. These tools include physical prototypes, user tests and interviews, and increasingly, human-product interaction simulations using digital human modelling (DHM) tools (Duffy, 2016; Scataglini and Paul, 2019). The aforementioned tools are employed at early stages of the PD process to proactively evaluate and optimize ergonomic aspects. In the context of the PD process, two key attributes that determine the effectiveness of such simulation tools are functionality and accuracy.

Functionality refers to the "degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions" (ISO/IEC 25010, 2023). In simulation tools, functionality relates to the software's ability to support relevant modelling tasks and replicate human-product interactions in a way that enables users to configure, execute, and analyse ergonomic scenarios effectively for informed decision-making.

Accuracy, conversely, refers to the “closeness of agreement between a test result and the accepted reference value” (ISO 5725-1, 2023). Within simulation, accuracy reflects how closely the software's outputs conform to the real or expected behaviour of human-system interactions, which is essential for generating reliable insights.

Functionality and accuracy are not inherent attributes of a product or system; rather, they are influenced by the design, validation, and implementation of tools. These elements can be improved through effective features and design practices (ISO/IEC 25010, 2023; ISO 5725-1, 2023). Evaluating the functionality and accuracy of simulation tools is essential to ensure relevant and trustworthy results. Such evaluations are typically conducted by researchers or tool developers, and the outcomes can inform ergonomics designers to adapt their processes or choose the appropriate tools to support effective decision-making.

2.2 DIGITAL HUMAN MODELLING

Consider the early stages of PD when designing or optimizing a car interior. At this stage, the car exists only as a digital CAD model. To evaluate how drivers of varying size and proportions might reach the controls, see over the dashboard, or operate pedals comfortably and safely, the engineer turns to a DHM tool.

DHM tools are meant to assist designers, engineers, and other stakeholders in evaluating and optimizing human-product interaction and ergonomic considerations early in the PD process. They are particularly useful when the product exists only as a virtual model (Wolf et al., 2020). As software systems, DHM tools model human characteristics and capabilities, either partially or fully, for further simulation (Wischniewski, 2013; Demirel et al., 2021). By offering digital representations of humans and their interactions with product models, DHM tools enables virtual assessments that reduce the reliance on physical prototypes and their corresponding development time and costs (Scataglini and Paul, 2019).

DHM tools differ from other CAD tools by incorporating a digital human model (often called a manikin, computer manikin, or avatar) that interacts with a digital product model within a digital environment, typically based on CAD models. Understanding product use contexts early in the PD process supports the identification and evaluation of potential design flaws related to human-product interaction. DHM tools can help reduce the risk that such flaws, such as poor reachability, visibility, or comfort, are only discovered in later development stages, when changes may be costly, time-consuming, or even unfeasible to implement. Using DHM tools early in the design process is commonly referred to as *proactive ergonomics* because it enables designers to run "what-if" scenarios in the early stages of the PD process (Ahmed et al., 2019). In such cases, DHM tools support designers and engineers by providing simulations, evaluations, and optimizations that help improve the design and reduce the risk of costly redesigns late in the PD process. Conversely, waiting to identify and resolve ergonomic issues until after they arise in later stages of the PD process, or when the product is produced, can be referred to as *reactive ergonomics*.

Perhaps the most significant benefit of using DHM tools for assessing human-product interaction is their ability to represent the diversity of human anthropometry by enabling the creation and use of manikins of different sizes and proportions in simulations and evaluations.

DHM tools help designers move beyond the common but limited practice of using one, two, or three predefined manikins, often representing a short woman, an average-sized man, and a tall man, to represent an entire population (Bertilsson et al., 2010). While this approach is sometimes assumed sufficient to represent user diversity, it can result in misleading conclusions and designs that inadequately accommodate many intended users. In reality, individuals may fall between or outside these predefined manikins, and anthropometric diversity spans many dimensions beyond stature, weight, and sex (Bertilsson et al., 2010; Holmes and Maeda, 2018). Relying on a small set of representative manikins risks excluding users whose body dimensions or proportions do not align with those selected, potentially resulting in products that accommodate the majority poorly or not at all (Daniels, 1952; Pheasant and Haslegrave, 2006).

Because human body size and shape vary greatly across individuals, it is difficult to capture the full range of anthropometric diversity of humans. However, this can be addressed in DHM tools by generating relevant representative manikins. These manikins are models representing human bodies based on anthropometric databases, which are collections of measurements of human body dimensions, including height, limb lengths, and body composition proportions from specific user groups such as military personnel, or industrial workers (Robinette et al., 2002; Casadei and Kiel, 2025). DHM tools allow users to define manikins based on specific anthropometric measurements, enabling the creation of digital human models that represent individual characteristics. While some tools offer a limited set of dimensions to modify, others allow full customisation of body dimensions, supporting ergonomics evaluations across a broader range of body types. Customisation of human models is typically accomplished parametrically by providing specific measurement values or percentile data. Examples of DHM tools used in the industry, which include customisable human models, are Siemens Jack (Raschke and Cort, 2019), Virtual Ergonomics by Dassault Systemes (Charland, 2019), Ramsis by Human Solutions (Wirsching, 2019), Santos (Abdel-Malek et al., 2019), IPS IMMA (Hanson et al., 2019), and EMA (Bauer et al., 2019). Customisable models provide an intuitive opportunity to understand the interaction of a diverse group of individuals with a product, promoting inclusiveness. For example, manikin families, i.e. predefined sets of digital human models, can be based on one or different population databases, allowing a designer to consider anthropometric diversity efficiently, reducing the risk of major design flaws that affect some users but not those most often considered (Brolin et al., 2019). The ability to test products in digital simulations also eliminates potential risks that real users could suffer during early design evaluations, favouring safety and a sustainable work environment (Settgast et al., 2022).

DHM tools can also support the analysis of biomechanical aspects of the human body for design assessments. Biomechanical human models offer a detailed representation of the musculoskeletal system, enabling static and dynamic human motion analyses. The calculation of these dynamic analyses includes internal stresses and muscle and joint forces. Such biomechanical analyses can be quantified and used as a metric for proactive ergonomic assessment and as an approach for enhancing the design decisions made by design engineers (Rasmussen and Christensen, 2005; Wagner et al., 2008). Examples of biomechanical DHM tools include AnyBody Modelling System (Damsgaard et al., 2006), OpenSim (Seth et al., 2018), and Santos (Abdel-Malek et al., 2019).

Another powerful feature of DHM tools is their capability of generating visual representations of humans interacting with products. This feature includes various aspects that help designers identify, understand, and communicate issues relating to human-product interaction. The first aspect is that DHM tools offer different graphical representations of manikins interacting with the digital environment, from stick figures to more realistic appearances, such as 3D models and animations (Lämkull et al., 2007). A second aspect is the ability to show appearance diversity, i.e. the DHM tool's ability to represent different-looking manikins (Kolbeinsson et al., 2021). Furthermore, a third aspect is the application environment, where the manikin interacting with the product can be viewed.

Even though existing DHM tools provide distinct advantages, they still have limitations. A common assumption in virtual ergonomics and DHM communities is that humans can be modelled and simulated. However, in practice, producing accurate simulations is difficult and not yet attainable because of the complexity of human motions and behaviour. Most DHM tools focus on physical ergonomics and cover limited cognitive and perceptual aspects of human behaviour. Also, human emotion, mental workload, and decision-making variability have still not been adequately included in current DHM tools (Demirel et al., 2021). Despite challenges in creating precise DHM simulations, defining a handful of expected action sequences or static postures within specific product interaction scenarios can often guide the enhancement and assessment of specific ergonomic aspects. As such, DHM users typically specify only key manikin postures. Then, DHM tools can predict manikin transitions between these postures, based on various prediction approaches (Zhu et al., 2019). Often, these predictions are "good enough" but do not necessarily reflect any specific actual human movement.

When DHM tools cannot simulate complex human postures and motions with sufficient accuracy, motion capture and extended reality (XR) technologies can be integrated to support the analysis (Zhu et al., 2019). Examples of DHM tools able to use motion data captured from real humans are AnyBody Modelling System (Damsgaard et al., 2006), Siemens Jack (Raschke and Cort, 2019), EMA (Bauer et al., 2019), and IPS IMMA (Hanson et al., 2019). The main limitation of DHM-integrated motion capture is that, although the recorded movements are highly accurate, aligning these movements and their implications with digital geometries and use contexts in the virtual environment can often be challenging (Qiu et al., 2014). Further, it is important to note that while recorded behaviour in DHM tools exhibits high accuracy, it primarily represents the individual being recorded. The extent to which this information can effectively represent other individuals, which is often required in design, needs to be carefully considered, especially when designing for a broader user base or catering to specific user groups with distinct characteristics. In addition, using these types of technology requires subject studies using a physical or digital prototype. Although physical prototypes are valuable for proactive assessment, they are often unavailable during early design stages of PD due to cost and time constraints. Similarly, interactive digital testing environments, such as those based on XR technologies, still require human participants for user testing, which remains a time-consuming and costly process (Ahmed et al., 2019). Consequently, using technology systems such as motion capture/XR to investigate human movement in the PD process comes with limitations, including potential data loss, limited simulation accuracy, time-consuming procedures, and limited representation of population diversity.

Further challenges arise because DHM tools currently offer limited support for simulating non-anthropometric or non-biomechanical factors that can influence human-product interactions. These factors include, for example, the impact of clothing, personal protective equipment, or gear worn by the user. Manufacturing, military, and similar organisations are particularly interested in the effect of safety equipment on operators or body armour on soldiers, affecting the operator/soldier and how they interact with vehicles and additional equipment. In addition to limitations with modelling clothing, most DHM tools represent objects as rigid, non-deformable bodies. As a result, commonly used tools in automotive design do not fully simulate the effects of deformable materials, such as those used in vehicle seating (Marshall and Summerskill, 2019). Such deformation can impact human-vehicle interactions, leading to unexpected issues related to posture, visibility, or reaching within the vehicle. Some simulation software, like Madymo (Simcenter Madymo, 2025), can model how deformable human bodies interact with elastic materials. However, these tools are mainly used in occupant safety simulations instead of early-stage ergonomics design. These limitations with clothing and rigid deformations complicate the work for those utilising DHM tools and anthropometric data in PD.

Another challenge involves integrating detailed biomechanical models into DHM tools. While most tools still rely on simplified models for estimating joint angles and torques, recent research has introduced more advanced musculoskeletal and dynamic models that simulate muscle activations and joint forces (Ratnakumar et al., 2022; Lamas et al., 2022; Nasr et al., 2024). However, these models remain computationally demanding and are not yet standard in commercial tools. Motion control accuracy is another concern, as simulation uncertainties in velocity and acceleration can reduce biomechanical validity (Cohen and Harrison, 2021). Moreover, current tools rarely account for physical limitations or variations, such as the use of prosthetics, reduced mobility, or asymmetrical body shapes, which limit their effectiveness in designing products that are inclusive and sustainable (Demirel et al., 2021; He et al., 2024).

Finally, while DHM tools can provide important insights into user-centred product design, many still face limitations in both functionality and usability. Functionality issues may include untrustworthiness or inadequate support for simulating realistic human movements, a lack of advanced evaluation methods, or limited integration with other design tools (Demirel et al., 2021). Usability challenges, on the other hand, relate to how these tools are used in practice, such as unstandardised procedures (Paul and Wischniewski, 2012), complex and unintuitive interfaces (Ranger et al., 2018), or time-consuming workflows that reduce efficiency in studying human interaction (Bertilsson et al., 2010; Wischniewski, 2013; Perez and Neumann, 2015). Improving both the functionality and usability of DHM tools would correspondingly enhance their accessibility for proactive ergonomic assessment (Högberg, 2009; Lämkull and Zdrodowski, 2020).

2.3 HUMAN MOTION PREDICTIONS

Specifying interaction models and inaccurate human motion predictions are long-standing challenges for DHM. Back in 1997, Broberg (1997) conducted a survey showing that the potential users of human simulation tools might not have sufficient ergonomic expertise to understand the predicted ergonomic outcomes provided by existing DHM tools. Further, the inability to predict human motions was a driver for the development of the HUMOSIM Laboratory at the University of Michigan in 1998. Some years later, Chaffin (2007) outlined significant issues to address to improve the human simulation functionality of DHM tools. He also highlighted the importance of generating accurate postures and motions, as the subsequent ergonomic and biomechanical analyses otherwise can lead to errors. Lämkull et al. (2009) compared DHM simulation outcomes, specifically predicted postures and task feasibility, with real-world observations from manual vehicle assembly tasks. They found that DHM tools provided accurate predictions for simple, standing, and unconstrained tasks. However, for more complex tasks involving asymmetric postures or constrained conditions, the simulations were less reliable, and expert evaluations were needed to interpret or adjust the results. Even though several authors have proposed different methods to fix or improve human-product interaction simulations (Wolf et al., 2020), it is still reported as a research trend and gap for future directions in recent years for DHM tools (Wischniewski, 2013; Zhu et al., 2019; Demirel et al., 2021; Hanson et al., 2021). The history and ongoing debates around human motion prediction in DHM research areas highlights the importance of the topic and its complexity.

Although current DHM tools provide posture and motion prediction functionalities, their accuracy remains limited. Consequently, users often need to make manual adjustments to attain more realistic simulation outcomes. However, as reported by Lämkull and Zdrodowski (2020), these manual adjustments may introduce several issues, including:

- Inaccurate human postures occur when manikins have to perform tasks interacting with the boundary of reach motion geometries.
- Collision avoidance features usually makes the manikin behave as if those particular geometries were "lethal to touch", which might be necessary for human interactions with harmful equipment, but it is not always the case.
- DHM users do not know the root cause of the awkward or inaccurate body postures, making it more challenging to adjust and define constraints to improve them.
- The procedure for generating human motions with task animation wizards is time-consuming. Often many sub-actions or sub-tasks must be created to make the movement of the manikin more accurate and less "robot-like".
- There is an uncertain bias involved in the results. Manual adjustments rely on DHM users' experience and subjectivity, which varies within and between ergonomics designers' teams.
- Need for validations. The uncertainty of simulation results introduces a need for validation with prototypes, user tests, or interviews to create good products.

In light of these limitations, the visualisation capabilities of DHM tools remain critical in ergonomics analysis, as they help designers to identify potential problems and risks when humans interact with different design solutions (Chaffin, 2007; Zhu et al., 2019). However, the usefulness of these visualisations is reduced when simulations are considered inaccurate or require excessive manual adjustments.

DHM tools must include methods to predict human-product interactions to become a proactive digital tool (Chaffin, 2005; Wolf et al., 2019). While previous studies propose models with different perspectives and terminology related to human-product interaction systems (Chaffin, 2002; Högberg et al., 2019; Wartzack et al., 2019), all of them have in common the following four entities: the human/user/worker, the product/machine/workplace, the environment, and the task/interaction. Figure 2.1 represents the relation between these common entities across different interaction systems.

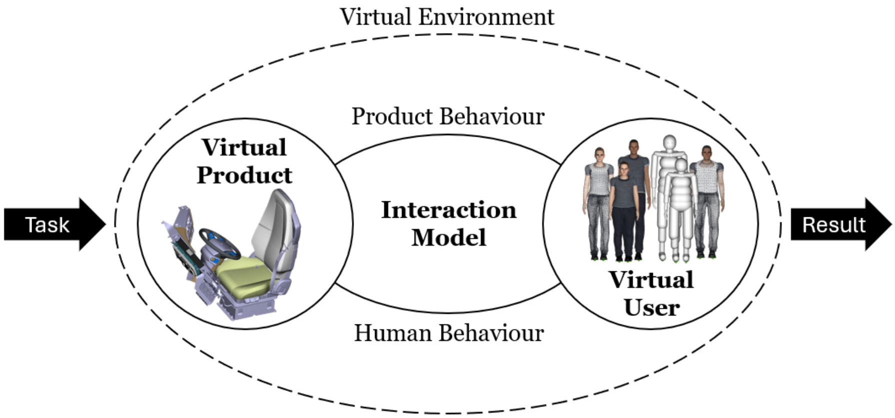

Figure 2.1. Virtual human-product interaction system. (Adapted from Wolf et al., 2020).

The human and product interaction occurs as a constant feedback loop. To properly evaluate the interaction, the prediction method should simulate the human motions accurately considering the entities mentioned above. Human motion prediction covers kinematic and dynamic aspects. Kinematics refers to the translational and rotational motion of points, bodies, and groups of bodies, focusing on the properties of motion without considering its causes, whereas dynamics refers to the study of the causes of motion and the rules governing it (Pentland and Liu, 1999; Aristidou et al., 2018). Therefore, the methods to predict human motions in DHM tools may consider one or both aspects.

Human-product interactions may take different forms, e.g., touching, reaching, looking, or sitting in particular designed contexts. A typical approach in DHM tools is to simulate such interaction by defining target positions on geometries or products for specific body parts of the manikin to move to. However, this is the only information provided to the DHM tool generally, which means that the rest of the manikin body has to be simulated to behave like a real human. The manual definition of each body part of a manikin when interacting with any product would be a very time-consuming and tedious process. Moreover, if all aspects of a human posture could be manually specified with complete accuracy, it would imply that the correct posture is already known in advance, which is rarely the case in design and simulation contexts. The main idea behind proactive methods is that human motions can be predicted accurately given a few input values.

Although posture prediction in DHM involves multiple layers, including behavioural aspects, timing, and obstacle avoidance, the problem can be simplified to the kinematic one by focusing on the skeletal configuration required to reach a target. In this reduced form, the problem resembles what is commonly referred to in robotics as an Inverse Kinematics (IK) problem. An IK problem involves specifying the target of an end effector(s) as input and aims to calculate the posture of a biomechanical model (or articulated model more generally) such that the end effector can reach the target position and conduct the given task. The posture of the articulated model is the output of the problem. Ideally, a solution to an IK problem will provide an appropriate joint configuration of the remaining (uncontrolled) joints of the articulated model for which an end effector(s) moves to the desired target position(s) as accurately, smoothly, and rapidly as possible (Aristidou and Lasenby, 2011; Aristidou et al., 2018). End effectors are specific control points that can include end joints, such as hands or feet, or inner joints, such as the knee or the elbow, of an articulated model, e.g. a human body or human manikin. IK problems may have a single unique solution or many, though they may also have no solution. However, many target positions will have infinite solutions for most chains with more than two degrees of freedom. The challenge of IK problems is identifying which from these multiple or infinite solutions is the correct one, i.e. the closest to a natural human posture. The definition of objectives and constraints, which are not always straightforward and known, is required to get closer to a natural human motion solution, selecting one solution from all the possible ones.

IK problems can be approached in different ways. Various motion modelling methods have been utilised over the years to predict how humans behave in terms of postures and movements. Several authors have attempted to categorise human posture and motion prediction approaches (Chaffin, 2005; Farahani et al., 2015; Wolf et al., 2020). While these approaches generally aim to reduce the number of infinite solutions down to one, they follow different methodologies. These methods fall into the following groups: *data-driven*, *optimization-based*, and *heuristic* methods. The selection of IK solvers mainly depends on the definition and particularities of the problem. Desired smoothness, computational cost, scalability to different models, and the possibility to apply constraints, are several parameters to consider in selecting solvers or methods since they vary across the different types. These methods can be used for kinematic or dynamic human posture and/or motion predictions.

Data-driven (or phenomenological) methods predict human motions based on previously examined observations. A large amount of kinematic data from experiments is statistically analysed to derive predictive models (correlation and regression models) of similar tasks to those captured (Farahani et al., 2015). This approach can also include reinforcement learning and artificial neural network methods. Data-driven methods solve the IK problem by allowing users to input key variables into a regression or a trained model to predict the remaining joint values. The main advantage of this approach is that it can generate postures and motions that appear realistic, because the models are built from large datasets of actual human movement, capturing patterns and variability observed in real-life behaviour. However, these approaches require considerable data to cover human anthropometrics and behavioural variability. Further, data-driven models are limited to predictions of cases that are similar to the data on which the model is based. Another limitation is the restricted ability to analyse individual differences between specific users, such as anthropometry or range of motion (Zhu et al., 2019; Wolf et al., 2020).

Optimization-based methods predict human motions using optimization algorithms. Optimized variables may include kinematic or dynamic objective functions such as joint angle deviation, discomfort, and energy consumption to optimize human motion predictions (Yang et al., 2005). Optimization methods predict the remaining joint values by attempting to fulfil some set of optimization criteria. In theory, the main advantage of using this approach is that it provides accurate predictions of specific movements without previous observations of similar movements. Moreover, it also allows for analysis of the influence of specific additional dimensions, such as external forces or individual characteristics of humans and extra-kinematic/dynamic constraints. While optimization-based methods can be more successful in novel cases than data-driven methods, they also introduce limiting assumptions that may or may not reflect actual human physical behaviour. These assumptions may include the duration of the movement, initial postures and motions, unrealistic optimizations, or constraint definitions. Further, predicting motions under objective functions can produce results that ignore human variability in favour of converging to an optimal solution. Also, the number of objective functions included can make it a highly iterative process with a corresponding increase in computation time and effort (Wartzack et al., 2019; Wolf et al., 2020).

Heuristic methods can be used to predict human motions without using complex calculations. Heuristic algorithms are generally composed of simple operations involving points, distances, angles, and lines in an iterative fashion, leading to an IK solution. Heuristic methods can also handle the IK problem with several end effectors, dividing the articulated model into smaller sub-sections and defining constraints to each joint independently rather than globally (Aristidou and Lasenby, 2011; Aristidou et al., 2018). The main advantages are the low computational cost of providing the solution quickly and their easy implementation and broad applicability to different problems. However, one of their main limitations is the potential for inaccurate motions occurring even with all the joint constraints fulfilled. Another limitation is integrating global constraints to meet spatiotemporal correlations with nearby joints. Cyclic coordinate descent (CCD) and forward and backward reaching IK (FABRIK) are two examples of IK solvers. Figure 2.2 exemplifies a complete iteration with a single effector of the FABRIK solution.

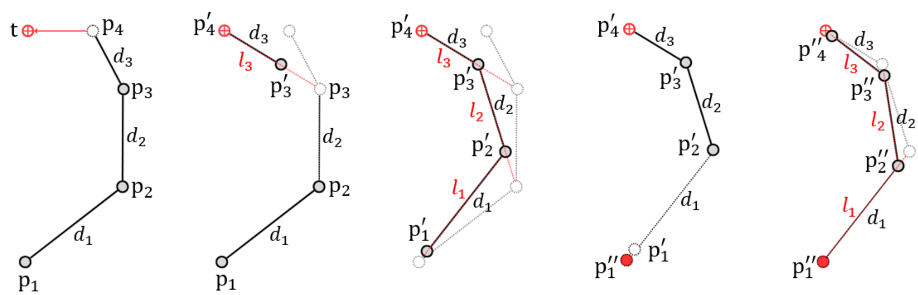

Figure 2.2 Complete iteration of FABRIK solver (Aristidou and Lasenby, 2011). p_4 is the end effector, t the target position, and p_1 is the root or initial position.

According to Wolf et al. (2020), a suitable interaction model for engineering design should cover the following requirements:

1. A proactive/predictive approach requiring little to no previous knowledge or ergonomic expertise.
2. A universally valid approach, which is applicable to the majority of products and use cases.

3. A time-efficient, standardized, and intuitive modelling procedure.
4. A comprehensive and straightforward modelling approach.
5. Integration between computer-aided engineering and DHM tools.

However, the authors also note that no existing human-product interaction model fully meets the ideal criteria for accurate and generalisable predictions in current DHM tools. Data-driven, optimization-based, and heuristic IK methods are currently implemented in various DHM tools individually or in combination (Wolf et al., 2020; Demirel et al., 2021). However, heuristic IK solvers are relatively new and are not widely used for DHM tools. Further research is needed to investigate the suitability of heuristic IK solvers, as well as to identify and define the critical task-specific constraints, such as those influencing driving postures, and evaluate how these constraints interact with different prediction approaches.

2.3.1 PREDICTIONS OF SEATED DRIVING POSTURE

When considering occupant packaging or interior vehicle ergonomics in the automotive development process, it is essential to accurately predict the initial static posture of drivers and passengers. Getting this initial posture right is crucial because the design and development of other ergonomic requirements, such as seated posture comfort, operating controls, and interior and exterior visibility, depend on it. Different seated driving posture models have been developed through both data-driven and optimization methods of human motion prediction.

Data-driven or statistical regression models predict the coordinates or body joint angles of specific positions of humans in the car interior. These regression models consider different parameters such as sex, anthropometry, age, or body symmetry, and vehicles such as vehicle class, seat designs (Reed et al., 2002; Park et al., 2016a). While these models share the same prediction approach, the data collection and analysis methodology varies and is not always adequately described (Schmidt et al., 2014). Other limitations are the prediction of individualities and the difficulty of developing regression models considering all human and vehicle parameters affecting the seated driving posture to cover human and vehicle variability.

Optimization processes aim to minimize expected discomfort by decreasing the deviations from neutral or optimum comfort angles. Over the years, several studies have investigated optimal driver postures and seating comfort (Rune et al., 1994; Porter and Gyi, 1998; Hanson et al., 2006; Park et al., 2014) providing general joint angle recommendations that can be used to define target angles for manikins or as reference points in ergonomic assessments. The main limitation of this approach is that the models do not currently include individual-specific comfort joint angles based on anthropometry variables (Brolin et al., 2020). These studies typically define a general set of preferred joint angles for the overall population, without differentiating between individuals based on anthropometric or behavioural variability. This limits the applicability of optimization methods, as real users may adopt different postures even in identical conditions, depending on body dimensions, preferences, or task strategies.

A notable example of a posture prediction system combining both data-driven and optimization-based methods is RAMSIS, which is widely regarded as the leading DHM tool for occupant packaging in the automotive industry (Wirsching, 2019). In practice, the system first uses empirical data from real driving posture studies to derive representative target postures, and then applies a constrained optimization process, where posture prediction is guided by both geometric task constraints (e.g., foot on pedal, hands on steering wheel) and posture probability functions derived from empirical studies (Park et al., 2011; Wirsching, 2019).

2.3.2 HUMAN MOTION PREDICTION IN OTHER FIELDS

Human motion prediction receives significant attention across various research fields beyond ergonomics within product and production development. Understanding and modelling human motions has garnered the interest of researchers from different disciplines, including robotics, animation, psychology, and cognitive science (Rudenko et al., 2020; Lyu et al., 2022). Research from other research fields targeting human motion analysis and prediction can provide a deeper understanding of human motion while providing knowledge about new methods and models for improving human motion simulations within DHM tools.

In cognitive science, researchers have recognized the significance of predicting human motions for a wide range of applications. They have sought to understand the underlying cognitive processes and decision-making mechanisms that drive human motion (Kelso, 1995; Warren et al., 2001; Lamb et al., 2017; Nalepka et al., 2019). By developing models that simulate human behaviour, these fields aim to enhance the predictive capabilities of human motion simulations.

One example of a dynamic model used in cognitive science is the Time-to-Collision (TTC) model proposed by Lee (1976). The TTC model considers information about the relative motion of objects to estimate the time remaining before a potential collision occurs. By incorporating this model into human motion prediction, researchers have been able to anticipate human reactions and movements in critical situations, contributing to safer designs and interventions (Lee, 1976; Kiefer et al., 2006). Furthermore, the Steering Dynamics Model (SDM) has been a subject of interest within cognitive science. The SDM demonstrates how one can move towards desired locations avoiding obstacles (Warren et al., 2001). The authors of these models believe there is a need to integrate dynamical and information-based approaches for more complex, adaptive behaviour depending on interactions with the environment while following IK algorithms. However, it is important to note that the SDM has primarily been developed in a two-dimensional (2D) space context, limiting its applicability in complex real-world scenarios.

Despite the advancements in the various fields, their potential contributions to DHM tools and ergonomics within product and production development contexts are yet to be fully realised. By considering the diverse range of studies conducted in cognitive science, which investigates human decision-making, perception, and motor control, the capabilities of DHM tools can be enriched. Integrating and developing existing models and knowledge into DHM tools can improve understanding and accuracy while accounting for human behavioural diversity, ultimately enhancing ergonomics in PD.

RESEARCH METHODOLOGY

3 RESEARCH METHODOLOGY

This chapter presents the research methodologies applied to address the three research questions guiding this thesis.

3.1 METHOD OVERVIEW

Design Science is the dominant research approach applied throughout the research described in this thesis. Design Science emphasises the development and evaluation of artefacts to tackle practical issues while generating knowledge about their impact (Hevner et al., 2004; Johannesson and Perjons, 2014). Artefacts, which may include constructs, models, methods, or instantiations, can involve developing an entirely new artefact or enhancing an existing one (Hevner et al., 2004). The artefacts in this thesis consist of methods and models that aim to improve the functionality and accuracy of DHM simulations in support of ergonomics-related decision-making during PD. This research was conducted in close collaboration with Swedish automotive companies, ensuring that the developed artefacts are grounded in current industrial challenges and consider multiple sources and stakeholders.

The following sections provide an overview of the different research methods used to address the research questions. The nature of each research question has guided the selection of methods. Detailed descriptions of how the methods were applied are presented in the methods sections of the respective appended papers.

3.2 INTERVIEWS

Interviews are a particular kind of conversation generally led by a researcher covering a specific topic of interest to generate material for research purposes. As a data generation method, interviews are particularly suitable for obtaining rich, detailed information, posing complex questions, and exploring feelings and experiences that are not easily observable. They can be divided into three different types: structured, which follow a fixed set of predetermined questions; semi-structured, combining predefined questions with the flexibility to explore emerging topics in greater depth; and unstructured interviews, allowing the conversation to evolve freely based on the participant’s responses (Oates, 2006; Meuser and Nagel, 2009).

In Paper A, a semi-structured interview study was done to help identify the gaps in current ergonomics designers' work practices. The interview study involved a list of themes and questions to be covered while offering flexibility for changes in the order and including additional questions if the flow of the conversation requires it (Ahmed, 2007). The ten participants involved in the study are considered experts following Littig's (2009) criteria since each had specialized knowledge in the specific domain of ergonomics PD/user-centred PD.

3.3 FIELDWORK

Fieldwork or field research is a qualitative method that aims to observe, interact with, and understand people while they are in their natural environment or, as in this case, doing their work activities. It relies on personal interaction or engagement between the researcher and those being researched, allowing for the observation of practices and challenges that might otherwise be inaccessible through other research methods (Pole and Hillyard, 2016).

Fieldwork was used throughout the development of this research to understand the needs and critical challenges of industry partners while developing the methods and models and receiving continuous feedback for refining them. This took form in continuous and engaged presence in meetings, workshops, and demos with associate companies contributing to Papers B, C, D, and E.

3.4 POSE DETECTION AND MOTION CAPTURE

Pose detection and motion capture are research methods used to collect information about the position or movement of humans and objects in space. Pose detection refers to capturing static configurations or postures. In contrast, motion capture refers to the technology used to track and record movements. It typically involves systems that measure positions, orientations, and velocities through various sensor technologies, including optical cameras with or without markers and inertial measurement units (IMUs). Motion capture systems enable the collection of accurate, objective, and high-volume observational data, enhancing traditional observational methods by offering quantitative insight into movement patterns and behaviours (Menolotto et al., 2020).

In this thesis, recorded posture and motion capture data provided by collaboration partners, Volvo Cars and Texas Tech University, was used to support the development and evaluation of posture and motion prediction models.

Driving posture data collected by Volvo Cars in two different vehicle models were used in Paper D and Paper E. Test participants, recruited from Volvo Cars staff, were instructed to adjust the seat and steering wheel to their preferred configuration before completing a 10-minute driving session in real traffic. The study included 104 participants for Vehicle A (collected in 2017) and 95 for Vehicle B (collected in 2018). These static postural data were collected using manual measurements and onboard camera systems, following an established industrial procedure, and were used as ground truth for evaluating posture prediction methods.

Paper F incorporated experimental data collected by Texas Tech University to evaluate motion prediction methods for reaching tasks. Ten right-handed participants performed four reaching tasks while assuming a seated driving posture. Upper-body motion was captured using a motion capture system with eight infrared cameras at a sampling frequency of 60 Hz, with retroreflective markers placed on the right upper body. These data were used as ground truth for evaluating motion prediction methods.

3.5 MODELLING

Modelling is the process of creating a model, a simplified representation of a real-world system. A model helps to understand how a system works, how it can change, and what effects these changes may have. It should capture the important parts of the system but still be simple to work with. In research, models are used to test ideas, predict behaviours, and support decision-making. Models can have different forms (conceptual, logical, etc.), but those intended for simulation software are typically mathematical. They are often built using data, assumptions, and theories about the system. A good model finds a balance between being realistic and being manageable (Maria, 1997).

This research involved various models to predict and simulate human postures and motions. The subsequent sections outline the different modelling approaches used.

3.5.1 CONCEPTUAL MODELLING

Conceptual modelling focuses on creating high-level representations of ideas, frameworks, or relationships within a system. These models are typically descriptive rather than mathematical. They help to structure problems, guide additional model development, and communicate complex ideas clearly. Conceptual models can later form the basis for more detailed mathematical or computational models (Maria, 1997; Robinson et al., 2015).

In this thesis, conceptual modelling was applied in Papers B, C, and E. In Paper B, the "Body Shape Alignment" concept was introduced to improve posture prediction by considering body surface contact, which facilitates a more controlled simulation of the manikins. In Paper C, a conceptual structure was presented for how simulation in DHM tools can be used to handle multiple, and sometimes conflicting, ergonomic goals. This structure was then implemented as a simulation-based optimization method. In Paper E, a framework was proposed to expand datasets using simulation, allowing the development of more tailored optimization models for different anthropometric profiles.

3.5.2 STATISTICAL MODELLING

Statistical modelling is based on analysing data to find relationships between variables. In this approach, models are built using observations from real-world measurements. The goal is to create equations that predict outcomes, such as a person’s preferred seating position based on their body proportions (Park et al., 2016a, 2016b).

Papers B, C, and D used statistical models to predict key measurements in driving posture (such as the H-point, eye-point, steering wheel position, or joint angles). In Paper B, statistical models were used to predict preferred seating positions (H-point to hip joint relationship). In Paper C, statistical models were used as cost functions to guide the optimization process. In Paper D, statistical models were compared against optimization-based predictions for driving seated postures.

3.5.3 OPTIMIZATION MODELLING

Optimization-based modelling aims to find the best possible solution for a given problem. This is done by defining one or several objectives, such as minimising discomfort, and applying mathematical methods to find the best configuration. The model uses constraints, such as joint limits or reachability, to ensure realistic solutions (Xiang et al., 2010; Yang et al., 2011; Zaman et al., 2021).

Papers C, D, E, and F used optimization models. In Paper C, a multi-objective optimization approach was applied to predict seat and steering wheel locations, considering multiple ergonomic objectives. This approach was implemented using the Non-dominated Sorting Genetic Algorithm II (NSGA-II), which is well suited for solving complex design problems with conflicting objectives (Deb et al., 2002). In Paper D, optimization models were compared against statistical models for predicting seated driving postures. In Paper E, optimization models were developed to tailor posture prediction to specific anthropometric groups. In Paper F, optimization was used to predict reaching motions and was evaluated against a heuristic method.

3.5.4 HEURISTIC IK MODELLING

Heuristic IK modelling involves using simple rules instead of complex equations for finding a configuration of joint angles that allows a body to reach a target point. These rules are based on observations of how humans usually move, such as preferring smaller movements or avoiding awkward postures (Aristidou and Lasenby, 2011).

In Paper F, a heuristic IK model was combined with a velocity and a path planning model to predict hand reaching motions.

3.5.5 BEHAVIOURAL DYNAMICS MODELLING

Behavioural dynamic modelling, which comes from cognitive science, focuses on modelling how people plan and perform actions over time. This approach uses mathematics, typically differential equations, to describe how movement decisions are made and updated as a person interacts with their environment (Fajen and Warren, 2003, 2007).

In Paper F, a behavioural dynamics model was used alongside a heuristic IK model to simulate hand reaching motions.

3.6 SIMULATION

Simulation is a research method used to imitate the operation of a real-world system over time by executing a model. While modelling refers to the creation of the system representation, simulation is the process of using that model to observe and analyse system behaviour under different conditions. Simulation is especially useful when it is difficult, expensive, or impossible to experiment directly with the real system. Through simulation, researchers can predict system behaviour and outcomes, evaluate alternatives, and support decision-making processes (Maria, 1997; Lorig, 2019).

In this thesis, simulation was used to develop and evaluate human posture and motion prediction in the context of product design to improve the functionality and accuracy of DHM tools. As described in Chapter 2, DHM aims to simulate how humans will interact with products during early phases of PD using virtual environments. DHM tools use different types of models to simulate how people interact with designs. In this research, the different modelling approaches presented in the previous section (statistical modelling, optimization-based modelling, heuristic inverse kinematics modelling, and behavioural modelling) were tested and evaluated using simulation.

3.6.1 SIMULATION EXPERIMENT

A simulation experiment is a planned series of simulation runs. The goal is to systematically change the input variables of a simulation model and observe the effects on the outputs. Simulation experiments are used to answer research questions, compare different system configurations, or evaluate model performance (Maria, 1997; Carson and Maria, 1997; Lorig, 2019).

Simulation experiments were conducted in Papers B, D, E, and F. Paper B used a simulation experiment to compare a new conceptual model against statistical models. In Paper D, different posture prediction methods were tested against real driving data. In Paper E, a simulation experiment supported the development of tailored optimization models. In Paper F, motion prediction methods were compared through simulation experiments evaluating reaching tasks.

3.6.2 SIMULATION-BASED MULTI-OBJECTIVE OPTIMIZATION

Simulation-based multi-objective optimization (SBMOO) is a specific type of simulation experiment that focuses on finding the best solutions considering multiple objectives at the same time. Instead of just observing how the system behaves, the simulation is combined with an optimization strategy that guides the search for the best input settings. The optimization process uses feedback from the simulation results to adjust and improve the input values over several iterations.

In Paper C, SBMOO was used to find the best seat and steering wheel positions by balancing multiple conflicting ergonomic objectives while considering constraints. As described in Section 3.5.3, this approach was implemented using a genetic algorithm (NSGA-II), which supported exploring optimal solutions across diverse anthropometric profiles.

3.7 DATA ANALYSIS

This research applied qualitative and quantitative data analysis methods to answer the research questions. Qualitative analysis helped to interpret data collected through interviews and field observations, while quantitative analysis was used to evaluate and compare model performance in simulation experiments against experimental data collected. The choice of method depended on the nature of the data and the research aim of each study.

3.7.1 QUALITATIVE ANALYSIS

Qualitative analysis focuses on understanding patterns, experiences, and behaviours based on non-numerical data. It allows for a deep exploration of complex human activities by systematically interpreting meanings, contexts, and structures in the data. Qualitative content analysis follows a rule-based but flexible process, combining systematic coding with interpretation. Categories are developed to organise and interpret the material, either starting from theory or emerging from the data itself. In this way, qualitative analysis balances openness to new findings with the need for structured analysis (Mayring, 2014).

Paper A used qualitative analysis, which involved semi-structured interviews with ergonomics designers in industry to understand current work practices, identify challenges, and highlight areas for improvement in ergonomics design. The collected data were analysed using a coding process, using predefined categories based on the interview structure, although the process remained flexible. New subcategories were created when important insights emerged that were not captured by the original categories. In addition, qualitative data from fieldwork activities with industry partners were also analysed. Notes, diagrams, and observations were reviewed to identify recurring patterns in workflow, tool usage, and ergonomic concerns in practice. These insights helped to frame the problem space and define requirements for the development of simulation models and methods in Papers B, C, D, and E.

3.7.2 QUANTITATIVE ANALYSIS

Quantitative analysis focuses on processing numerical data to assess relationships, differences, and model performance. Several types of quantitative analysis were applied in this research.

3.7.2.1 Statistical and Error Analysis

Descriptive statistics (mean and standard deviation), error metrics (absolute error, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE)), and inferential statistics (Correlation coefficient) metrics were used to assess the overall accuracy of posture and motion predictions. Additionally, the Bland-Altman method was applied to evaluate agreement between predictions and experimental measurements. This method plots the difference between two sets of values against their average, allowing for the identification of systematic bias and calculation of the limits of agreement (Giavarina, 2015; Doğan, 2018).

These statistical and error analysis metrics were applied in Papers B, C, D, E, and F to quantify the prediction error and strength of the relationship between variables. The Bland-Altman method was used in Paper D to evaluate how predicted driving postures aligned with experimental data.

3.7.2.2 Dynamic Time Warping (DTW)

Dynamic Time Warping (DTW) is a method for comparing time-series data. Unlike traditional point-by-point comparison measures, DTW can align two sequences that may vary in timing or speed, making it sensitive to the overall shape of the motion (Senin, 2008; Cleasby et al., 2019).

In Paper F, DTW was used to evaluate the shape similarity between predicted and experimental motion trajectories. The application of DTW is a novel innovation arising from this thesis work within the context of ergonomics and DHM, with the aim of improving accuracy measurements and comparisons between experimental and simulated motions. The application of DTW also allowed for novel insights into trajectory shape relationships that are not provided by common statistical and error analyses.

3.7.2.3 Variability Analysis

Variability analysis was conducted to evaluate how well the predictive models reproduced the natural differences observed between participants. Instead of only looking at average errors between predicted and experimental data, this analysis compared the variability across all subjects within the experimental data and then compared it to the variability calculated across the subjects for the simulated data. Pairwise comparisons of RMSE and DTW values were used within each group (experimental data and model predictions) to assess modelled variability.

This method helped to understand if the simulation models could replicate the natural variability observed in human behaviour. Variability analysis was introduced and applied in Paper F.

3.8 SUMMARY OF RESEARCH DESIGN

To summarise the methodological approach applied in this thesis, Table 3.1 provides an overview of the research methods, data sources, and analysis techniques used to address each research question. This cross-comparison reflects how the methodology evolved in response to each research objective's specific challenges and data characteristics as well as the combination of qualitative and quantitative approaches applied throughout the research.

Table 3.1. Overview of research methods and analysis techniques applied in the thesis, per appended paper.

Papers	Research Method	Data Analysis
Paper A	Interview Fieldwork	Qualitative Analysis: Coding and category analysis
Paper B	Fieldwork Modelling (Conceptual) Simulation experiment (Conceptual model vs Statistical model for posture prediction)	Quantitative Analysis: Mean, Standard Deviation, Correlation, MAE, RMSE
Paper C	Fieldwork Modelling (Optimization) SBMOO for occupant packaging design	Quantitative Analysis: Number of constraint violations, Number of unacceptable joint angles

Paper D	Fieldwork Simulation experiment (Statistical model vs Optimization model for posture prediction)	Quantitative Analysis: Mean, Standard Deviation, Correlation, MAE, RMSE, Bland-Altman Analysis
Paper E	Fieldwork Modelling (Conceptual) Simulation experiment (Optimization model vs Anthro-Optimization model for posture prediction)	Quantitative Analysis: Mean, Standard Deviation, RMSE
Paper F	Modelling (Heuristic) Simulation experiment (Optimization model vs Heuristic model for motion prediction)	Quantitative Analysis: Accuracy: RMSE, DTW Variability analysis: RMSE, DTW

3.9 ETHICAL CONSIDERATIONS

The research area of virtual ergonomics includes both social and applied science, which means that human subjects are directly involved in this research. To ensure ethical integrity throughout the project, the research followed the ethical guidelines established by the Human Factors and Ergonomics Society (Code of Ethics HFES, 2020). Special care was taken to protect participant rights through informed consent, secure data management, and remaining aware of potential risks such as bias in modelling or the misinterpretation of results. Issues related to research misconduct were proactively avoided by maintaining transparency and traceability in all development and evaluation stages. Ethical considerations were addressed at different stages of the project, particularly during:

- Data collection through interviews and motion capture systems (with attention to informed consent, privacy, and participant safety)
- The development of predictive models based on empirical data (with awareness of possible limitations in accuracy)
- The simulation of human behaviour (acknowledging that predictions may not represent the full variability of human motions)

While the aim of this research was to advance DHM tools through the development and evaluation of methods and models that enhance the functionality and accuracy of human posture and motion simulations, not all factors influencing human-vehicle interaction could be captured. It is important to acknowledge that the developed models may not include all possible postures or motions, and that atypical behaviours or outlier cases might have been excluded from the simulation scope.

RESULTS

4 RESULTS

This chapter presents and summarises the six appended papers, which form the thesis results. Before summarising each appended paper, Figure 4.1 illustrates how the papers connect to the overarching research questions (RQs) and objectives (Os). The research progressed sequentially: Papers A and B first addressed RQ1, providing insights into current practices and challenges in applying DHM tools for ergonomics design. These findings highlighted limitations in simulation accuracy and motivated a shift in focus toward RQ2 and RQ3. RQ2 is supported by Papers B and C, which explore improvements to simulation functionality. RQ3 is split into two objectives: O3 (posture modelling) and O4 (motion modelling). O3 is addressed in Papers D and E, and O4 is addressed in Paper F. Additionally, the concept model introduced in Paper B was reused and developed further in Papers C, D, and E (represented by dashed lines in Figure 4.1). Results from Paper D also informed the work in Paper E, showing a progression in the research flow.

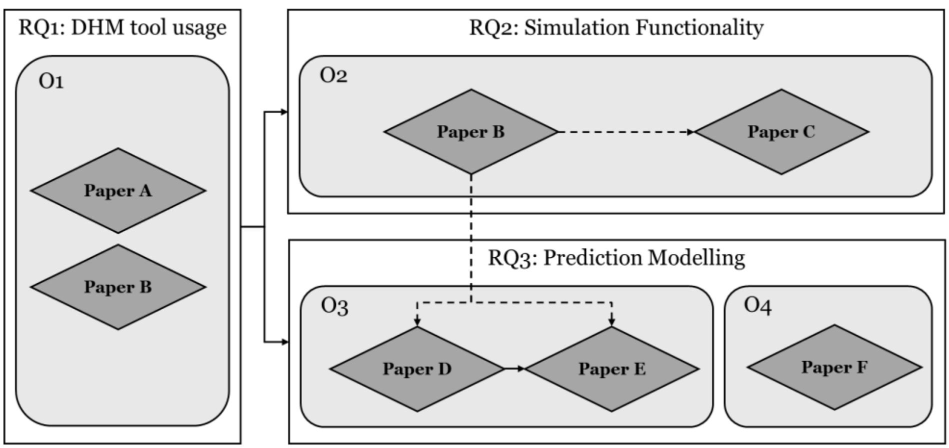

Figure 4.1. Connection between appended papers to research questions (RQs) and objectives (Os).

Each appended paper is summarised by describing its aim, the methods used, the key results obtained, and the significance of the study within the context of the field and the thesis work.

4.1 PAPER A

Perez Luque, E., Brolin, E., Högberg, D. & Lamb, M. (2022). Challenges for the Consideration of Ergonomics in Product Development in the Swedish Automotive Industry – An Interview Study. Proceedings of the Design Society (DESIGN2022), 2, 2165–2174. https://doi.org/10.1017/pds.2022.219

Paper A aimed to understand how ergonomics is considered in the Swedish automotive PD process. Specifically, in identifying how ergonomics designers work, which tools and methods they use, what challenges they face, and what their ideal future tools would be.

The study was based on semi-structured interviews with ten ergonomics designers from seven different Swedish automotive companies. Participants were considered experts, with at least five years of experience in ergonomics design in PD. Each interview, lasting between 60 and 75 minutes, followed a structured guideline covering objectives, workflows, tools, ideal tool features, and challenges. The collected data were transcribed and analysed following a categorisation and codification procedure.

The study found that the main objective of ergonomics designers is to achieve an appropriate balance between different criteria affecting the design. This balance must be achieved both between ergonomics aspects (such as comfort, posture, and visibility) and between ergonomics and other PD aspects (like safety, aesthetics, and manufacturing). Their workflow follows the typical stages of PD processes: from defining requirements, making predictions and analysis, verifying, validating, and communicating results. The most common tools were physical prototypes, DHM tools, and virtual reality (VR) to assess ergonomics aspects such as visibility, reachability, and comfort. However, DHM tools were described as time-consuming, difficult to use, and often providing inaccurate human posture predictions. Participants envisioned an ideal tool that seamlessly integrates DHM, VR, and motion capture, and provides more reliable behavioural predictions, objective ergonomics evaluation methods, and better usability. Lastly, several challenges were reported, including tool-related and interdepartmental communication issues.

Based on the interview results, four main gaps were identified for future improvement of DHM tools: human behaviour predictions, simulation tool usability, ergonomics evaluation methods, and seamless integration between different systems.

This study highlights limitations in current practices for integrating ergonomics early and effectively into vehicle design. It reveals a strong industry need for more reliable, efficient, and user-friendly simulation tools, emphasising the importance of improving DHM technologies to better support human-centred vehicle design processes.

4.2 PAPER B

Perez Luque, E., Brolin, E., Lamb, M. & Högberg, D. (2022). Simulation of Hip Joint Location for Occupant Packaging Design. Proceedings of the 7th International Digital Human Modeling Symposium (DHM2022), 7, 34.
https://doi.org/10.17077/dhm.31742

Building on the limitations reported in Paper A, particularly regarding manual adjustments and alignment issues in DHM simulations, Paper B aimed to investigate a specific challenge in DHM tools regarding driving seated posture prediction: the lack of a clear and standardized relationship between vehicle reference points (such as the H-point) and human biomechanical joints (specifically the mid-hip). The H-point, widely used as a standardized reference in vehicle design, describes a theoretical intersection of a reference occupant's torso and thigh lines (Gkikas, 2016). However, the H-point only approximates the location of the human anatomical hip joint (mid-hip) and does not fully capture human variability.

This misalignment between vehicle design standards (e.g., SAE J826 defining the H-point) and human biomechanical modelling limits the accuracy of simulations using DHM tools and presents limitations in current practices. It was observed through workshop discussions with automotive companies that DHM users typically follow the same standard procedure: once the manikin family has been defined to represent the target user group, 1) the manikin family assumes a default driving posture, generally defined by the DHM software following specific angles according to a particular study; 2) constraints are set to fulfil basic requirements such as "feet on the pedals" and "hands on the steering wheel"; 3) finally, manual adjustments, typically in the mid-hip and/or torso, are often applied to better align the manikin's body with the seat. As reported in Paper A, this third step is commonly introduced by practitioners themselves, outside of standard tool procedures, and may introduce variability and potential bias in posture prediction results. The same process is typically repeated for the remaining manikins in the manikin family.

To address challenges in alignment procedures, a conceptual method named Body Shape Alignment was developed as an alternative, aiming to reduce subjectivity while considering the occupant packaging standards. The Body Shape Alignment method uses a virtual copy of the physical H-point machine (HPM) as a template (SAE J4002, 2022). In the virtual environment, the H-point of the HPM is first aligned with the Seat Reference Point, and then the manikin body meshes are aligned to the HPM template surfaces.

The Body Shape Alignment method was compared to a commonly used Statistical Prediction method (Park et al., 2016b), which relies on regression models to predict mid-hip locations relative to vehicle reference points. Both methods, the Statistical Prediction method and the Body Shape Alignment method, were compared using 14 realistic human body meshes that represent a wide range of body mass indexes (BMIs) from the BioHuman project (UMTRI HumanShape™, 2020). These meshes were generated from CAESAR anthropometric data (Robinette et al., 2002) and simulated within a virtual car cabin to analyse the mid-hip to H-point analysis offsets.

The Body Shape Alignment method showed a more consistent and realistic positioning of manikins across different BMIs compared to the Statistical Prediction method. In particular, the Statistical Prediction method often resulted in unrealistic gaps between the seat and manikins with higher BMI, which would require manual corrections. Although promising for helping to predict mid-hip to H-point location of the automotive seat in a standardized way, the Body Shape Alignment still had limitations, such as potential subjectivity during manual alignment steps and dependency on the availability and accuracy of high-quality human body meshes.

This paper highlights that current DHM methods inadequately address the relationship between human anatomy and seat design reference points, particularly across diverse body types. The proposed Body Shape Alignment method represents a step toward standardizing and improving seated posture simulations, potentially reducing manual adjustments, improving repeatability, and enhancing communication of ergonomics requirements during the automotive PD process.

4.3 PAPER C

Perez Luque, E., Iriondo Pascual, A., Högberg, D., Lamb, M. & Brolin, E. (2025). Simulation-based Multi-objective Optimization Combined with a DHM tool for Occupant Packaging Design. International Journal of Industrial Ergonomics, 105, 103690. https://doi.org/10.1016/j.ergon.2024.103690

Paper C aimed to investigate how combining simulation-based multi-objective optimization (SBMOO) with a DHM tool can improve the occupant packaging design process. The paper explored whether this method could better address human variability and conflicting ergonomic objectives in vehicle interior design, while reducing the manual, time-consuming, and subjective nature of current practices.

The SBMOO method was applied to a fabricated occupant packaging design scenario. It focused on optimizing five variables: seat adjustment range (fore-aft and vertical location), steering wheel position (fore-aft and vertical location), and seat bottom angle. In addition, four design constraints were applied: head-to-roof distance, knee-to-vehicle distance, thigh-to-steering wheel distance, and down vision angle.

A family of 14 digital manikins, generated using the anthropometric module in IPS IMMA, was defined to represent a wide range of anthropometric characteristics. The manikins were created based on the boundary ellipsoids method developed by (Brolin, 2016), which enables the representation of anthropometric variation of several key dimensions. Each manikin had a different combination of body dimensions, resulting in individual sets of optimal joint angles according to the statistical models developed by Park et al. (2016b). These predicted optimal joint angles, along with acceptable deviation ranges defined by root mean squared error (RMSE) values from the same study, were used to formulate the objective functions. For each manikin, the goal was to minimize the number of simulated joint angles that fell outside the acceptable range. The evaluated joint angles included hip-to-eye, head, neck, thorax, abdomen, pelvis, thigh, and knee.

The genetic algorithm NSGA-II (Deb et al., 2002) was applied to automatically and iteratively generate and evaluate 1000 design variants through simulations in the DHM tool IPS IMMA. Each design variant was evaluated through a structured simulation process. For every variant, 25 equidistant seat positions within the seat adjustment range and 3 different torso angles (0°, 15°, and 30°) were defined. This resulted in 75 posture configurations per manikin, and with 14 manikins, a total of 1,050 sub-simulations were performed for each design variant. Each sub-simulation evaluated both posture acceptability (based on joint angles) and fulfilment of the design constraints.

The SBMOO method demonstrated the ability to find occupant packaging design solutions that achieved a balance between conflicting ergonomic objectives while considering constraints. The method enabled the automatic exploration of a wide design space considering anthropometric diversity at an individual level. Rather than requiring users to simulate each combination manually, the SBMOO method allows ergonomics designers to define the design problem (specifying objectives, constraints, and key variables) and delegate the exploration of the design space to the optimization algorithm, testing thousands of design variants automatically, efficiently, and systematically. This showed that the SBMOO method could improve the effectiveness of the design process by reducing manual simulations and adjustments, enhancing simulation traceability, and minimizing subjectivity, which could result in allowing ergonomics designers to focus more on evaluating results and making informed decisions.

This study demonstrates that integrating SBMOO with DHM tools can make occupant packaging design processes more effective and systematic. It shows that it is possible to automatically balance several ergonomic aspects for a diverse population, rather than relying on manual trial-and-error adjustments. Although the study used a fabricated scenario, it provided a potential foundation for future work. The method could be extended to real vehicle design cases, and with objectives representing additional non-ergonomic features. Thus, possibly supporting automotive companies to optimize ergonomics and safety aspects in a concurrent fashion at early virtual stages in the PD process.

4.4 PAPER D

Perez Luque, E., Brolin, E., Nurbo, P., Lamb, M. & Högberg, D. (2025). Comparison of Driving Posture and Position Prediction Methods for Occupant Packaging Design. Under Review Process in the International Journal of Industrial Ergonomics.

Paper D compared two methods for predicting seated driving postures in vehicle occupant packaging design: a Statistical Prediction Method (SPM) and an Optimization Prediction Method (OPM). The objective was to evaluate how well each method predicts key driver position variables (H-point, eye-point, and steering wheel position) compared to real-world measurements, and to explore how their strengths and weaknesses vary depending on vehicle configuration.

Driving posture and position data were collected from 199 participants in a user study conducted by Volvo Cars in two vehicle models with different seat heights. Test participants were instructed to adjust their seat and steering wheel positions to preferred settings during repeated driving sessions in real traffic situations. Key points such as the H-point, eye-point, and steering wheel position were measured and used as ground truth to evaluate the two posture prediction methods.

The SPM used a cascade modelling approach based on linear regression equations, predicting first the steering wheel position, then the H-point, and finally the eye-point, based on driver anthropometrics and vehicle measurements.

The OPM, implemented in the DHM tool IPS IMMA, predicted postures by optimizing joint angles toward ergonomic comfort while considering geometric constraints.

Predictions from both methods were evaluated against the collected real-world data using metrics like average error, correlation, MAE, and RMSE for the key driver points along both x (fore-aft direction) and z (vertical direction) axes.

The comparison between the two prediction methods showed that their performance varied depending on the specific prediction point and the vehicle configuration. Table 4.1 shows a summary of the best-performing method for each key driver point and vehicle.

Table 4.1. Best performing method for key driver points and vehicles.

Prediction point	Vehicle A	Vehicle B
H-point x	OPM	OPM
H-point z	SPM	SPM
Eye-point x	OPM	SPM
Eye-point z	OPM	SPM
Steering wheel x	OPM	OPM

OPM predicted H-point x and Steering wheel x better across both vehicles. In addition, OPM also performed better for predicting Eye-point x and Eye-point z in Vehicle A, making it the more accurate method for predicting Vehicle A, which had a lower seat height configuration. In contrast, SPM demonstrated better performance in predicting H-point z for both vehicles. Additionally, SPM showed better predictions for both Eye-point x and z in Vehicle B, indicating that it is the most accurate method for this type of vehicle, i.e. which had a higher seat height configuration.

This paper points out important directions for future improvements in posture prediction methods, such as refining optimal joint angle models for different user groups and vehicle types, and better capturing human variability in posture prediction. These insights are essential for improving the accuracy of DHM tools in vehicle ergonomics.

4.5 PAPER E

Perez Luque, E., Brolin, E. & Nurbo, P. (2024). Improving Occupant Packaging Posture Prediction through Integration of Simulation and Real Data. Accepted and presented at the IEA2024 Conference and awaiting Proceedings Publication.

Paper E developed a process to enhance driver posture prediction in DHM tools by integrating real-world driving posture data with simulation. The objective was to address the limitations caused by incomplete datasets and improve the predictive accuracy of seated driver postures, particularly by tailoring predictions to specific population groups.

The proposed process combining real-world with simulation data involves three main steps: (1) collecting anthropometric and driving position data (seat position, steering wheel position, and eye-point) from test drivers; (2) generating manikins matching each participant's anthropometry; and (3) positioning the manikins on their measured preferred positions along with general constraints such as feet on pedals and hands on the steering wheel. From each positioned manikin, body joint angles and distances to specific vehicle geometries are extracted, contributing to a larger and more complete dataset for posture prediction models' improvement.

The proposed method was evaluated by considering data collected from 49 test drivers in an SUV model. The extracted data from the process were analysed and used to create three new driving posture strategies (set of optimal angles) based on stature groups: short, medium, and tall. These stature-based strategies were then compared to the default "Seated Car" strategy of IPS IMMA to assess improvements in posture prediction.

The analysis revealed significant differences in preferred joint angles across stature groups, particularly at the hip, knee, wrist, and upper torso (C6C7) joints. The new stature-based strategies demonstrated improved predictive accuracy for short and tall drivers compared to the default strategy, especially in vertical positioning (z-axis) of the H-point and eye point.

However, the existing default strategy, “Seated Car,” performed slightly better for the medium-stature group, suggesting that the current model aligns well with average-sized drivers. The findings indicate that improving posture prediction is possible by customizing optimal joint angles according to stature, and possibly other anthropometric measurements, instead of depending on a general model.

This study introduces a practical approach to enhancing posture prediction models by integrating real-world data with simulation, even when only partial datasets are available. The process provides a promising method for developing more customized and accurate posture prediction models, contributing to safer and more comfortable vehicle designs. Further refinements, such as improving manikin body meshes, considering seat properties, and validating across more vehicle types, were recommended to strengthen the approach.

4.6 PAPER F

Perez Luque, E., Lamb, M., Lee, S. & Yang, J. (2025). Predicting Human Reaching Motions: A Comparative Evaluation of Optimization and Heuristic Methods. Journal manuscript ready for submission.

Paper F compared two predictive approaches for modelling human reaching motions in 3D space: an optimization-based method and a heuristic method. The study aimed to evaluate how accurately each method predicts reaching motions across key components: spatial path, velocity, and arm joint configurations, as well as how well they capture natural inter-subject variability.

The experimental data collected included ten healthy participants. Participants performed four different reaching tasks while seated, such as reaching for a rear-view mirror, seatbelt, gear shifter, and passenger seat, mimicking movements typically performed while seated in a car. The right-arm movements were recorded with a motion capture system using ten reflective markers placed on anatomical landmarks.

The optimization-based method (OPM) predicted movements by optimizing a cost function balancing minimal joint displacement and maximal end-effector velocity (Yang et al., 2004). On the other hand, the heuristic method combined three rule-based models: the Steering Dynamics Model (SDM) for spatial path planning (Fajen and Warren, 2003), an adaptive lowpass velocity model (Gottlieb et al., 1989), and the FABRIK inverse kinematics solver for joint angle computation (Aristidou and Lasenby, 2011), which is referred to as Steering-FABRIK Method (SFM).

Both methods were evaluated using two complementary metrics: RMSE to assess the magnitude of error, and DTW to assess shape similarity. Motion components were analysed separately for velocity profiles, end-effector paths, and arm joint configurations. Additionally, inter-subject variability was analysed to understand how well each method reproduced natural differences between individuals.

The comparison revealed distinct strengths for each method depending on the motion component. Table 4.2 summarizes the best-performing method for each motion component based on accuracy and modelled variability.

Table 4.2. Summary of the best-performing prediction method for each motion component based on RMSE and DTW.

Motion component	Accuracy (RMSE & DTW)	Modelled Variability (RMSE)	Modelled Variability (DTW)
Path Planning	SFM	SFM	SFM
Velocity	SFM	OPM	SFM
Arm Configuration	OPM	OPM	SFM

- Path: SFM consistently outperformed OPM in predicting spatial paths, showing lower errors in RMSE and DTW, and better representing variability across participants.
- Velocity: SFM achieved higher accuracy in replicating velocity profiles and peak velocities, and better captured variability in the shape of individual velocity curves. OPM, however, better captured the inter-subject variability in velocity magnitude.
- Arm Joint Angle Configuration: OPM achieved better accuracy in predicting joint angles, with lower RMSE and DTW values. It also preserved more variability in joint magnitude, while SFM better captured variability in movement patterns.

Overall, SFM was more effective for predicting the diversity and shape of human reaching behaviours, whereas OPM provided more precise and biomechanically consistent joint predictions.

This study provides a comprehensive and nuanced evaluation of human motion prediction methods, showing the trade-offs between accuracy and behavioural diversity. It shows that assessing predictive methods using magnitude-based (RMSE) and shape-aware (DTW) metrics provides a more complete understanding of their behaviour. These findings offer valuable insights for further improving specific aspects of the motion models and selecting the most appropriate one depending on the application's focus.

DISCUSSION

5 DISCUSSION

This chapter discusses the main findings of the thesis and reflects on how they contribute to answering the overarching research questions.

5.1 ANSWER TO RESEARCH QUESTIONS

The following section discusses how the summarised papers' results relate to the research questions and to what extent the research questions have been answered. The correspondence between the papers, the research questions (RQs), and the objectives (Os) introduced in Section 1.2 is shown in Table 5.1.

Table 5.1. Papers' contribution to research questions and objectives.

	RQ1	**RQ2**	**RQ3**	
	O1	O2	O3	O4
Paper A	X			
Paper B	X	X		
Paper C		X		
Paper D			X	
Paper E			X	
Paper F				X

5.1.1 RQ1: HOW ARE DHM TOOLS CURRENTLY USED TO ADDRESS ERGONOMICS IN PD PROCESSES?

O1: Characterize the use of DHM tools for addressing ergonomics during the PD process.

Paper A shows that ergonomics designers across automotive companies employ DHM tools in PD to make predictions, analyses, and validations during internal and external balancing. *Internal balancing* refers to achieving a balance between various ergonomic factors that impact the end user's experience, whereas *external balancing* involves achieving a balance between ergonomic and non-ergonomic considerations during the PD process. Although DHM tools are mainly beneficial for aiding a proactive design process and their ability to represent human anthropometric diversity, among others, they also have some key limitations affecting the workflow of PD. Literature highlights the lack of cognitive modelling in simulations, poor usability, and inaccurate postures and motions as some of the main limitations within DHM tools. Paper A provides evidence that these limitations have not yet been adequately addressed in professional practice and identifies four gaps in the DHM field and within the Swedish automotive industry concerning DHM tools: human behaviour predictions, simulation tools' usability, ergonomics evaluation methods, and seamless system integration. These research gaps motivated the development and focus of RQ2 and RQ3 of this thesis to focus on simulation functionality and human motion predictions, respectively.

Paper B focuses specifically on occupant packaging design and provides a deeper understanding of crucial limitations that DHM users face when it comes to simulations of initial driving seated posture. A core problem is the lack of a relationship between the seated reference points used by vehicle designers and the human kinematic models used in DHM tools. Additionally, the mesh appearance of manikins in DHM tools can further complicate the accuracy of posture prediction, as it may lead to inaccurate collision volumes that affect how the manikin interacts with the seat geometry.

The lack of standardized methods and the unrealistic body meshes pose a significant challenge on their own. However, when combined, they introduce even greater and more unpredictable errors. This means that even when predictions between human body models and standards guidelines appear accurate, ergonomics designers still rely on their subjective visual assessments to modify and quantify the driver's seat position until the manikin body shape and automotive geometries look appropriately aligned.

Results from Paper A and Paper B motivate the need to develop consistent methods and models that provide simulation functionalities and accurate results that support decision-making in occupant packaging design.

While Paper A only considers automotive companies in the Swedish industry, the results are considered generalisable since they were corroborated in the literature, as shown in Chapter 2.

5.1.2 RQ2: HOW CAN SIMULATION FUNCTIONALITY OF DHM TOOLS BE IMPROVED TO SUPPORT DECISION-MAKING IN PD PROCESSES?

O2: Develop methods to improve the functionality of DHM simulations for human-product interaction in PD processes.

Quantifying the initial seated driving posture is crucial for ergonomics and safety evaluations, as the design and development of other ergonomic requirements, such as seated posture comfort, operation control, and visibility, depend on it. New methods to improve simulation functionality and facilitate decision support in automotive ergonomics are presented in Papers B and C.

Paper B discusses current approaches for achieving initial driving postures in DHM tools and highlights key limitations related to the lack of standardized procedures and the unrealistic representation of manikin body shapes. To address these issues, Paper B introduces the Body Shape Alignment, a method designed to standardize human model alignment with seated reference points based on body surface contact. An alternative method to estimate the hip location in seated postures is the regression model method proposed by Park et al. (2016b), which predicts the mid-hip to H-point relationship based on body dimensions and vehicle layout. While this statistical approach provides quick predictions and accounts for factors such as age and BMI, its performance has limitations for individuals with higher BMI, as shown in Paper B. Instead, the Body Shape Alignment method utilizes direct geometric surface alignment, allowing for better adaptability across a broader range of body shapes without relying on population averages.

The Body Shape Alignment method was further developed after its initial publication by defining additional constraints to enable more consistent and automatic alignment between the HPM template and the manikin (Figure 5.1). The Body Shape Alignment method reduces subjectivity, promotes consistency across users, and improves the initial positioning of digital manikins in virtual environments. However, the method still relies on simplified assumptions, such as fixed seat-manikin interaction and a lack of detailed body and seat deformations. Further development could increase its accuracy by considering how different body types interact with soft seat surfaces.

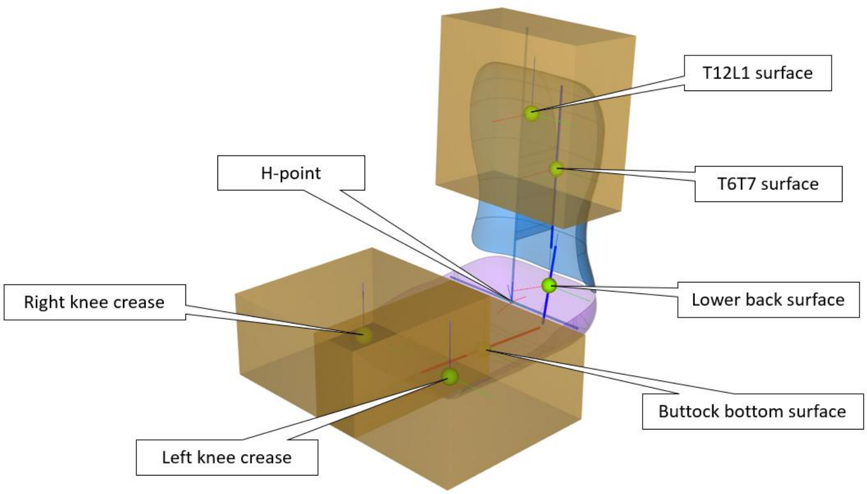

Figure 5.1. Constraints for the Body Shape Alignment method.

Building on this development, Paper C proposes an SBMOO method to enhance DHM functionality for occupant packaging design further. This method enables ergonomics designers to simultaneously consider multiple objectives, such as minimising joint angle deviations or maximizing comfort, and to account for constraints such as minimum clearance distances, vision angles, and reachability requirements. Compared to previous occupant packaging studies that applied single-objective optimization approaches (Parkinson and Reed, 2006) or conducted optimization procedures outside of DHM tools (Parkinson et al., 2007), the proposed SBMOO method enables simultaneous evaluation of multiple objectives directly within a DHM environment. Additionally, anthropometric diversity can be incorporated into the evaluation by using a manikin family, where each manikin's characteristics may be treated as either objectives or constraints, depending on the study setup. By applying this new method to performing DHM simulations, assisted by the use of genetic algorithms, some of the undefined subjectivity and issues commonly associated with traditional DHM simulation procedures are addressed.

The SBMOO method demonstrated in Paper C allows for automatic exploration of a vast solution space without requiring extensive manual simulation setups. It also enables individual tailoring by incorporating regression models as objective functions, thereby considering preferred joint angles and acceptable variability ranges for each manikin separately. Thus, this method supports the possible representation of human diversity and variability during simulation. While Paper C uses the regression models by Park et al. (2016b) as an example, the method is adaptable, meaning that industry users could implement their own regression models or preferred joint angle databases as objectives or constraints, depending on their target populations and design needs.

In conclusion, this thesis presents two methods that support potential improvement of simulation functionality within DHM tools: the Body Shape Alignment method and the SBMOO-based method. Together, they contribute to addressing critical challenges by enabling standardized manikin positioning, consideration of human diversity, and the balancing of multiple ergonomic and design requirements in occupant packaging design. Notably, the Body Shape Alignment method, first introduced in Paper B, was later used as a fundamental step in the studies developed in Papers C, D, and E, demonstrating its broader potential beyond its initial introduction. While there is still room for further development, such as validating the efficacy of the solutions generated through SBMOO, the results so far show that these methods offer a step forward in creating more reliable and consistent simulations supporting the decision-making for human-product interaction in PD.

5.1.3 RQ3: HOW CAN HUMAN POSTURE AND MOTION PREDICTION MODELS BE IMPROVED AND EVALUATED FOR DHM TOOL IMPLEMENTATION?

Improving and properly evaluating human posture and motion prediction models is essential to increasing the functionality and accuracy of DHM tools in PD processes. Advancing simulation accuracy not only enhances ergonomic assessments but also supports better decision-making early in the process.

O3: Develop and evaluate human posture prediction models to increase simulation accuracy in PD processes.

Paper D contributes to this objective by comparing two different seated driving posture prediction methods: SPM and OPM. The study highlights the value of comparative analyses, which remain rare in the DHM field but offer valuable insights into model strengths, limitations, and differences in underlying assumptions. Establishing comparisons between prediction models, in addition to comparisons against experimental ground truth data, creates a clear reference that can expose model-specific weaknesses and opportunities for improvement.

Compared to Reed et al. (2000), one of the few notable studies comparing posture prediction methods in the context of occupant packaging, Paper D advances this work by using a larger and more diverse participant sample, applying more evaluation metrics, evaluating multiple prediction points including both H-point, eye location, and steering wheel location, and validating models against real in-vehicle data instead of mock-ups.

The findings from Paper D show that neither approach was clearly better. Rather, their performances varied depending on the vehicle configuration, predicted point, and participant anthropometry. Notably, prediction errors tended to be lower for medium-sized participants and increased for shorter and taller participants. These findings indicated that posture prediction models based mainly on average body sizes may fail to generalise accurately across a diverse population. Following the analysis, Paper D also proposes specific directions for improvement in posture prediction methods. In addition, understanding the differences in core model functioning, such as the cascade structure of SPM versus the cost functions of OPM, was also identified as critical for guiding future model refinements.

Building upon these observations, Paper E introduces a new process to address insufficient datasets. The process involved developing stature-based posture prediction strategies by integrating real-world driver posture data into DHM simulations. By creating tailored strategies for short, medium, and tall drivers, Paper E demonstrates improved posture prediction accuracy for short and tall groups compared to the default strategy in the DHM tool used in the study. While the study in Paper E focused on stature-based posture prediction strategies, the proposed approach is generalisable and could be applied to other variables such as BMI or gender, offering new avenues for enhancing accuracy in specific target users in occupant packaging design.

Beyond model development, Paper D also emphasises broadening evaluation practices for posture prediction models. In addition to traditional error metrics such as MAE or RMSE, Paper D introduced Bland-Altman plots to assess the agreement (bias and limits) between predicted and measured postures and the consistency of errors (possible patterns or trends). This multi-metric evaluation provides a deeper and more comprehensive understanding of model performance.

In summary, improvements in human posture prediction models for DHM tools need to address both model development and evaluation aspects. Development efforts can focus on capturing human diversity more accurately, refining posture prediction assumptions, and adapting prediction strategies based on key anthropometric variables. Evaluation benefits from incorporating multiple complementary metrics to assess not only error magnitude but also other aspects, such as systematic bias or error patterns. Comparative studies, as performed in Paper D, combined with targeted model refinements, as demonstrated in Paper E, provide a solid pathway toward more accurate and reliable simulation models in DHM tools.

O4: Develop and evaluate human motion prediction models to increase simulation accuracy in PD processes.

Following a similar approach to posture prediction, Paper F contributes to O4 by comparing two distinct motion prediction methods for human reaching tasks in 3D space: an optimization-based method (OPM) and a heuristic method (SFM). By comparing predictions both against experimental ground truth data and against another predictive method, it becomes easier to identify specific areas where models succeed or where further improvements are needed.

Regarding model development, Paper F introduces the heuristic method (SFM), which integrates three rule-based sub-models to predict reaching motions in 3D space. The SFM method combines an adaptive lowpass velocity model, the SDM, which was formulated for 2D space but is extended here to 3D, and the FABRIK IK solver to obtain full arm postures. The integration of these sub-models offers a novel and efficient approach for simulating human reaching motions. Although the SFM method builds on existing components, the work presented in Paper F demonstrates their extension and combination in a new way, contributing to exploring alternative modelling approaches in motion prediction.

In addition to model development, Paper F places a strong focus on expanding evaluation practices. The study introduces the use of DTW as a complementary metric to traditional RMSE for time series motion evaluation. While RMSE evaluates the magnitude of errors between predicted and observed data, DTW measures shape similarity, providing a more complete picture of prediction performance (Senin, 2008; Cleasby et al., 2019). Although well-established in fields such as speech recognition and pattern analysis, DTW has rarely been used for evaluating human motion predictions involving multivariate time series, such as 3D path planning and 7D joint angle configurations.

Using both RMSE and DTW together allows a more comprehensive evaluation: a method may achieve low magnitude of errors (good RMSE) but still perform poorly in replicating the shape of motion trajectories (high DTW), revealing specific aspects of model behaviour that would otherwise remain hidden.

Furthermore, Paper F also evaluates the models' ability to replicate the inherent inter-subject variability observed in human reaching motions. In addition to comparing each model's predictions to experimental data, the study assesses how well each method reproduces the natural variability observed between individuals performing the same tasks. This was achieved by calculating pairwise comparisons (e.g., subject 1 to subject 2, subject 1 to subject 3, etc.) within the experimental data and within the model predictions. By comparing the variability observed in real experimental data with the variability modelled by each prediction method, the study assesses how well the models preserve natural differences in movement strategies. This approach offers valuable insights into whether a model replicates an average movement or whether it can realistically reproduce the diversity of human behaviour, an important consideration when simulating human motions in DHM applications.

In summary, the work presented in Paper F highlights that improving human motion prediction models for DHM tool implementation requires not only exploring new modelling approaches but also expanding evaluation strategies that go beyond simple accuracy measures. Together, these steps provide a deeper understanding of model behaviour and guide future improvements for achieving more realistic and reliable human motion simulations within DHM tools.

5.2 CONTRIBUTION

This thesis primarily contributes to advancing theoretical knowledge in the field of DHM and within the context of occupant packaging design. Through the work presented in the appended papers (Papers A-F), new methods and models have been developed and evaluated to improve the functionality and accuracy of human posture and motion simulations.

New approaches have been proposed to address key limitations in DHM tool usage, improve human model prediction accuracy and simulation functionality, and strengthen model evaluation through comparative analyses and complementary metrics. Although the main contribution is theoretical, the outcomes, such as the method for standardized manikin positioning (Body Shape Alignment in Paper B) and the use of SBMOO in DHM (in Paper C), offer practical applications that can support more consistent, human-centred design processes in industrial PD. Overall, the research provides foundations for advancing DHM tool development while offering adaptable methods for practical implementation.

CONCLUSION

6 CONCLUSION AND FUTURE WORK

This chapter summarises the main conclusions of the research and proposes directions for future work.

6.1 CONCLUSION

This thesis has aimed to advance DHM tools by developing and evaluating methods and models that enhance the functionality and accuracy of human posture and motion simulations, supporting decision-making when addressing ergonomics in PD processes. Through the study of current practices, the development of new approaches, and the improvement of evaluation strategies, several conclusions can be drawn from the research presented:

- DHM tools are widely used to support ergonomics evaluations during PD processes. However, key limitations persist related to usability, simulation accuracy, and the integration of human behavioural diversity, highlighting the need for improved methods.
- The Body Shape Alignment method was developed to standardize the initial seated driving posture of manikins in DHM tools, reducing subjectivity and improving consistency in simulations involving occupant packaging design.
- A SBMOO method was proposed to systematically balance several ergonomic and non-ergonomic requirements while accounting for human anthropometric diversity, enabling more comprehensive and proactive occupant packaging evaluations.
- Comparative studies between posture prediction methods (SPM and OPM) were conducted, revealing how performance varies across different anthropometric groups and identifying limitations when predicting postures across a diverse population.
- A stature-based strategy process was introduced to refine posture prediction models, demonstrating improved accuracy for short and tall individuals and illustrating how tailoring strategies to user characteristics can enhance DHM simulations.
- Motion prediction models were compared and evaluated using both RMSE and DTW, highlighting the importance of considering both the magnitude of errors and shape similarity when assessing prediction model performance.

- The evaluation of modelled inter-subject variability provided further insights into how well prediction models replicate the diversity of human motion, an essential consideration for enhancing human behavioural variability of DHM simulations.
- Although the primary contributions of this thesis are theoretical, the methods developed, particularly the Body Shape Alignment and the SBMOO method, offer practical potential for application in industrial contexts to support more accurate, consistent, and human-centred design processes.

6.2 FUTURE WORK

While this research contributes to advancing simulation functionality and prediction models' accuracy within DHM tools, several opportunities remain for further development and exploration. Future work could build upon the methods and findings presented in this thesis to strengthen their applicability, extend their scope, and enhance their integration into industrial and research contexts.

First, further validation of the Body Shape Alignment and SBMOO methods with ergonomics designers would be valuable to assess their impact on simulation consistency, decision-making, and the PD workflow. If these methods prove successful in practice, integrating and automating them could significantly improve the usability of DHM tools. Additionally, the Body Shape Alignment method could be refined by considering tissue deformation effects at the contact interfaces between the manikin and seat geometries, particularly for individuals with varying anthropometric characteristics such as higher BMI or body mass.

Second, the applicability of the developed methods (Body Shape Alignment and SBMOO) beyond the automotive domain presents an attractive direction for future research. Exploring their use, for example, in construction or agricultural vehicles, public transport, or cockpit design, could test the generalisability of the approaches and broaden their impact on human-centred design practices across industries. In addition to vehicle-related contexts, these methods could also be valuable in other domains involving seated or constrained work environments, such as medical equipment design, control room workstations (e.g., air traffic control or emergency response centres), industrial assembly stations, or office furniture development.

Third, further development of prediction models could focus on extending the consideration of human diversity and variability. Integrating different anthropometric and biomechanical factors, such as body mass index (BMI) or body roundness index (BRI), gender, and age, could lead to more personalised and accurate simulations. Exploring cognitive and behavioural sources of variability could also offer new perspectives on posture and motion prediction. In parallel, refining posture prediction strategies using larger, more diverse datasets, and optimizing motion prediction models by fine-tuning model parameters would continue to strengthen the models' robustness.

A specific opportunity for model development relates to obstacle avoidance in reaching motions. Within this thesis work (during a six-month ERASMUS exchange scholarship, Autumn 2024 to Spring 2025), experimental data were collected at the University of Antwerp using the state-of-the-art MOVE4D body motion scanning system (Move4D, 2024), capturing human reaching tasks with obstacle interference. This data will be used to extend the behavioural SDM (Fajen and Warren, 2003) by developing and validating the obstacle avoidance component, thus expanding the capabilities of the heuristic prediction model, SFM, introduced in Paper F.

In terms of evaluation, future work could expand beyond RMSE and DTW by investigating alternative time series comparison metrics, such as Fréchet distance (Eiter and Mannila, 1994; Cleasby et al., 2019), broaden the set of metrics to improve sensitivity to different motion characteristics and support advanced analyses, such as sub-trajectory clustering of distinct movement phases like reach, grasp, or return phases. Enriching the evaluation framework would offer deeper and more nuanced insights into model strengths and weaknesses.

Finally, the application of the developed methods and models in new domains offers promising future directions. Posture prediction models could be tested in vehicle safety design contexts, such as HBM (human body models – virtual crash test dummies) positioning or seatbelt fit assessment (Takeuchi et al., 2024). Similarly, motion prediction models could support the development of human-robot collaboration systems, assistive robots (Hossein Sadat Hosseini et al., 2024), or exoskeleton technologies (Ren et al., 2019; Souza et al., 2024), where accurate simulation of reaching and interaction motions is critical.

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APPENDED PUBLICATIONS

PAPER A

INTERNATIONAL DESIGN CONFERENCE – DESIGN 2022
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Challenges for the Consideration of Ergonomics in Product Development in the Swedish Automotive Industry – An Interview Study

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Abstract

This paper presents an interview study aiming to understand the state of the art of how ergonomics designers work in the vehicle development process within the Swedish automotive industry. Ten ergonomic designers from seven different companies participated in the interview study. Results report the ergonomics designers' objectives, workflow, tools, challenges, and ideal work performance tool. We identify four main gaps and research directions that can enhance the current challenges: human behavior predictions, simulation tool usability, ergonomics evaluations, and integration between systems.

Keywords: vehicle, ergonomics, human-centred design, simulation-based design, digital human modelling

1. Introduction

Studying how a product will affect the potential end-user is essential in product development (PD). Such human-product interaction studies are typically done at relatively late stages of the design process, even though there are benefits to taking action regarding user-related aspects at earlier design process stages (Schröppel et al., 2021). Obstacles towards more proactive ergonomics measures are found to be lack of time, knowledge, methods and tools for consideration of ergonomic issues together with a lack of cooperation and communication between project stakeholders (Broberg, 1997; Falck and Rosenqvist, 2012). Ergonomics design methods and objectives can be applied in many different contexts or application areas, making the ergonomics designers' objectives, workflow, and tools to vary widely. This makes it difficult to identify the kinds of challenges ergonomics designers face and what tools or tool improvements might enhance their ability to meaningfully contribute to the design process at the early stages of PD. Ergonomics designers can use various tools to proactively assess ergonomics aspects at early stages of PD, including physical prototypes, user tests and interviews, and human-product interaction simulations using digital human modelling (DHM) tools. However, these tools and methods can have limitations such as being expensive, time-consuming, difficult to use, and subjective, making it hard to apply insights acquired to other aspects of the design process.

DHM tools are meant to support that product designs fit the users' requirements as soon as possible in the design process when the product exists only as a virtual model, reducing the need for physical prototypes and their corresponding development time and costs (Scataglini and Paul, 2019; Demirel, Ahmed and Duffy, 2021). DHM tools enable to test digital humans (a.k.a. manikins) interacting with a digital product model in a given digital environment, which help to assess and reduce unhealthy or uncomfortable conditions due to design features. Using DHM tools early in the design process is a form of proactive ergonomics assessment since it enables designers to run “what-if” scenarios in the early design phases (Demirel, Ahmed and Duffy, 2021). DHM tools are widely used in the automotive industry for occupant packaging. Occupant packaging consists of organizing space for people and parts of the vehicle to satisfy transport, a clear example of a human-centred design activity (Bhise, 2016). Ergonomics often develop significant constraints or requirements on vehicle design to fulfil the users' interaction needs. Understanding such human-vehicle interactions for developing new vehicles models early in development is crucial to the design process. While computer-aided design (CAD) tools are used to perform most design work, DHM tools are often set apart which creates a challenge to incorporate user aspects through proactive tools in the product development process (PDP). Needs from previous and current research projects motivated to conduct this study (MOSIM, 2018; SVE, 2020; ADOPTIVE, 2021). The development of virtual tools and the consideration of human-centred aspects in product and production development is an important research area for the Swedish industry's (Made in Sweden 2030, 2017). This study aims to understand the state of the art of how ergonomics is addressed in the vehicle development process within the Swedish automotive industry. Our aim is divided into several objectives: identifying ergonomics objectives, methods, tools, the "ideal" tool for the ergonomics designers' work, and identifying challenges for enhancing them. We anticipate that this insight can lead us to see the research gaps in the field and direct future research.

2. Methodology

2.1. Participants and data collection

Semi-structured interviews of active ergonomics designers were used to gain a deeper insight and address the research aim. Ten ergonomics designers from seven Swedish automotive companies participated in the interview study. All of the participants involved in the study are considered experts according to the criteria suggested in Littig (2009), with at least five years of experience and specialized knowledge in the specific domain of ergonomics PD/user-centred PD. Experts with different roles, experiences, and positions within their companies were enrolled in the interview study to gather different perspectives. While all the companies included in the study are from the automotive sector, they focus on different types of vehicles such as cars, trucks, buses, and construction vehicles. The size of the companies varies from 1,000 to 56,000 employees.

Semi-structured interviews, lasting between 60-75 minutes, were held online via a video meeting tool (Zoom). The interview topics were pre-determined and specified by an interview guide consisting of open and closed questions and follow-up questions. All the interviews were video and audio recorded for data analysis. The interviewer also took notes during the interviews to record additional information about the context, clarifications, and observations during the online meetings. The interview topics and purpose of the study were described to the participants, and they were given the opportunity to consent to participate in the study as per Swedish law and the Declaration of Helsinki. The interviews were conducted between May and June 2021.

2.2. Interview guideline

An interview guideline was developed to direct the interview conversation to discussions covering the research objectives and ensure consistency and comparability across interview sessions (Meuser and Nagel, 2009). The interview was divided into five phases. Phase one aimed to create a comfortable environment for the interviewees. The study's goal was explained to the participants, the consent form reviewed, and the interviewees introduced themselves. The second phase focused on understanding the main goals and objectives of the interviewees' position and their departments within the product development process (PDP). Phase three addressed the workflow or methodologies that ergonomics designers follow to achieve those goals and objectives. Phase four involved identifying all of the tools used by ergonomics designers in their work. Phase five consisted of imagining and describing an "ideal" tool to support their work. At the end of the interview, participants could give feedback, additional comments and ask questions.

2.3. Data analysis

The deductive procedure described by Mayring (2014) was followed for analyzing the qualitative content. This procedure consists of four steps: 1) data preparation; 2) definition and revision of categories; 3) data codification, and 4) interpretation of results. All the recorded material was transcribed for further analysis in the first step. Discussion categories were based on the research objectives and interview phases of the interview guide: Objectives, Workflow, Tools, Ideal tool, and Challenges (Mayring, 2014). Next, the transcribed interview content was coded, and the data were sorted into a table to compare information across participants and discussion phases. Following this, a horizontal analysis of the data made it easier to identify the patterns and reported challenges.

3. Results and interpretation

3.1. Objectives

When discussing their aims and objectives, most of the interviewees (n = 8) used the word "balance" to describe the purpose of their work. According to one interviewee, *"The ultimate goal or objective we have is to balance things instead of having the best one thing because if they are the best in one thing, they are bad in all the others and we have to find a balance between all of them"*. The data analysis of the interview results made it possible to identify and divide the different "things" they have to balance into internal and external balancing relationships. Internal balancing involves balancing among the various distinct ergonomics aspects affecting the end-user. External balancing involves balancing between ergonomics and the various non-ergonomic aspects in PD (safety, aesthetics, manufacturing, materials, etc.). The process of balancing involves identifying the specific limits of identified requirements as well as the relative ranking and/or weight of each requirement. Following the funnel concept developed by Clark and Wheelwright (1993), the PDP has three steps: Predevelopment, Development, and Production (Figure 1). As illustrated in Figure 1, ergonomics designers primarily perform internal balancing during the predevelopment phase and external balancing during the development phase.

Defining what counts as an appropriate balance (both internal and external) is critical for most designers. In most cases, designers adhere to a set of company-defined requirements or objectives to assess the success of internal and external balancing processes. The priority and definition of ergonomics requirements vary by company. Some focus on the study of physical and cognitive ergonomics by different groups, while others address physical and cognitive ergonomics by the same ergonomics group. In ergonomics, "*the requirements are closely connected to the experience of the customer and product user*." Visibility, driver sitting posture, roominess, comfort, life use, and ingress-egress are some examples of ergonomics aspects covered during internal balancing by the fulfilment of requirements. Regardless of the ergonomics aspect, requirements may include ergonomics standards as specified by regulatory bodies and/or internally defined ergonomics standards derived from internal company research and design history. Legislative bodies may also set ergonomics requirements (e.g., visibility or other safety standards). All of these requirements are assembled into a set of company-defined requirements. Thus, designers indicated that by accomplishing the company-defined requirements, legal and acceptable products are realized, ensuring no physical damage to users and a minimum standard for comfort and user experience for end-users.

Along with ergonomics requirements, non-ergonomics requirements are often defined and may cover many aspects of product design, including aesthetic, safety, production, and material standards. The requirements that apply to the design process can also vary depending on the PD area and level. For example, roominess and visibility are considered more heavily in assessing the balance in the predevelopment phase. In contrast, cost and manufacturing limitations are given more weight later in the development phase. Applying requirements to assess balance can be complex for certain ergonomics aspects, so some designers indicated that they report their results by "*describing consequences instead of certain decisions"* in ergonomics. Instead of focusing on how each requirement was met and balanced, they report the design consequences of fulfilling or not fulfilling the requirements.

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3.2. Workflow

Figure 1 presents a typical ergonomics designers' workflow characterization based on the interviews. Such characterization and the terminology consist of an author's interpretation. However, interviewees validated it in a workshop session weeks later to the interview sessions. This workflow describes the steps of ergonomics designers through the different product design phases, where the most relevant steps are the internal and external balancing (Figure 1). Not all of these steps involve ergonomics designers. However, based on interview comments, ergonomics designers consider the decisions made at most of these steps to impact or constrain the ergonomics of the product. For example, when the PDP begins with the project or product definition, this work is typically done by the marketing team. The type of vehicle, intended use and users, and styling are likely to affect ergonomics requirements and decisions. However, ergonomics designers do not typically have a significant role in this step. They have a more prominent role in the following steps, though their role may vary depending on numerous factors.

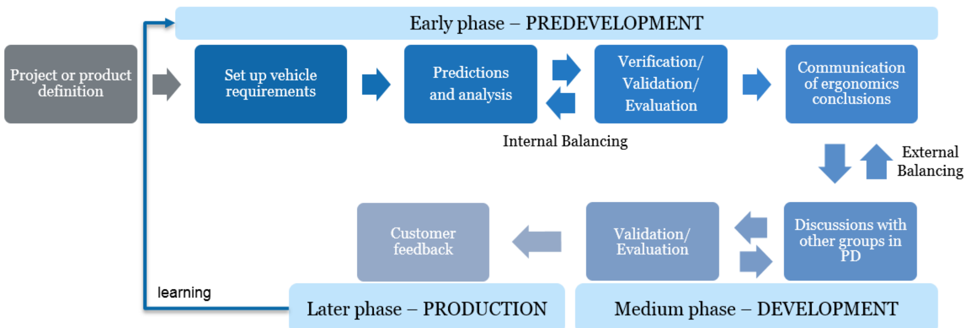

Figure 1. Ergonomics designers' work in the product development phases

Some projects aim to develop an entirely new product from scratch, though it is more common for a new product to be built on or modified by a previous or existing product. That means ergonomics designers would start directly in the external balancing at the development phase. This limits the ability of ergonomics designers to test new methods and explore novel designs. Instead, ergonomics designers often must work from existing designs and requirements, adapting and improving what is already there. Regardless of how novel a new product is, once defined in the first step of the development process, the ergonomics group sets and prioritizes the vehicle requirements accordingly. Once the ergonomics requirements are defined, ergonomics designers make predictions using simulation tools, previous studies, user testing, and analyses relating to the ergonomics aspects as a part of the internal balancing process. These predictions may be based on designs and models in progress and reference designs when the product is an update to an existing/previous product. As a part of this step, ergonomics designers ensure that ergonomics requirements, especially those with higher priority, are considered and ranked or weighted to achieve good ergonomics and user experience according to the vision defined by the product definition. This balancing process is accomplished iteratively by verifying, validating, and evaluating the consequences of various weightings and prioritizations of candidate requirements. Following the definitions from ISO standards (ISO 11064-7, 2006), verification involves testing and confirming that the product meets the defined requirements and may be done by simulation or user testing. Once verified, the ergonomics designers build prototypes and engage in user studies or make simulations to validate, which consist of confirming that meeting the requirements has the expected consequence or intended application (ISO 11064-7, 2006). Finally, the results of the verification and validation processes are evaluated to determine how well the requirements are met and how well meeting the requirements results in the expected user experience/ergonomics outcomes. Designs, simulation methods, and occasionally requirements are updated until evaluation indicates that a desirable ergonomics balance has been achieved. The ergonomics requirements results achieved during these processes are then communicated to the rest of the areas involved in the PD. Here in the medium phase,

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the development of the product starts. Ergonomics designers discuss and test different possible solutions until a balance between ergonomics and the rest of the PD areas is achieved. All these new possibilities are validated and evaluated again before the later phase of production starts. At this point, the work of ergonomics designers is considered finished. The final products are subjected to user tests, which provide input information for the ergonomics team to learn from for future projects (Figure 1).
The time to perform the entire PDP can vary from 3 to 10 years, depending on the type of vehicle and project definition. While changing or improving a specific feature of a product can be done quicker, developing a new product from scratch requires more time since many more requirements and ergonomics aspects need to be considered and tested. The time involved in identifying and balancing ergonomics requirements is affected by the length of the PD cycle and the size of the ergonomics design team. Interviewees reported a different number of people involved in the ergonomics groups, ranging from 3 to 19 persons. Typically, no single person on an ergonomics team is specialized in all the required tasks, and thus different individuals have different skills and areas of expertise in the process such as simulations, interviews, or user tests.

3.3. Tools

Table 1 indicates the types of design, development, analysis, and measurement tools mentioned in the interviews and the number of interviewees that indicated using each. In the interview context, a tool was presented as an instrument that can enhance the efficiency of a task (Martin, 1997). While this definition might lead to understanding technology or methods also as tools, they were included since interviewees reported them as instruments enhancing a task. These tools are used during the internal and external balancing to make predictions, analyses, and the corresponding validations.

Table 1. Used tools in the ergonomics product development

Tool	Common use descriptions	Freq.
Physical prototype	Physical prototypes allow the designers and users to interact with an idea or concept and test for ergonomic issues. Ergonomics designers also test the impact of product features on ergonomic aspects like roominess.	10
DHM tool	DHM tools simulate human product interactions in a virtual environment. Ergonomics designers test several ergonomic aspects (visibility, driver position, and reachability, among others) with manikins of different anthropometry.	10
Virtual Reality	Virtual reality uses computer technology to create an immersive simulated environment. Ergonomics designers test different ergonomic aspects of the vehicle cabin, mainly visibility, roominess, and user behaviour in different environments.	7
CAD software	CAD software is a technology for the design and technical documentation of 2D and 3D physical components. They are used for studying clearance considering the interior vehicle geometries. Rendering techniques are also used for the study of light reflections (visibility).	6
Camera	Cameras capture visual images. They are used for study documentation.	5
Motion capture	Motion capture consists of recording the movement of people or objects. It is used for studying specific tasks like the ingress-egress of vehicles.	4
3D Scanner	3D scanners measure the 3D surface shape of physical objects. They are used to capture the eye point position, body posture, and size to make validations. They are also used for comparisons of competitor products.	3
Driving simulator	Driving simulators allow creating a driving experience by placing the drivers in virtual environments with a physical control interface. They are used to analyze ergonomic aspects and driving behaviours in different vehicle models scenarios.	3
User tests	User testing of actual products provides the evaluation of developed products. Interviews, statistical analysis, and correlations are some of the methods used.	3
Vehicle Compliance	Complimentary CAD software for automotive compliance design. Ergonomics designers use it to ensure visibility standards.	2
Image-editing software	Editing and manipulation software helps ergonomics designers with user behaviour analysis (cognitive aspects).	1

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3.4. Ideal tool

The interview ended with a focus on identifying key features of an ideal tool for ergonomics designers. All the interviewees indicated that such a tool should be easy to use and flexible across various human anthropometries. Some visualized an improved DHM tool as the ideal tool, while others preferred a tool overlapping the digital and physical world to allow for user tests with physical prototypes. Interviewees who indicated that their ideal tool would be a DHM application agreed that such a tool would need improved usability, more accurate predictions, and additional functions relative to existing DHM tools. Suggested additional features included predictions of forces and seamless integration with motion capture and virtual reality (VR). Interviewees who indicated a preference for a combined digital-physical tool described a tool capable of combining basic physical prototypes with more detailed digital models, named VR or augmented reality (AR) tool as an example. This would allow for changing various details of the digital representation while maintaining some physical interactions. Such a setup would also allow for more flexibility in user testing different design concepts and environments.

Beyond the ideal tool's form (hardware/software), most interviewees indicated that a good integration between systems (motion capture - DHM - VR/AR) would be important. The interviewees agreed that the ideal tool would allow them to study and analyze a candidate user's entire movement or activity sequence with different measurement and simulation methods. The same visualization and analysis tool could be used for both simulation and direct measurement. This would also allow for direct comparisons between simulated and real data. Thus, they could import motion capture recordings from VR or real user-product interaction into the imagined DHM tool and see the recorded motion applied to different anthropometries and body shapes. In essence, interviewees expect from such a DHM tool more reliable testing and analysis of different ergonomic aspects while considering human physical diversity. Relatedly, interviewees also suggested integrating the ergonomics tool with VR to immerse people from other departments in the product under development to help them see and experience the ergonomics insights first hand. Having ergonomics evaluation methods is another feature suggested by some for the ideal tool. Ergonomics evaluations would rate how good or bad different ergonomic aspects are for the end-users, decreasing the need for validating predictions based on simulations and accelerating the design iterations. Lastly, some of the participants also mentioned the ability to consider cognitive aspects in the process. Some of the interviewees indicated that cognitive aspects of users likely have a large impact on physical product interactions. As such, they believe that the ability to consider cognitive aspects in the ideal tool could contribute to a more proactive and holistic ergonomics design process.

3.5. Challenges

The ergonomics designers were also asked about the different issues or challenges they face during the PDP within the Swedish automotive industry. The indicated challenges are categorized and described in Table 2, along with the number of interviewees identifying each challenge. The reported challenges could be divided into interdepartmental communication issues and tool development issues. Both groups affect the workflow of PD.

4. Discussion

4.1. Ergonomics in product development

With this interview study, we aimed to understand the state of the art of the Swedish automotive industry when considering ergonomics within the PDP. Several interviewees described a workflow that is similar to the design process proposed by Cross (2021). Cross' four-stage model, consisting of the main activities: exploration, generation, evaluation, and communication, can be interpreted in the context of ergonomics design. During exploration, ergonomics designers seek to understand the project definitions and define and balance requirements. Generation includes producing proposals for solving and fulfilling the product requirements. Ergonomics designers feed into this stage by producing, evaluating predictions of ergonomics requirements and balancing them as applied in simulations or results from user tests. Evaluation includes verifying and validating the predictions and evaluating results relative to the project definitions and previously selected internal balances. Finally, communication involves both

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Table 2. Identified challenges in ergonomics product development

	Challenge	Description	Freq.
Tool development issues	DHM tool gaps	Inaccurate predictions of human posture require the need of validations.	10
		DHM tools usability, they are not easy to use, time-consuming.	10
		Limited measures of human posture and environment interactions.	3
	Evaluation methods	Lack of driver ergonomics evaluation methods.	4
	Need of ergonomics knowledge	CAD and design knowledge and skills are insufficient. Knowledge and experience in ergonomics are needed as well.	3
	Willingness for simulation work	Difficult to find the people willing to do the simulation work due to the usability of these tools and reliability.	2
	Tools providing objective data	Ergonomics requirements can depend on subjective assessment by ergonomics designers. Objective methods are needed.	1
	Workflow steps - consuming iterations	Ergonomics designers have to validate all the predictions from either simulations or user tests information, which can be time-consuming and slow the process down.	1
Communication issues	Difficulty reporting ergonomics conclusions	Ergonomics designers often struggle to report and explain the ergonomic issues to other departments. They have to communicate these issues easily enough for a non-ergonomics expert without losing key insights that come with specialized knowledge.	10
	Ergonomics consideration early in the PDP	There is a need to consider human aspects early in the projects to give feedback to the rest of the PD areas. However, ergonomics designers may be involved later in the process.	3
	Planning vs. flexibility	The ergonomics group needs to plan activities but keep flexibility. Planning always changes. It is difficult to find a balance.	3
	Interdepartmental communication	Misunderstandings between departments about the project or product definition in the predevelopment phase can lead to mismatched ergonomics requirements.	2

reporting and discussing ergonomics issues to other PD areas. While several studies indicate a lack of design method's applied in engineering design practice (Vredenburg et al., 2002; Birkhofer, Jänsch and Kloberdanz, 2005; Geis et al., 2008; Eisenmann and Matthiesen, 2020), within the companies represented by the interviews, there seemed to be some adherence to company defined design methodologies for ergonomics PD which map onto more traditional academic PDP models.

While ergonomics designers' work and workflow are built on general product design methodologies, interviewees highlighted the differences in considering and studying ergonomics in PD and production. In PD, the study of ergonomics relies a lot on subjective predictions and interpretations of user-product interaction. In the context of vehicle design, the subjectivity is due, in part, to a lack of established evaluations methods for vehicle ergonomics. In contrast, in production ergonomics, the assessment of physical workload is typically defined by established ergonomics evaluation methods. One interviewee explained, *"we understand more the cognitive capacities that assembly line workers can do in labour, whereas I would say here* [in PD] *we are not as worried about how frequent they are doing things in a vehicle, it is more about finding comfort."* As described by one of the interviewees, *"ergonomics in PD is softer, it is more luxury,"* it is about improving human-product interactions focusing on comfort and user experience. These comments imply that ergonomics evaluation methods are not independent of the general design context and that currently, most objective ergonomics evaluation methods are designed for production design.

Identifying objective ergonomics assessments for product design may be complicated because companies, at least those involved in the study, had very different approaches to ergonomics in PD. One notable distinction is that some of them consider both physical and cognitive aspects together, whereas others separate these ergonomics domains. This kind of division can, for example, change the definition of the term comfort between companies and even across ergonomics design groups within companies. While ergonomics designers focusing on physical aspects may define comfort in terms of biomechanical

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loads, those combining cognitive and physical ergonomics considerations may further define comfort according to the feeling of safety experienced by the end-user.

4.2. Main gaps as improvement opportunities

Based on the interviews, we identify four gaps or research directions for improving ergonomics designers' work based on the reported challenges and desired ideal tool, focusing on DHM tools.

4.2.1. Human behaviour predictions

The lack of realistic and accurate human predictions in DHM simulation tools leads to time-consuming validation during PD. This makes ergonomics designers rely on their own experience for predictions and simulations, which introduces an unspecified level of subjectivity and bias into the results. Sometimes ergonomics designers manually specify postures that may provide the insights needed but are difficult to reproduce. In other cases, designers specify many postures (as a way of simulating a movement sequence), and then DHM tools can predict manikin transitions between these postures, often based on some definitions of ergonomics optimization (Zhu, Fan and Zhang, 2019). Both of these processes can be difficult, time-consuming, and subjective. Often these predictions are "good enough" but do not necessarily reflect actual human movements (Lämkull and Zdrodowski, 2020; Demirel, Ahmed, and Duffy, 2021). More reliable human product interaction predictions would accelerate the consideration of ergonomics in the PDP, resulting in better user experiences and helpful input for other departments, such as vehicle safety. Human product interaction predictions, including driving postures, may be improved by improving current posture prediction methods or identifying new posture prediction methods within DHM tools.

4.2.2. Simulation tool's usability

While DHM tools can provide critical human factors insights into PD, they have limited usability. According to various studies, some DHM tools in engineering design may be unstandardized, complicated to use, not trustworthy, or time demanding systems for the development of the study of human interactions (Ranger, Vezeau and Lortie, 2018; Lämkull and Zdrodowski, 2020). Interviewees involved in this study echoed these earlier findings and stated that current simulation tools are time-intensive, requiring ergonomics designers to spend considerable time using them. The time-intensity of these tools reduces the willingness of others outside and inside of ergonomics design groups to learn the tools. This can hinder the development of cross-departmental communication channels based on existing DHM tools. Improving the usability of DHM tools would likely make them more accessible for proactive ergonomics assessment and correspondingly increase the number of DHM tool users within companies (Högberg, 2009). This could, in turn, reduce the time it takes to perform both internal and external balance processes.

4.2.3. Ergonomics evaluation methods

As mentioned above, ergonomics evaluations methods typically assess ergonomics in production development or running production. However, there is a lack of corresponding objective methods within vehicle development, and evaluation relies on internal validations based on each company's internal user testing and simulations. These simulations provide company-specific and often subjective visualization and measurements of user-product interactions. New methods and improvements to existing methods for objectively evaluating different vehicle ergonomics aspects (driving sitting posture, comfort, reachability, among others) would accelerate the ergonomics designers' work by reducing the time spent on validations and increasing the quality (e.g., in the form of increased reliability and validity) of the evaluations performed as well. Once new or improved objective methods are available, integrating these methods within DHM tools would assist ergonomics designers in finding successful design solutions at stages of the design process where the vehicle is only available as a CAD model. Such functionality would further make it easier to report ergonomic issues to other actors within PD by having the ergonomics evaluation methods to clarify the degree to which a specific system or component design does not fulfil ergonomic requirements. Further, objective methods in simulations would make it easier and quicker to compare potential changes, both during a specific product design process and across design processes.

4.2.4. Integration between tools

The integration between tools such as motion capture, DHM, and VR/AR, was commonly mentioned as a feature in an ideal tool. Studying the same case of user-product interaction across different design tools would improve studies and analysis performance, especially relating to communication of ergonomic issues to other departments in PD. *"The best communication is to have each person that's involved in the project experience the ergonomic issue."* That could be done by deploying simulated cases within the virtual world (e.g., using VR). In addition, importing motion capture data into DHM tools would be another way of improving human behaviour predictions and representations by building on behaviours recorded during actual user interactions with previous models or physical mock-ups. Motion capture data may provide data for building up knowledge about human behaviour. The main limitation of motion capture is the lack of integration with CAD software, high costs for high accuracy systems, and inflexibility for setting up the systems in the user or the use context (Demirel, Ahmed and Duffy, 2021). So future development in this area may be particularly valuable for supporting the PDP.

4.3. Possible limitations and future follow up

The data generation and analysis were done by a sole author, which might involve bias in the study to some extent. However, the preliminary results were discussed during a workshop with most interviewees and other researchers for validation. In order to avoid influence in the interviewees' answers, the questions were formulated as open as possible. Still, there might have been some differences in how the interviewees define different concepts, such as verification and validation. Further discussions will continue in future research project workshops when aspects such as how to handle unfulfilled requirements and long term strategies to incrementally change the design could be covered.

5. Conclusion

Results from the interview study with ten ergonomics designers show that balancing between ergonomic aspects and with other product development aspects of the product design process are their primary objectives. In addition, the involved companies follow design methods when addressing ergonomics in PD. The most reported tools were physical prototypes, VR, and DHM tools for analyzing ergonomics aspects in the vehicle development. The reported challenges are divided into interdepartmental communication issues and tool development issues, both affecting the workflow of ergonomics designers in PD. Lastly, we identify four main gaps and research directions that can enhance the current challenges that ergonomics designers face: human behaviour predictions, simulation tool usability, ergonomics evaluations, and integration between systems. While addressing these challenges is no small task, we hope that understanding what challenges active ergonomics designers face can shape and direct future research and development of DHM and other design tools to support the development of more ergonomic design effectively.

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PAPER B

Simulation of hip joint location for occupant packaging design

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Abstract

DHM tools have been widely used to analyze and improve vehicle occupant packaging and interior design in the automotive industry. However, these tools still present some limitations for this application. Accurately characterizing seated posture is crucial for ergonomic and safety evaluations. Current human posture and motion predictions in DHM tools are not accurate enough for the precise nature of vehicle interior design, typically requiring manual adjustments from DHM users to get more accurate driving and passenger simulations. Manual adjustment processes can be time-consuming, tedious, and subjective, easily causing non-repeatable simulation results. These limitations create the need to validate the simulation results with real-world studies, which increases the cost and time in the vehicle development process. Working with multiple Swedish automotive companies, we have begun to identify and specify the limitations of DHM tools relating to driver and passenger posture predictions given predefined vehicle geometry points/coordinates and specific human body parts relationships. Two general issues frame the core limitations. First, human kinematic models used in DHM tools are based on biomechanics models that do not provide definitions of these models in relation to vehicle geometries. Second, vehicle designers follow standards and regulations to obtain key human reference points in seated occupant locations. However, these reference points can fail to capture the range of human variability. This paper describes the relationship between a seated reference point and a biomechanical hip joint for driving simulations. The lack of standardized connection between occupant packaging guidelines and the biomechanical knowledge of humans creates a limitation for ergonomics designers and DHM users. We assess previous studies addressing hip joint estimation from different fields to establish the key aspects that might affect the relationship between standard vehicle geometry points and the hip joint. Then we suggest a procedure for standardizing points in human models within DHM tools. A better understanding of this problem may contribute to achieving closer to reality driving posture simulations and facilitating communication of ergonomics requirements to the design team within the product development process.

Keywords: hip joint, H-point, seated reference point, simulation, digital human modelling

Introduction

Occupant packaging primarily aims to accommodate an intended range of drivers and passengers within the vehicle. However, this primary aim also constitutes one of the significant challenges, accommodating a maximal group of target users. Ideally, a vehicle will be designed to meet occupant needs while considering human diversity (Gkikas, 2016). Digital human modelling (DHM) tools have been widely used to analyze and improve vehicle occupant packaging and interior design in the automotive industry. However, these tools still present some limitations for this application. Current human posture and motion predictions in DHM tools are not accurate enough for the precise nature of vehicle interior design, hence typically requiring manual adjustments from DHM users to get more accurate driving and passenger simulations (Bhise, 2016; Brolin et al., 2020). Manual adjustment processes can be time-consuming, tedious, and subjective, easily causing non-repeatable simulation results. These limitations create the need to validate the simulation results with real-world studies, which increases the cost and time involved in the vehicle development process (Lämkull & Zdrodowski, 2020).

Working with multiple Swedish automotive companies, we have begun to identify and specify the limitations of DHM tools relating to driver and passenger posture predictions given predefined vehicle geometry points/coordinates and specific human body parts relationships. Two general issues frame the core limitations. First, human kinematic models used in DHM tools are based on biomechanics models that do not provide definitions of these models in relation to vehicle geometries. Second, vehicle designers follow standards and regulations to obtain key human reference points in seated occupant locations. However, these reference points can fail to capture the range of human variability. DHM tools aim to represent digital human models (manikins) within detailed CAD environments for analyzing and evaluating human interactions. Moreover, a lack of clear connections between occupant packaging guidelines and human biomechanical knowledge creates a limitation for ergonomics designers using DHM tools as they are not necessarily experts on all the nuances in both the guidelines and biomechanics. Perez Luque et al. (2022) reported that one of the main issues in vehicle development is the lack of accurate and reliable driving task simulation predictions. The lack of understanding and standardized procedures to make the manikin adopt the initial driving posture realistically means that ergonomics designers rely on their perception or expertise for quantifying driver seated positions and the consequential occupant packaging analyses.

This paper describes the relationship (or lack of it) between seated reference point in seat geometries and the hip joint for driving simulations, and presents an approach to sit manikins in a virtual environment considering geometric reference points and human body shape.

Problem description

Understanding and quantifying the initial seated posture is crucial for ergonomic and safety evaluations because the design and development of other ergonomic requirements such as seated posture comfort, operating controls, and interior and exterior visibility depend on it. Moreover, it is imperative to consider a diverse population with different body types and preferences to ensure a good fit for end-users (Gkikas, 2016). Successful vehicle interior design requires involvement insights from various fields, including biomechanics, ergonomics, engineering, and design. However, there is a lack of standardized methodologies connecting the different fields.

Ergonomics designers in automotive companies follow standards and legislation within occupant packaging and vehicle design. Standards, such as SAE and ISO, provide recommended tools and practices for defining key reference points, which specify the relative positions of the occupants with vehicle components. One of the essential reference points is the H-point (sometimes called the hip point). The H-point describes a theoretical intersection of a reference occupant's torso and thigh lines (Gkikas, 2016). This means that the H-point simulates but does not precisely represent the human mid-hip joint location and its variability across people. The location of the H-point relative to a physical seat is commonly determined using the H-point machine (HPM-II) (Figure 1), which will be called HPM in this paper, and which can be physical or digital (Reed et al., 1999; Yang et al., 2014; ISO 20176, 2020). Vehicle manufacturers define a vehicle design-specific H-point, the seated reference point (SgRP), as a fundamental reference point for determining seating location for occupant packaging and vehicle dimensions (Bhise, 2016). The SgRP enables correlation between physical and virtual environments, and provides a consistent method for comparing vehicles. While fixed seats have only one H-point position (the H-point and SgRP is the same point), adjustable seats have more than one H-point location. All these H-point locations are mapped and described in the seat movement envelope, hence representing an area rather than a point. The representation of the H-point in the virtual environment can be used as a reference point to position manikins since it does not say how they sit, but rather where they sit. However, using such standards does not ensure the consideration of human diversity and variability sufficiently.

An alternative to starting with design standards is to investigate human body angles in driving situations to determine expected human driving postures. Over the years, many authors have followed this approach from the biomechanical and ergonomic design fields, mainly focusing on values of subjective comfort, human structure and functions in driving situations (Schmidt et al., 2014). However, the results of these works are typically not specified in terms of actual vehicle reference points, making it difficult or impossible to apply the results in current design contexts without significant effort. Moreover, authors often consider and observe variability in posture among different factors, including gender, anthropometric measurements, age, symmetry, seat design, vehicle model, and driving venue (Reed et al., 2002; Schmidt et al., 2014; Park et al., 2016a). While these studies focus on human driving posture and body joint angles, they rarely define the relation of those angles to the driver seat geometries, limiting their current influence on DHM tools and design processes in general.

Figure 1. Physical HPM-II model (ISO 20176, 2020)

Park et al. (2016) have developed a data-based prediction model for passengers considering body dimensions, age, and gender from humans with reference to vehicle layout measurements. This model follows a “cascade” approach, in which the most relevant variables are predicted first, followed by less important variables. One component of this model is a regression model, which predicts the mid-hip location to the H-points of the automotive seat. The main limitations of this study include the fixed position of the backrest angle, the use of a non-naturalistic laboratory setting and the limitation to participants in the USA. Reed et al. (2019) expanded this model for pelvis position and rotation in the automotive seat by including data from highly reclined postures in automotive seats. They concluded that the spine and pelvis posture changes as the torso reclines in an automotive seat.

While existing human posture studies have been used to develop data-driven and optimization methods, some additional considerations complicate predicting hip joint locations related to vehicle geometries. The appearance of the manikins in DHM tools can lead to different and/or inaccurate predictions when it is tied to collision volumes determining the boundaries of the manikin relative to the seat geometry (Lämkull et al., 2007; Lämkull & Zdrodowski, 2020). Thus, even with possibly accurate predictions of where the mid-hip should be placed, ergonomics designers may modify the manikin's position and posture until the manikin body shape looks appropriately aligned to the automotive seat and realistically represents a human-vehicle interaction. Brolin et al. (2020) introduced DHM functionality which uses the mid-hip to H-point relationship from Park et al. (2016) together with constraints and adjustment ranges of vehicle components to statistically predict seated driving postures. Even though the results were promising, initial comparisons of such predictions with data from user tests showed some differences, which indicates that further research is needed.

In this paper, we compare the regression models from Park et al. (2016) with an approach to seat manikins in driving environments using more realistic human body meshes with a wide range of body mass index (BMI).

Approaches for initial seated driving posture

Several DHM tools are currently used for occupant packaging and automotive design like Ramsis, Santos, Jack Siemens, and IPS IMMA. While these tools are based on different modelling and prediction methods for defining the seated driving posture, they all mostly follow the same procedure for adopting the initial seated driving posture. A *DHM standard procedure*, identified in discussions with companies, consists of the following steps: First, DHM users make the manikin or manikin family assume the driving posture, which is generally defined by the DHM software following specifying angles according to a particular study. Second, constraints are set to fulfil basic requirements and get an initial seated driving posture consistent across manikins and simulations. These constraints or requirements are typically defined in the feet (e.g. heels should take up support, right foot on the accelerator pedal), grip points on the steering wheel, eye or mid-eye vector to define the head direction, and top of the head to ensure the manikin stays within a particular space. Finally, manual adjustments are often made. Typically, the mid-hip, torso or the automotive seat position are manually adjusted and constrained to get postures visibly fitting the seat geometry, following the ergonomics designer criteria. Notably, this DHM procedure has a limitation of not having any clear and direct relationship relating the mid-hip point and the H-point and the adjustment range of the seat. So, even if the manikin has "optimal angles", we would not know how or where to place it in the automotive seat.

As an attempt to fill this gap, in this paper, an approach is presented as an alternative to the previous ones to be able to sit manikins realistically and consistently while considering the standards and legal requirements for occupant packaging, a wide range of BMI, and the mid-hip and H-points relationship in driving environments. This approach was compared to the *Statistical Prediction* approach proposed by Park et al. (2016), which showed some issues for particular cases as mentioned in the literature.

Method

Figure 2 summarises the main steps of the *Body Shape Alignment*, the proposed approach in this paper. The first step consists of loading a virtual template of the HPM in the digital environment and positioning it in the vehicle direction. Next, the H-point of the HPM template should be aligned to the SgRP of the driver seat as well as the HPM template surfaces to the driver seat surfaces. Once the HPM template is correctly aligned to the seat, the manikin should be aligned to the HPM template surfaces. As the last step, the coordinates of the SgRP, H-point and mid-hip point could be extracted from DHM software to analyze the relationship or offset between these different variables further.

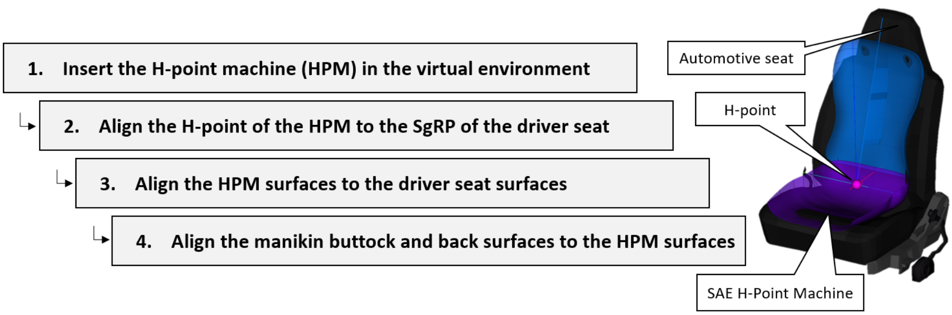

Figure 2. Body Shape Alignment approach.

Comparison procedure

The Statistical Prediction and Body Shape Alignment approaches were used to sit manikins in virtual driving environments using a wide range of BMI human body meshes. The automotive vehicle model was a Volkswagen Beetle from the training repository of Siemens Jack (Siemens, 2017). The human body meshes were obtained from the BioHuman website (*UMTRI BioHuman*, 2022) which provides 3D manikin meshes based on statistical analyses of high-resolution laser scans. The human body meshes were used for the comparison of seated driving approaches due to their closer to reality appearance. A manikin family of 7 females and 7 males was considered in the simulations. The anthropometric measurements of the manikin family were generated from two three-dimensional boundary ellipsoids, with a confidence level of 90% (Brolin et al., 2012). One ellipsoid for each sex, based on stature, body weight, and sitting height. The anthropometric data was taken from the CAESAR data set (Robinette et al., 2002). In addition to an average manikin case for each sex, six manikin cases were defined at the ends of the three axes of each of the two ellipsoids. US population was selected on the BioHuman website to generate the meshes since the regressions models of the Statistical Prediction from Park et al. (2016) are done considering the US population. 40 years was the defined age for all the manikins. Table 1 describes the anthropometric measurements of the manikin family. It should be noted that the generated test manikins span a more extensive range of BMI values compared to the sample from Park et al. (2016), which might affect the results. However, the measurement combinations of the generated test manikins are realistic and could be found within the CAESAR data set, which motivates the use of these more extreme test manikins.

Table 1. Anthropometric measurements of the manikin family cases.

Manikins	Body weight (kg)	Stature (mm)	Sitting height (mm)	BMI (kg/m^2)	Manikins	Body weight (kg)	Stature (mm)	Sitting height (mm)	BMI (kg/m^2)
Case F1	65	1639	865	24	Case M1	83	1775	926	26
Case F2	95	1814	949	29	Case M2	124	1974	1018	32
Case F3	50	1475	785	23	Case M3	61	1604	836	24
Case F4	47	1678	893	17	Case M4	62	1818	959	19
Case F5	115	1600	837	45	Case M5	122	1734	892	40
Case F6	64	1695	838	22	Case M6	82	1834	899	24
Case F7	66	1584	891	26	Case M7	85	1719	953	29

Results

Figure 3 shows the human mid-hip to the H-point of the automotive seat locations using the Statistical Prediction and Body Shape Alignment approaches. Figure 4 shows the initial driving posture of the human body meshes by using the Statistical Prediction and the Body Shape Alignment. It can be seen how the meshes in the former are more spread on the x-axis compared to the second approach (Figure 4).

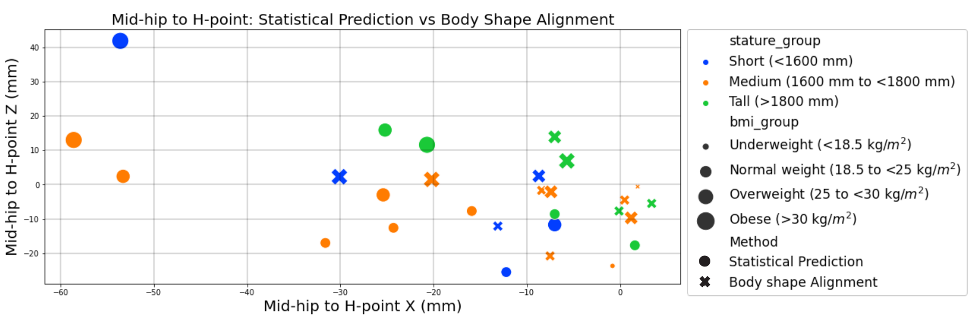

Figure 3. Mid-hip to H-point: Statistical Prediction vs Body Shape Alignment.

The most evident differences in the mid-hip to the H-point location were observed concerning the BMI. Figures 3 and 4 show how manikins with higher BMI move forward in the x-axis with the Statistical

Prediction approach. This might not be seen as a problem at first. However, the gap between the manikin with higher BMI and the seat is significantly evident when observing the human body meshes in such a position. That would lead to manual adjustments, making the statistical prediction not accurate. On the other hand, it can be seen how the spread in the x and z-axis is not as wide with the Body Shape Alignment as with the Statistical Prediction.

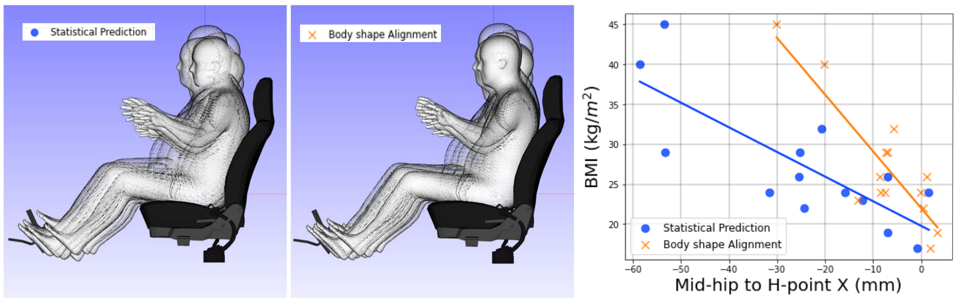

Figure 4. Initial seated driving posture: Statistical Prediction (left) and Body Shape Alignment (right).

Discussion and Conclusion

The presented Body Shape Alignment is an procedure for seated driving posture. Although the results show that obtaining the initial seated driving posture is possible, it needs further development. In its current state, it predicts the mid-hip to the H-point location of the automotive seat in a standardised way considering standards and legal requirements and human body shape variability. However, this initial procedure might involve subjectivity in the alignments. This can be fixed by further defining appropriate constraints between the automotive seat, HPM, and the manikins or body shapes meshes in the virtual environment. In this way, if you move the seat, the H-point's location and the mid-hip point's prediction will also move. The human body meshes used in the comparison were calculated with sex, stature, BMI, SH/S, and age from the BioHuman framework. However, people with the exact anthropometric measurements could also have different body shapes. Fit people could have BMI rates of overweight people due to the larger muscle mass. That is not reflected in the presented study. In addition, changing the body joint angles of the meshes was not possible since it is defined in the 3D mesh generation from the BioHuman framework. Going further, the use of the body shape alignment approach for getting a proper initial driving posture relies on, and therefore requires, accurate human body meshes within DHM software. The accuracy of the body shape alignment approach could be further advanced by implementing models regarding seat foam and human buttock deformation (Wang et al., 2021). The mid-hip location prediction on the human body shapes could also be further advanced, e.g. by considering the study from Brynskog et al. (2021), in which detailed pelvis geometry is predicted with overall anthropometric variables. While the presented approach seems to have consistent results across different anthropometries, more research is needed to know, for example, if this approach applies to non-US populations and other types of vehicles.

Figure 4 shows that the more considerable differences between mid-hip to H-location with different approaches come as the BMI increases. One reason could be the limited representation of people with higher BMI values in the developed regression models compared to the manikin family used in this study. In addition, measurement errors could occur since the mid-hip is a problematic and challenging point to identify. At the same time, the differences could have been due to the estimations of the mid-hip point in the human body meshes used in this study. While previous studies have found mid-hip locations typically forward of the H-point (Reed et al., 2002; Park et al., 2016a, 2016b), it can be seen in this study that various mid-hip locations are slightly rearward of seat H-point, as shown in Reed et al. (2019). Delving deeper into these mid-hip to H-point differences, we should consider the postural diversity within a population. Then, what can be viewed as a postural variety and an error? What can be considered an accurate initial driving posture? The Statistical Prediction approach includes root mean square error (RMSE) that could be used to represent human diversity. However, while statistical predictions can be beneficial, we should also consider that such values are determined for the specific conditions and population of that study and might be limited to use in other conditions, vehicle types, and different populations. The driving posture was different in the past and will be different in the future, especially considering the introduction of autonomous vehicles and new concepts of transportation (Yang et al., 2019). Simulation and evaluation of different sitting postures and non-driving related activities are becoming critical for developing future vehicles in regards to ergonomics and safety. When modelling human-product interactions, the main challenge comes with the need to be able to predict any possible interaction (in existing and future vehicles) accurately and realistically. Further research is required to identify and define suitable interaction models for engineering design covering a universal valid approach.

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PAPER C

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Simulation-based multi-objective optimization combined with a DHM tool for occupant packaging design

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ABSTRACT

Occupant packaging design is usually done using computer-aided design (CAD) and digital human modelling (DHM) tools. These tools help engineers and designers explore and identify vehicle cabin configurations that meet accommodation targets. However, studies indicate that current working methods are complicated and iterative, leading to time-consuming design procedures and reduced investigations of the solution space, in turn meaning that successful design solutions may not be discovered. This paper investigates potential advantages and challenges in using an automated simulation-based multi-objective optimization (SBMOO) method combined with a DHM tool to improve the occupant packaging design process. Specifically, the paper studies how SBMOO using a genetic algorithm can address challenges introduced by human anthropometric and postural variability in occupant packaging design. The investigation focuses on a fabricated design scenario involving the spatial location of the seat and steering wheel, as well as seat angle, taking into account ergonomics objectives and constraints for various end-users. The study indicates that the SBMOO-based method can improve effectiveness and aid designers in considering human variability in the occupant packaging design process.

1. Introduction

In vehicle design, occupant packaging is a layout problem involving selecting locations for components such as pedals, seats, and steering wheel in the vehicle cabin under certain geometric constraints, e.g., general vehicle cabin dimensions. A general aim of occupant packaging processes is to identify vehicle cabin configurations that meet defined accommodation levels for expected vehicle occupants, or to optimize accommodation levels when targeted levels cannot be met due to conflicting requirements. Generally, accommodation measures the number of individuals who can find a comfortable posture to perform all required transportation tasks (Roe, 1993; Gkikas, 2016; Bubb et al., 2021). Component location is critical because it is the primary constraint on occupant postures, affecting both satisfaction and safety (Scott et al., 1996; Parkinson and Reed, 2008; Smith et al., 2015; McMurry et al., 2018; Reed et al., 2019; Gao et al., 2022; Jeong et al., 2023). Considering a diverse population with different body types and behavioural preferences is imperative to ensure a good fit, safety, comfort, and accessibility for targeted end-users. However, diversity consideration introduces one of the most significant challenges in occupant packaging, often requiring design engineers and ergonomics designers to consider several interdependent and sometimes conflicting factors and objectives, such as addressing roominess and reachability for people of different sizes while maintaining the vehicle's safety, aerodynamics, and overall styling (Wolf et al., 2022). The consideration of interdependent and conflicting factors occurs throughout the development process via balancing activities until the design is fully defined. For that, designers aim to find the best balance ("compromise") that enhances the vehicle's performance without sacrificing the ergonomics and safety of individuals with varying anthropometrics (Bhise, 2016; Kikumoto et al., 2021). The balancing activities involve various tools or methods, including physical prototypes, user tests and interviews, computer-aided design (CAD), and digital human modelling (DHM) tools to investigate or simulate the interaction between the occupants and the vehicle interior (Gkikas, 2016; Perez Luque et al., 2022a). Although CAD and DHM tools have significantly advanced the design process by enabling design and simulations in a virtual world, the use of such tools still involves issues, such as time-consuming procedures and occasionally providing questionable simulation results, which in turn, can lead to a need for costly re-design iterations or validation studies (Wolf et al., 2020; Parida et al., 2020; Demirel et al., 2021).

Previous studies identify that ergonomics designers typically follow a

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standard procedure when using DHM tools in occupant packaging design (Green, 2000; Hanson et al., 2006, 2008; Gkikas, 2016; Perez Luque et al., 2022b). The standard procedure typically involves two main activities: 1) The digital manikin(s) is (are) set to an initial posture, typically a default posture based on predefined body joint angles; 2) Constraints are then defined to ensure basic requirements are fulfilled, e.g., placing feet on the pedals and hands on the steering wheel. Depending on the DHM tool, the sequence of these steps may vary, but typically, the essential activities remain the same. Next, the DHM tool automatically positions the manikin(s) in a posture that is considered optimal, i.e., the most likely seated posture according to the DHM tool's prediction method, while fulfilling the defined constraints. To represent anthropometric diversity, the DHM standard procedure is generally performed using a set of manikins, collectively often referred to as a *manikin family*, where each manikin has a systematically defined combination of anthropometric measurements (Brolin et al., 2019).

Since occupant packaging design is a complex act of finding a successful balance between many kinds of requirements, there is a need for methods that can assist design engineers and ergonomics designers in considering multiple objectives and constraints in a simultaneous optimization process. Simulation-based multi-objective optimization (SBMOO) is a process that combines digital simulations with optimization methods, with the aim of finding superior solutions to design problems involving multiple, potentially conflicting, objectives. In the SBMOO process, the design problem is modelled in a simulation tool, and relevant conflicting factors are defined as objective functions (Carson and Maria, 1997; Lorig, 2019). Recent studies have shown the potential of SBMOO in manufacturing contexts (Barrera Diaz et al., 2021; Guo et al., 2021; Lind et al., 2023). In regards to vehicle design, SBMOO has been applied to performance and component's design (Xu et al., 2004; Nandi et al., 2015; Holjevac et al., 2020; Astaneh et al., 2022; Wang et al., 2024) as well as vehicle structure design from a safety point of view (Gu et al., 2013; Dai et al., 2023; Gao et al., 2024). However, the application of SBMOO methods to occupant packaging design is still relatively unexplored. More traditional optimization techniques have been applied to solve vehicle occupant packaging problems, in which optimization techniques are used as a stochastic posturing method to consider anthropometric and postural variability (Parkinson and Reed, 2006; Parkinson et al., 2007). However, these approaches do not incorporate optimization techniques combined with DHM simulations. SBMOO has recently been combined with DHM tools for production design contexts (Iriondo Pascual et al., 2021). The framework in Iriondo Pascual et al. (2021) for performing SBMOO consists of three main phases: 1) Problem definition and creation of the optimization model; 2) Optimization process; and 3) Presentation and selection of results.

This paper investigates the potential advantages and challenges of using a SBMOO method combined with a DHM tool in occupant packaging design, following the framework in Iriondo Pascual et al. (2021). The aim is to investigate and discuss how such an approach can improve the effectiveness of the occupant packaging design process, addressing two main challenges common in current process: limited support for balancing multiple and conflicting objectives, and the tendency to perform limited analyses due to time-consuming procedures.

2. Method

This section describes how Phases 1 and 2 in the SBMOO framework in Iriondo Pascual et al. (2021) were set up and run to solve a fabricated occupant packaging design scenario.

2.1. Problem definition and creation of the optimization model

2.1.1. Problem definition

One crucial aspect concerning driver accommodation in occupant packaging is selecting the appropriate size and position of the seat adjustment range. The issue of sizing and placing adjustment ranges becomes more complicated for ergonomics designers if both the seat and the steering wheel are adjustable. Very large adjustment ranges for all components would accommodate almost all drivers. However, adjustability ranges are typically limited by other design considerations, including costs, space, and safety. Thus, the seating adjustment range specification is often a critical step for ergonomics designers in occupant packaging design.

A design scenario was created to investigate the application of SBMOO in occupant packaging design. The scenario involved determining the location of the seat adjustment range centre, the position of steering wheel, and the angle of the seat bottom. To achieve this, the location specification of these elements was divided into five variables: fore-aft and vertical location of the seat adjustment range centre, fore-aft and vertical location of the steering wheel centre, and angle of the seat bottom. These constitute the optimization *variables* in the design scenario (black arrows in Fig. 1). The variables have a limited range within which they can be optimized (grey areas in Fig. 1). The optimization problem is constrained by vehicle geometries such as the dimensions and locations of the roof, pedals, dashboard, and windscreen. The seat adjustment range, which has a fixed size area, is also part of the constraints (light blue delimited area in Fig. 1). A set of manikins with different anthropometric measurements were considered, each having unique optimal joint angles for a seated driving posture (clarified in Section 2.2.3). In this way, each manikin's combination of optimal body joint angles represents one *objective* of the optimization problem.

The design scenario illustrates an optimization problem in which comfort-related postural aspects (manikin's combination of body joint angles) are considered when optimizing the seat adjustment range centre position, steering wheel position, and seat bottom angle. In addition, four design constraints are added to the optimization problem (red arrows in Fig. 1): head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle, which are measured and evaluated against predefined targets along with the comfort-related postural aspects.

2.1.2. Model creation (DHM tool)

The DHM scene (Fig. 2) was modelled in a research version of the DHM tool IPS IMMA version 2023.R2 (IPS IMMA, 2023). A family of 14 manikins (7 female, 7 male), each with different anthropometrics (Table 1), was defined using the CAESAR data set (Robinette et al., 2002). Using existing manikin definition functionality in IPS IMMA, anthropometric data for each manikin was generated from two 3D boundary ellipsoids (one ellipsoid per sex) with a confidence level of 90% (Brolin et al., 2012), i.e. assuming a 90% accommodation level for the targeted population given by the CEASAR data set. Three body dimensions were selected as key dimensions: stature, body mass, and sitting height. This selection was based on the body dimensions used in the regression equations in Park et al. (2016). In addition to central cases (Manikins 1 and 2 in Table 1), representing an "average" manikin per sex, six boundary cases were defined at the ends of the three axes of each of the two ellipsoids, each representing test subjects with extreme but realistic combinations of the key anthropometric variables (Manikins 3 through 14 in Table 1). In IPS IMMA version 2023.R2, anthropometric data, including weight, is treated as normally distributed when defining the multidimensional ellipsoids and boundary cases. However, weight is positively skewed in the CAESAR data set and therefore the percentile values reported here, calculated based on standardised Z-values, are not identical to the actual percentile values of the CAESAR population.

An original Volkswagen Beetle (Type 1) vehicle CAD model was used to represent a relatively small cabin size (Fig. 2) and to provide a generic environment to test and demonstrate the SBMOO approach. This CAD model can be found in the training repository of Siemens Jack (Siemens, 2017). The origin (0,0) of the scene, consisting of the CAD geometry imported in the DHM tool, is located in the centre of the vehicle in x direction and on ground level in z direction (Fig. 2). Measurements in x

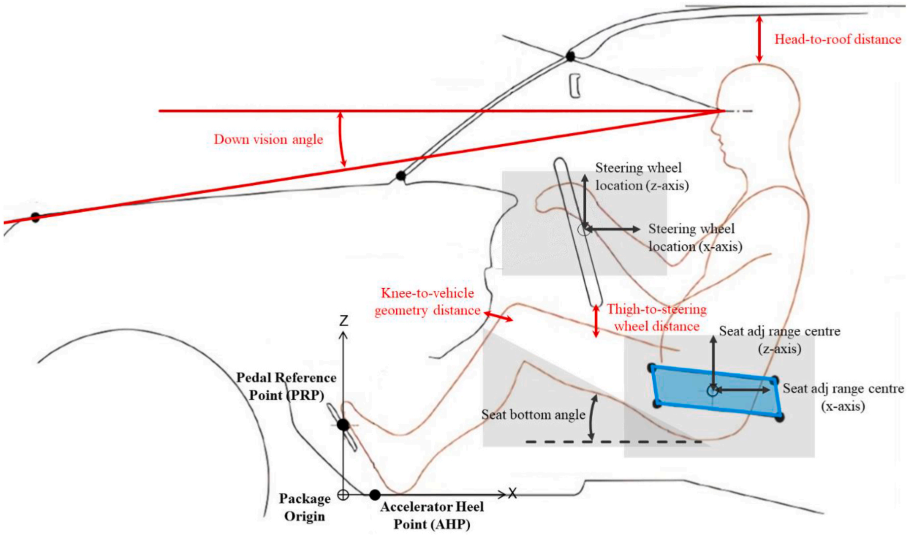

Fig. 1. Optimization definition: variables (black arrows); variables' range limit (grey areas); fixed size area of seat adjustment range (light blue); design constraints (red). Image based on Parkinson and Reed (2006). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Manikin family anthropometric measurements.

Manikin	Sex	Stature	Z	Percentile	Sitting height	Z	Percentile	Weight	Z	Percentile
1	F	1639	0.000	50.0	865	0.000	50.0	66	0.000	50.0
2	M	1776	0.000	50.0	925	0.000	50.0	85	0.000	50.0
3	F	1601	−0.535	29.6	839	−0.732	23.2	94	2.028	97.9
4	M	1732	−0.563	28.7	892	−0.836	20.2	114	1.732	95.8
5	F	1694	0.757	77.6	838	−0.759	22.4	65	−0.096	46.2
6	M	1833	0.718	76.4	898	−0.686	24.6	83	−0.132	44.7
7	F	1806	2.315	99.0	947	2.257	98.8	85	1.368	91.4
8	M	1960	2.326	99.0	1016	2.260	98.8	114	1.732	95.8
9	F	1678	0.535	70.4	891	0.708	76.0	38	−2.073	1.9
10	M	1819	0.540	70.5	957	0.787	78.5	55	−1.817	3.5
11	F	1584	−0.772	22.0	892	0.736	76.9	67	0.050	52.0
12	M	1719	−0.727	23.4	952	0.663	74.6	86	0.048	51.9
13	F	1472	−2.330	1.0	783	−2.284	1.1	47	−1.414	7.9
14	M	1592	−2.338	1.0	834	−2.284	1.1	56	−1.757	4.0

Note: F = female; M = male; Stature in mm; Sitting height in mm; Weight in kg.

and z direction from the Scene origin to the Packaging origin, are given in Fig. 2. The Body Shape Alignment approach in Perez Luque et al. (2022b) was followed to make manikins adopt an initial seated driving posture within the virtual environment. Six so called attachment points were used in IPS IMMA to align manikin meshes to a virtual template of the H-point machine (SAE J826, 2021; SAE J4002, 2022) in addition to positioning the manikins' hands on the steering wheel, and right foot on the accelerator. These attachment points limit the potential movements of the manikins to a two or three-dimensional defined range (a line, a plane, or a box). Fig. 3a illustrates the six attachment points (T12L1 surface, T6T7 surface, lower back surface, buttock bottom surface, and right and left knee crease) used to align the manikin mesh to the H-point machine template, referring some of the attachment points to internal joints and others to mesh surfaces (see Perez Luque et al., 2022b for more details). Once the attachment points are defined, the DHM tool defines a final posture following a quasi-static posture prediction method for each manikin that fulfils the defined attachment points (Bohlin et al., 2012; Delfs et al., 2013).

2.2. Optimization process

2.2.1. Optimization method

Candidate design variants are identified through an iterative process of simulation and optimization, where each design variant consists of a different seat adjustment range centre, steering wheel position, and seat bottom angle. For a given design variant, simulations produce data for objective functions, which are used to evaluate the current design variant (clarified in Section 2.2.3). The genetic optimization algorithm NSGA-II was selected for its efficiency in multi-objective optimizations, its suitability for handling a range of correlated objectives, and the ability to incorporate objectives and constraints of both integer and real number nature in optimizations (Deb et al., 2002). An initial random population of 50 design variants was generated within the constraints of the vehicle cabin to seed the NSGA-II algorithm. Then, iteratively, the algorithm evaluated the data obtained from each simulation from the previous population, and then generated a new child population. The NSGA-II algorithm specifies variable values for new design variants by

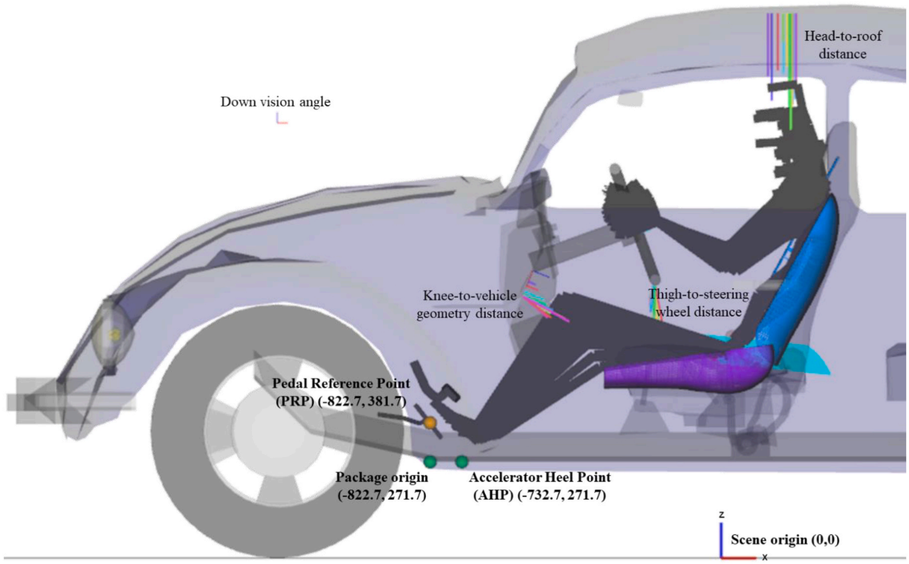

Fig. 2. Scene in the DHM tool: vehicle geometries, manikin family seated in a driving position (stick figure visualisation), and design constraints (head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle). Measurements in mm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

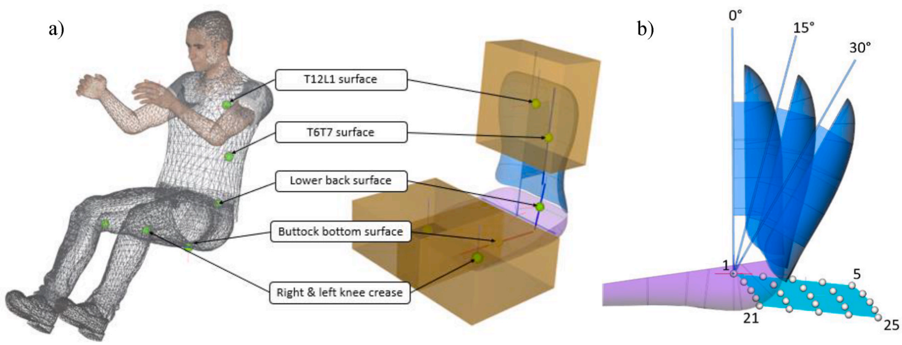

Fig. 3. a) Attachment points of the H-point template for the Body Shape Alignment approach; b) 25 seat adjustments (white spheres) inside the seat adjustment range (light blue shape) and three torso angles (0, 15, and 30°). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

applying crossover, mutation and selection operators (Deb et al., 2002) (Table 2). NSGA-II continued this process for 1000 iterations. Each design variant consisted of a different seat adjustment range centre, steering wheel position, and bottom seat angle. All 14 manikins were simulated for each design variant, including new values of seat adjustment range centre, steering wheel position, and bottom angle, using a multi-simulation approach (described in Section 2.2.2).

Table 2
Optimization method configuration.

Optimization method	NSGA-II algorithm
Population size	50
Child population size	50
Crossover probability	0.95
Mutation probability	0.05
Tournament size	2
Maximum iterations	1000

2.2.2. *Simulation method*

As described, each design variant was evaluated by simulation in the DHM tool, and then the evaluation results were provided to the NSGA-II algorithm to continue the optimization process. Each design variant, with a defined seat adjustment range centre, steering wheel position, and seat bottom angle, provides many possible configurations for vehicle occupants to set the seat position and back angle within the seat adjustment range, which consequently affects their driving posture. In order to simulate these possible configurations within the adjustment ranges in a DHM tool, 25 equidistant seat positions covering the entire seat adjustment range (white spheres in Fig. 3b) and three torso angles (0, 15, and 30° in Fig. 3b) were specified. The seat adjustments (25) combined with the torso angles (3) were considered in each design variant, resulting in 75 sitting postures per manikin (14) in each design variant.

A multi-simulation approach was used to evaluate the 75 sitting postures, running multiple sub-simulations for each of the 75 sitting postures for the 14 manikins. Therefore, 1050 sub-simulations conform to a complete simulation for one design variant. The simulation method step in the optimization framework in Iriondo Pascual et al. (2021) can be represented as a flowchart of sub-steps to perform the sub-simulations (Fig. 4). Once the design variant was automatically set up in the DHM tool, the first seat adjustment of the seat adjustment range was set. For each seat adjustment, each manikin was automatically seated using three torso angles (0, 15, and 30°).

Simulation values of each manikin and sub-simulation were stored for later analysis. Iteratively, results from the 1050 sub-simulations were evaluated in the next steps of the optimization process. This simulation method step (Fig. 4) was performed for every design variant to generate evaluation data to feed the optimization algorithm.

2.2.3. Driver posture evaluation

As the basis for quantifying comfort and inclusion, joint angle values of the manikins simulated in the DHM tool were compared with joint angle values from the regression models in Park et al. (2016). Based on sex, age, anthropometrics, and vehicle measurements, these regression models predict hip-to-eye vector, head, neck, thorax, abdomen, pelvis, right thigh, and right knee angles (Park et al., 2016). To explain posture variation, Park et al. (2016) give the root mean squared error (RMSE) for each joint angle. In this study, these RMSE values were used to specify an acceptable range of deviation between the manikin joint angles in the simulations and those predicted by the regression models (pRMSE). A simulated joint angle was considered acceptable ("OK") if the difference between the simulated and the model predicted value was not greater than pRMSE for that joint. If a joint angle deviation was greater than its corresponding pRMSE, the joint angle was considered unacceptable ("NOT OK"). Thus, joint angle acceptability was defined for each joint of each manikin individually in terms of an acceptable range as opposed to an optimal value. The number of acceptable and unacceptable joint angles were counted to quantify the overall acceptability of a manikin posture, providing an objective function to evaluate driver postures for the design variants generated in the optimization process.

There was a joint angle offset between the skeleton definitions provided by IPS IMMA (Hanson et al., 2019) and used in this study, and the skeleton definitions provided by the HumanShape framework (UMTRI HumanShape, 2020) and used in the Park et al. (2016) study. As a result, joint angles provided by IPS IMMA could not be directly compared to the results which were used in the body angle equations for posture evaluation. This body joint angle offset was compensated by measuring and incorporating the differences between the skeleton definitions in the equations for the posture evaluation. Offset values for the compensation were based on differences measured between joint angles for a neutral posture with arms resting at the side (n-pose). Table 3 lists the identified and incorporated offsets for each manikin and joint, including the hip-to-eye, head, neck, thorax, abdomen, pelvis, thigh, and knee.

2.2.4. SBMOO definition: mathematical modelling

In the mathematical definition of the design scenario, manikin anthropometries and postures were considered in the optimization objectives along with the design constraints. The design constraints were specified as mandatory minimum distances between the manikin and the vehicle interior, which are typically considered in occupant packaging design. In this case, these included: head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle (Fig. 2). There were 14 optimization objectives, i.e. one evaluation of total number of unacceptable angles per manikin (numbers of "NOT OK"). The indices and parameters of the design scenario, with the addition of outputs to calculate the number of design constraint violations, are given in Table 4, and measurement limits of the constraints, with assumed realistic limit values, are given in Table 5.

The total number of unacceptable joint angles for each manikin in each seat adjustment for one design variant (f_{m_t}) was determined by calculating the absolute difference between the angle value from the simulated manikin and the predicted angle for the manikin and comparing it to the pRMSE (r) in Park et al. (2016) for a specific joint angle i and anthropometrics of a manikin m.

$$f_{m_t} = \sum_{i=1}^{I} \left[\left| a_{i_{m_t}} - p_{i_{m_t}} \right| > r_{i_m} \right] \quad (1)$$

f_{m_t} represents the number of angles that deviate more than r_i from the regression model predicted value for manikin m in adjustment t. Prac-

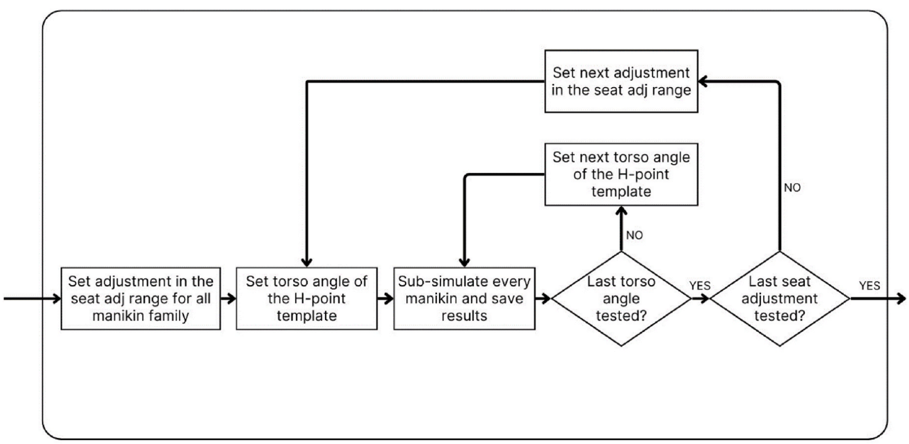

Fig. 4. Multi-simulation approach with sub-simulations performed for each occupant packaging design variant.

Table 3
Skeleton offset per manikin and joint angle (degrees).

Manikin	Hip-to-eye	Head	Neck	Thorax	Abdomen	Pelvis	Thigh	Knee
1	4.16	−27.06	−11.43	4.21	−0.36	8.1	2.79	−8.59
2	7.31	−24.28	−3.26	9.89	−2.22	14.04	5.25	−10.63
3	2.12	−29.09	−14.04	−0.52	4.92	6.68	0.82	−6.27
4	5.5	−25.64	−7.64	6.01	2.3	12.87	2.71	−9.64
5	3.92	−27.14	−10.99	1.95	4.32	7.78	0.86	−7.48
6	6.68	−24.17	−3.7	7.71	0.57	13.63	2.83	−7.94
7	−0.44	−32.51	−17.87	−5.36	1.33	4.22	−1.06	−3.59
8	3.1	−28.45	−11.25	1.3	4.27	10.47	0.87	−9.1
9	1.6	−30.45	−15.27	−1.47	2.39	5.42	0.99	−7.33
10	4.99	−26.74	−6.97	5.03	0.4	11.94	3.67	−11.68
11	0.14	−31.43	−17.8	−4.15	4.09	4.44	0.81	−5.78
12	3.79	−28.02	−10.86	2.88	2.92	10.61	2.57	−11.92
13	1.83	−29.72	−14.59	−1.05	3.64	5.87	0.71	−6.56
14	5.2	−26.24	−7.36	5.48	1.26	12.34	3.1	−10.86

Table 4
Indices, parameters, and simulation outputs of the design scenario.

Indices	Parameters	Simulation outputs
$i = 1.I$ Angle index	$I = 8$ Total number of body joint angles	$a_{i_{m_t}}$ Simulated body joint angle values from manikin
$m = 1.M$ Manikin index	$M = 14$ Total number of manikins	$p_{i_{m_t}}$ Predicted body joint angle values from manikin
$j = 1.J$ Design constraint index	$J = 4$ Total number of design constraints	$c_{j_{m_t}}$ Design constraint measurement
$t = 1.T$ Adjustment index	$T = 75$ Total number of adjustments	
	r_{i_m} pRMSE of each body joint angle of manikin	

Table 5
Constraint limits of the design scenario.

Design constraints name	Design constraint limit
Head-to-roof distance (mm)	$l_1 = 50$
Knee-to-vehicle geometry distance (mm)	$l_2 = 50$
Thigh-to-steering wheel distance (mm)	$l_3 = 50$
Down vision angle (°)	$l_4 = 13$

tically, f_{m_t} represents the number of unacceptable joint angles for a given manikin for each seat and torso angle adjustment. The solutions were also evaluated for the design constraints violated. For each manikin, m, and adjustment, t, a violation was identified when the measurement value of the design constraint c_j was less than the design constraint limit l_j. The total number of violations h_{m_t} was calculated:

$$h_{m_t} = \sum_{j=1}^{J} \left[l_j \geq c_{j_{m_t}} \right] \tag{2}$$

The complete set of adjustments evaluations, S, including the violated constraints and the optimization objectives for each manikin m was defined:

$$S_m = \left\{ h_{m_t}, f_{m_t} \right\} \tag{3}$$

The result of constraints violated for each manikin was calculated as the minimum value of h_m^{Min} in S_m:

$$h_m^{Min} = Min\{ h_{m_t} | m_t \in S_m \} \tag{4}$$

Then, to calculate the number of unacceptable angles, the solution with the minimum unacceptable angles, f_m^{Min}, was selected from the solutions that fulfilled $S_m = h_m^{Min}$:

$$f_m^{Min} = Min\left\{ f_{m_t} | m_t \in \left\{ S_m = h_m^{Min} \right\} \right\} \tag{5}$$

The resulting minimum constraints violated of each manikin, h_m^{Min}, and minimum unacceptable angles of each manikin f_m^{Min} were given to the optimization algorithm to evaluate the design variant.

3. Results

This section describes results of the SBMOO method, which corresponds to Phase 3 in the framework in Iriondo Pascual et al. (2021).

3.1. Presentation of results

During the optimization process, 1000 design variants were iterated (generated), with each iteration performing 1050 sub-simulations. Convergence, which is the point where the iterative process stabilises by producing design variants that no longer significantly improve the specified objectives, was reached at 550 iterations (i.e., design variants generated). The number of objectives in the design scenario was 14, therefore, the Pareto front had 14 dimensions. Due to the complexity of data representation for a solution space of 14 dimensions, additional metrics showing single values as results of each iteration/design solution were used to analyse the results.

Fig. 5 shows the decision space (value limits for the optimization variables) and the solution space (each grey line represents a design variant of the occupant packaging design problem) using a parallel coordinate plot chart. In a parallel coordinate plot, each design variant corresponds to a line and each column represents a different optimization variable. These lines intersect with the vertical axes, one for each variable, and the position of each line along these axes conveys the value of the corresponding variable. In Figs. 5 and 6, the columns represent the five optimization variables and their corresponding value limits to represent the design space.

Next, the results of the design variants can be filtered to facilitate the solution selection process. In Fig. 6, filters are applied to highlight design variants that meet predefined criteria to represent the solution space. This is to illustrate how data filters can support decision processes, where colours are used to highlight variants that meet the filter criteria. The defined filter criteria include solutions that: have the minimum number of unacceptable angles for the entire manikin family (orange), have the minimum number of unacceptable angles per individual manikin (blue), and fulfil both filters (magenta). The first filter helps identify variants that have the best overall outcomes. The second filter helps identify the worst individual outcome for a given design variant. The third filter helps identify the best design variants' results considering the overall and individual outcomes (first and second filters combined). Different filters can be particularly useful as they allow identifying variants where results vary at an overall or individual level.

First, design variants that violated the design constraints were eliminated, and then the filters were applied. The filter value for the maximum unacceptable angles for the entire manikin family (orange) was specified as 56, equivalent to 4 unacceptable angles per manikin,

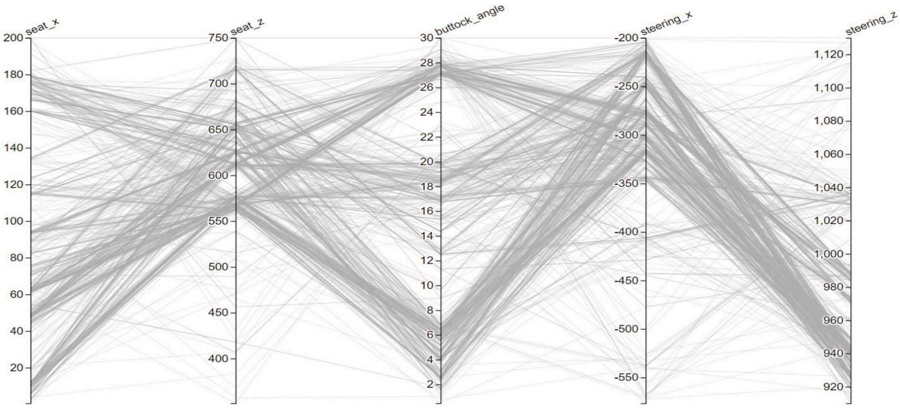

Fig. 5. Design and solution space without filters.

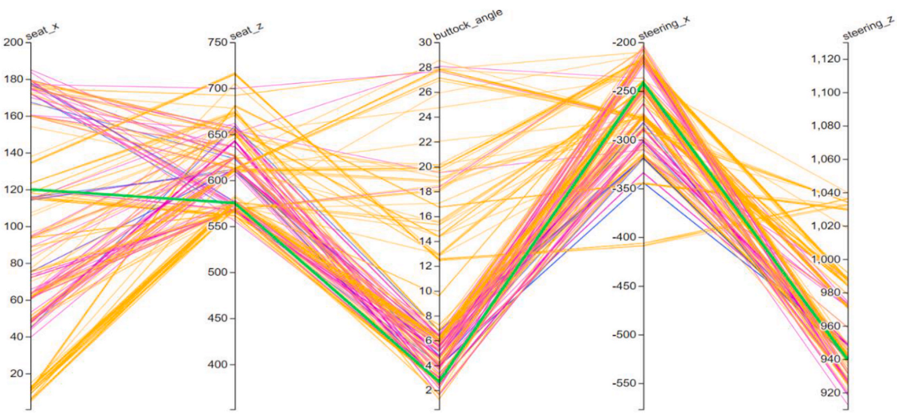

Fig. 6. Filtered solution space: max of unacceptable angles for manikin family = 56 (orange); max of unacceptable angles per individual manikin = 5 (blue); solutions fulfilling both filters (magenta); selected design variant for analysis (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and the filter value for the maximum unacceptable angles per individual manikin (blue) was specified as 5.

Fig. 7 illustrates the filtered solution in 2D plots representing a side view of the cabin as in Fig. 2, in which the seat adjustment range centre and the steering wheel position are visible following the same colour filtering as in Fig. 6. Each design variant is represented by two points in the plot, one representing the steering wheel position (top left cluster of points) and another representing the seat adjustment range centre (bottom right cluster of points) in the x- and z-axis. Regarding both the seat adjustment range centre and the steering wheel position, design variants fulfilling both filters (highlighted in magenta) are located in specific areas and not along all the tested areas across both dimensions. The steering wheel solutions are low along the z-axis, favouring an increased visual angle for drivers, and the seat adjustment centre is often farther back along the x-axis to avoid a knee collision with the vehicle geometry. This pattern is likely caused by the design constraints in the design scenario, particularly the down vision angle and the knee-to-vehicle geometry distance.

3.2. Selection of solution

In practice, an ergonomics designer may want to select specific design variants for more detailed examination. To illustrate this process, one design variant of the design scenario was chosen as an example following the third filter and focusing on the minimum number of unacceptable angles at the overall and individual level. More detailed information about the design variant and simulation results are given in Table 6. The selected design variant is highlighted in green in Figs. 6 and 7.

Table 7 corresponds to an in-depth analysis of the postures of the 14 manikins for the selected design variant, describing the adjustment index (or indices) of the seat adjustment range and torso angle value(s) providing the least unacceptable angles in the chosen design variant. In this case, the number of violated constraints is 0, but it is possible to include design variants with constraint violations at this stage if detailed information about those violations is of use. Fig. 8 illustrates the posture of manikins 1 and 2 in the adjustment index 16(15°) and 21(30°) of the selected design variant.

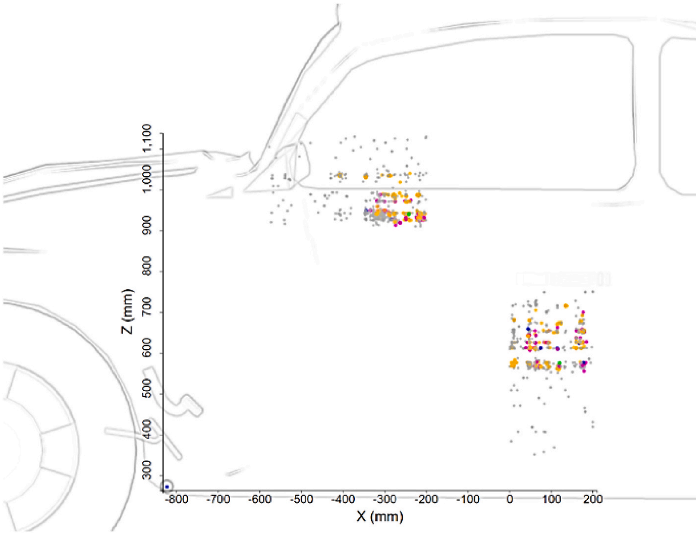

Fig. 7. Filtered solutions for seat adjustment range centre (bottom right cluster of points) and steering wheel position (top left cluster of points). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 6
Design information for selected design variant fulfilling both filters.

	Design information
Solution iteration/Design variant	#659
Seat adjustment range centre location in x-axis (mm)	120
Seat adjustment range centre location in z-axis (mm)	575.3
Steering wheel location in x-axis (mm)	−241.7
Steering wheel location in z-axis (mm)	939.8
Seat bottom angle (°)	2.7
The total sum of unacceptable angles per manikin family	38
Maximum unacceptable angles per manikin	4

Table 7
Manikin posture in-depth analysis for the selected design variant.

Manikin	Best seat adjustment index (1–25) & torso angle (0°/15°/30°) (fewest unacceptable angles)	No. of unacceptable joint angles	Names of unacceptable joint angles for the first best seat adjustment index and torso angle
1	16(15°), 21(15°)	2	hip-to-eye, neck
2	21(30°)	2	neck, pelvis
3	5(0°), 5(15°), 5(30°), 10 (0°), 10(15°), 10(30°), 15(0°), 15(15°), 15 (30°), 16(15°), 21(15°)	4	hip-to-eye, thorax, thigh, knee
4	21(30°)	3	hip-to-eye, neck, thorax
5	16(15°)	2	hip-to-eye, neck
6	5(30°), 9(30°)	3	neck, thigh, knee
7	17(15°), 21(15°), 22 (15°)	3	hip-to-eye, neck, thorax
8	22(30°)	2	neck, pelvis
9	21(15°)	1	abdomen
10	5(30°)	3	abdomen, thigh, knee
11	8(0°)	4	neck, abdomen, thigh, knee
12	16(30°), 21(30°)	3	hip-to-eye, neck, thorax
13	2(15°), 4(0°), 4(15°), 4 (30°)	3	neck, thigh, knee
14	7(30°)	3	neck, thigh, knee

4. Discussion

Occupant packaging activities are crucial for optimizing vehicle cabin space, ensuring comfort and safety for a wide range of occupants. The proposed SBMOO method combines DHM simulation and multi-objective optimization, using a genetic algorithm, to automatically generate and evaluate design solutions to find a successful one. The study was done to explore how SBMOO can enhance the effectiveness of occupant packaging design, addressing issues such as limited support for conflicting objectives and the tendency to limit analyses due to time constraints.

Firstly, the SBMOO method shows, when applied in the design scenario, how it can assist ergonomics designers to identify occupant packaging design solutions that balance multiple objectives while reducing constraint violations. This departs from the conventional, less capable, single-objective optimization approach commonly used in occupant packaging. Secondly, the SBMOO method makes it possible to consider variability across a wide range of dimensions (manikin, optimization variables, etc.) by automatically running many simulations to consider possible variations and providing interpretable results to ergonomics designers. As a result, SBMOO enables the consideration of targeted end-users with more granularity, for example, by considering the optimization of each manikin as its own objective. The ability to automatically evaluate the design space also minimises the subjectivity in the simulation process, improving simulation traceability and repeatability. In this way, design engineers and ergonomics designers can rather focus their expertise and experience on evaluation and decision-making processes, potentially increasing the effectiveness of the occupant packaging design process. Depending on the design scenarios, various features, objectives, and constraints can be tested and different evaluation methods can be deployed while facilitating the consideration of human anthropometric and postural variability in driving postures, enriching the design process.

The study illustrates the application of a framework introduced in Iriondo Pascual et al. (2021) and evaluates the results relative to the statistical prediction models specified by Park et al. (2016) based on a fabricated design scenario. Although the design scenario is fabricated, it

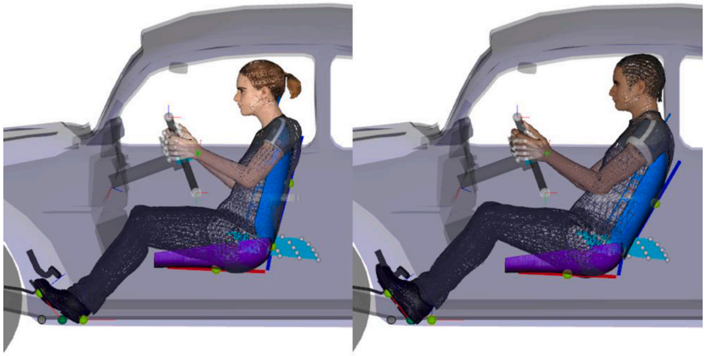

Fig. 8. Design scenario results visualisation: Manikin 1 – Adjustment 16(15°); Manikin 2 – Adjustment 21(30°). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

still includes actual design variables that are important to consider in vehicle ergonomics design processes, and represents critical aspects of driver accommodation. The ranges of the design variables are exaggerated to cover a larger area than usual. However, the range of the torso angle would have been more realistic with a smaller range, e.g. from 5° to 25° with 10° increments, instead of 0°–30° with 15° increments. Still, it is important to emphasise that the SBMOO approach can be extended to address various design challenges within occupant packaging, and by using other methods and evaluation metrics (Gao et al., 2021, 2022). In this case, the regression models in Park et al. (2016) were selected to have a posture evaluation method that considers anthropometric and postural variability across vehicle occupants. However, the regression models in Park et al. (2016) do not include any joint angle prediction equation for the upper limbs, which could be a limitation when predicting driver postures and using these prediction models to evaluate manikin postures, as in this study.

Despite the advantages and promising potentials of using the SBMOO approach for occupant packaging design, it is important to highlight that some specific challenges and limitations are closely tied to the capabilities of the DHM tool used. Current DHM tools provide a wealth of functions for predicting human posture and motion, but no contemporary DHM tool adequately considers or predicts all relevant aspects of human behaviour and form. For instance, the skeleton definition and body shape of the manikins in the version of IPS IMMA used in this study do not fully represent the diversity of potential human deformations that exist during real life driving scenarios. Moreover, cognitive factors, e.g. decision-making and mental workload, cannot be included in predictions because they are not a feature included in the predictive model used by IPS IMMA (Hanson et al., 2019; IPS IMMA, 2023). While these limitations are inherent in the software used, they also reflect broader challenges in the field of DHM (Demirel et al., 2021). Moreover, dealing with numerous variables and complex scenarios requires significant computational resources, which can also present challenges in terms of scalability and accessibility.

In summary, this study investigates using a SBMOO approach in occupant packaging design, aimed at finding design solutions that optimize driver postures for a diverse end-user group. While our method demonstrates efficacy, its efficiency, specifically in terms of time to complete an SBMOO analysis, has not been evaluated in this study and could be considered in future work. Future research challenges include comparing and evaluating the proposed method to currently used methods and exploring alternative optimization algorithms that may provide improved efficiency without compromising the quality of results. The proposed method could offer an alternative to more manual current approaches, potentially reducing the effort of engineers such as the challenges involved with manual methods (Perez Luque et al., 2022a). Further, the comparison between occupant packaging design tasks with current methods could be done by measuring factors like engineering man-hours. Additionally, although this study focuses on driver postures, different design scenarios tailored to passengers or diverse vehicles could be tested, such as trucks with unique occupant packaging requirements like stair-climbing, pedal placement, adjusting seat dimensions, or configuring control interfaces. Further research could also explore dynamic and time-based dependent evaluations, extending the application of SBMOO to larger manikin families and diverse vehicle types. This would enhance the accommodation of more diverse end-user groups and further increase the effectiveness of the work conducted by design engineers and ergonomics designers in the occupant packaging design process.

In conclusion, this study investigated and demonstrated the efficacy of using a SBMOO method combined with DHM tools to improve the effectiveness of occupant packaging design. The findings highlight the potential of SBMOO to assist ergonomics designers in balancing multiple conflicting objectives while considering human anthropometric and postural variability. These contributions provide valuable insights and a robust foundation for advancing the occupant packaging design process, thereby ultimately enhancing comfort and safety for diverse user populations.

CRediT authorship contribution statement

Estela Perez Luque: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aitor Iriondo Pascual:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Dan Högberg:** Writing – review & editing, Supervision. **Maurice Lamb:** Writing – review & editing, Supervision, Methodology. **Erik Brolin:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ergon.2024.103690.

Data availability

Data will be made available on request.

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PAPER D

Comparison of driving posture and position prediction methods for occupant packaging design – A case study

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Abstract: Accurately predicting driving postures and positions is crucial for occupant packaging design to ensure that vehicle interiors accommodate the intended and diverse driver populations. Digital human modelling tools typically play a critical role in occupant packaging design by providing functionality to predict the likely postures of different drivers in virtual environments. This paper presents a case study comparing two driving posture and position prediction methods: a statistical-based and an optimization-based method. The methods were compared against real-world data collected from two car models with different seat heights, involving 199 participants whose seat, steering wheel, and eye-point positions were measured. Patterns observed in the data, including method-specific deviations in fore-aft and vertical predictions, highlighted limitations in posture prediction accuracy and motivated this comparison. Application of these methods to real vehicles is rarely studied, and this case illustrates their performance in practice. The statistical method aligned better in the vehicle with a higher seat configuration, while the optimization method performed better in the lower seat configuration. Methodological insights based on the tested cases are outlined, providing guidance for refining posture prediction methods in DHM tools for occupant packaging design.

Keywords: occupant packaging design, digital human modelling, simulation, statistical regression, optimization

Introduction

Designing a vehicle interior to accommodate a wide range of drivers and passengers is the primary aim of occupant packaging design. Accurately predicting the seated posture and drivers' positions is crucial for effective occupant packaging design because it allows designers to optimize the vehicle interior for comfort, control accessibility, and visibility, among other ergonomic aspects, within the constraints of the vehicle geometry (Bubb et al., 2021). This predictive capability is particularly important as modern vehicles evolve to meet increasingly diverse user needs and expectations (Parida et al., 2020). Without accurate posture prediction, designers would have to rely on their own experience, previous designs, or extensive physical prototyping. These approaches are time-consuming, costly, and risk the accommodation of the target population (A. Wolf et al., 2020; Demirel et al., 2021; Perez Luque et al., 2022).

Integrating posture prediction capabilities into CAD and simulation tools in the design process provides designers with decision-support methods to develop vehicles that aim to meet ergonomic requirements. Further, to ensure inclusive, comfortable, and safe designs, human variability must be considered when predicting driver postures and key positions of their bodies, such as the H-point and eye-point. Variability can be related to differences in anthropometrics, behaviour, and individual preferences (P. Wolf et al., 2022; Jeong et al., 2023). For instance, shorter people typically sit more forward and higher within the seat adjustment range, while taller people usually prefer more rearward and lower positions within the seat adjustment range (Jonsson et al., 2008; Peng et al., 2018). Moreover, people with similar anthropometrics may behave and have different preferences, making them adopt substantially different driving postures (Peng et al., 2018). Consequently, driver prediction models should reflect these human differences in driving posture. Different driving posture models have been developed through data-driven and optimization approaches. Data-driven or *statistical* regression models generally predict the coordinates or body joint angles of specific positions of humans in the car interior. These regression models may consider different parameters from humans, such as sex, anthropometry, age, or body symmetry, and vehicles, such as vehicle class, seat designs, or driving venues (Reed et al., 2002; J. Park et al., 2016a, 2016b; Wu et al., 2022). In contrast, *optimization* methods predict human postures using optimization algorithms. It is often assumed that comfort or optimal joint angles are achieved when the torques across the joints are passively balanced. Consequently, postures can be predicted by assuming that individuals select postures that allow as many joints as possible to be close to the optimal angle (Yang et al., 2005; Hanson et al., 2006; Schmidt et al., 2014; S. J. Park et al., 2015). Thus, optimization works by minimizing deviations from the optimal angles within a search space usually restricted by defined constraints, such as roof height and available seat adjustment range, to achieve a desired prediction result. This process involves three essential components: determining the set of optimal joint angles, identifying the cost functions associated with deviations from the optimal posture, and determining how these costs are traded off or optimized (Reed et al., 2000).

Vehicle designers have traditionally used two- and three-dimensional accommodation tools provided by SAE International (formerly known as the Society of Automotive Engineers) for occupant packaging design. These tools help determine the layout of recommended seat positions, head contours, eye ellipses, and hand-reach envelopes to optimize driver ergonomics. However, due to the increasing complexity of occupant packaging design, designers have supplemented these traditional tools with three-dimensional human simulation tools over the past few decades, also known as digital human modelling (DHM) tools (Scataglini & Paul, 2019). DHM tools offer a broader framework for analyzing occupant packaging design, including tasks like ingress and egress, compared to SAE guidelines (Bubb et al., 2021). However, for posture prediction, SAE practices incorporate postural variability unrelated to driver or vehicle-specific variables, whereas the accuracy of DHM tools relies on the underlying assumptions of their predictive models, which can limit their accuracy and represent a general limitation in their application. An example of a DHM tool that has implemented a data-driven approach is Jack (Raschke & Cort, 2019), which uses a so-called cascade modelling approach, based on laboratory experiments with US subjects in specific vehicle package set-ups to predict driver positions (Reed et al., 2002). Another example of a data-driven approach implementation is a parametric CAD accommodation model for Fixed Heel Point (FHP) driving position, which provides geometric boundaries for equipped soldiers in military vehicles (Huston et al., 2024). In addition, Brolin et al. (2020) demonstrated that statistical prediction methods can be implemented in a DHM tool, which, at its core, has an optimization-based prediction approach. However, developing regression models that account for all human and vehicle parameters affecting seated driving posture is challenging for a data-driven approach due to the inherent complexity and numerous variables involved in driving situations, which must be accurately captured during data collection experiments. An example of DHM tools using an optimization approach is RAMSIS (Wirsching, 2019), with optimal angles based on a study by Seidl (1994). Limitations of the optimization approach include that the prediction models are not as accurate as the data-driven models for specific cases (Reed et al., 2000) and do not currently include anthropometry-specific comfort joint angles (Brolin et al., 2020).

While various posture prediction models have been proposed, few studies have explored how such models perform when applied in real vehicle configurations under industrial conditions. A notable exception is Reed et al. (2000), who compared prediction methods in a lab mock-up setting. However, with the growing integration of DHM tools in automotive workflows, it remains useful to examine how existing prediction approaches behave in specific contexts.

This paper presents a case study conducted in an automotive industrial setting to compare two existing prediction methods, a statistical regression-based method (SPM) and an optimization-based method (OPM), for seated driving posture. Rather than aiming to generalize to all vehicle types or populations, the objective is to examine how each method performs in relation to real-world data from two vehicles with distinct seat configurations. Through this comparison, the case study aims to provide insights into the strengths and limitations of each approach when applied in a specific design scenario. This case-based approach contributes to understanding how DHM prediction methods function in practice and supports reflection on their selection and application in vehicle interior design.

Method

Data collection context

This case study was conducted in collaboration with an automotive manufacturer, using real-world data collected in two Volvo vehicle models. The vehicles represent contrasting seating configurations relevant for occupant packaging design considerations: Vehicle A featured a lower seat configuration, H30=265 mm and L6=563 mm, and Vehicle B a higher seat configuration, H30=310 mm and L6=540 mm (all vehicle dimension definitions according to SAE J1100, 2009) (Figure 1). The test participants were asked to adjust the seat and steering wheel positions to find their preferred driving position before and during a driving session of 10 minutes in a real traffic situation. This was repeated three times, and the seat position and steering wheel position data were collected after the third driving session for each test participant. Before each trial, the seat was returned to an extreme or "not usable" position to trigger the participant to re-adjust.

Figure 1. Volvo car models used in the user study: Vehicle A (left) and Vehicle B (right).

The data collection followed an established industrial procedure, used internally at the company to evaluate occupant packaging layouts and to examine how posture prediction methods that are typically developed and evaluated in lab environments perform under actual usage conditions. The procedure for collecting H-point, eye-point, and steering wheel point data builds upon the approach described by Lejon & Thorsén (2013). Three-seat adjustment motors were employed to determine each test participant's H-point based on the seat position and adjustment settings. The seat tolerances specific to each car model were also considered when determining the H-point. The mid-eye position was captured with a photo from a fixed side-view camera in the car while participants were driving straight forward during the third driving session. Lastly, the steering wheel position was manually measured in the cars for all test participants.

Test participants were recruited from Volvo Cars staff, and they were not involved in the study's planning, design, or execution. In total, 104 participants were involved in collecting data for Vehicle A (study conducted in 2017), and 95 participants were involved in collecting data for Vehicle B (study conducted in 2018). The anthropometric data of the participants is described in Table 1.

Table 1. Summary of anthropometric data of participants involved in the case study.

	Vehicle A				Vehicle B			
Gender	**Females**		**Males**		**Females**		**Males**	
Subjects	44		60		39		56	
Anthropometric measurements	**Mean**	**SD**	**Min**	**Max**	**Mean**	**SD**	**Min**	**Max**
1 Stature without shoes	1730	128	1460	1981	1735	150	1460	2030
2 Weight (kg)	72	17	44	148.5	74	20	44	138.5
3 Waist circumference, standing	857	110	665	1260	868	127	630	1270
4 Hip width, standing	351	23	300	424	353	27	300	422
5 Upper arm length	353	31	300	428	355	39	295	454
6 Elbow-fingertip	455	40	374	550	456	45	365	560
7 Sitting height	920	58	802	1045	920	66	802	1106
8 Hip width, sitting	383	26	334	478	385	31	335	481
9 Buttock to front of knee	601	51	496	756	606	59	496	756
10 Buttock to back of knee	486	43	407	615	489	50	405	615
11 Knee height sitting, without shoes	531	53	439	667	538	60	440	699
12 Foot width without shoes (left)	94	9	74	113	95	9	74	114
13 Foot length without shoes (left)	249	22	206	304	250	26	189	312

Note. In millimetres (mm) unless otherwise noted.

Participants were selected to reflect a broad range of body measurements rather than to represent a population sample, with emphasis on including both shorter and taller individuals. As the collective anthropometric diversity of the participants did not follow a normal distribution, they were divided into three groups based on height: Short (1460–1634 mm), Medium (1635–1795 mm), and Tall (1796–2030 mm). Each group comprised approximately 1/3 of the total number of participants, ensuring a balanced representation of different statures. No separation was made between male and female participants regarding height distribution, as the analysis focused on stature regardless of gender.

Statistical Prediction Method (SPM)

The driver posturing process involves several parameters where both the steering wheel and seat can move independently of each other. In which order this process is done might vary between vehicles and drivers, i.e. it is not certain that the steering wheel position is set before the seat is positioned or vice versa. The positioning of the steering wheel and seat is rather done iteratively, more or less in relation to each other. In the study of this paper, the intention is to predict the position of all adjustable parts of the driver interior within given constraints, such as adjustment ranges of the steering wheel and seat. To implement a statistical prediction method that includes both prediction of steering wheel position and seat position, a specific sequence of these predictions needs to be decided. The statistical prediction method (SPM) uses two different prediction models in a sequence of three steps: 1) first, a prediction of the fore-aft steering wheel position is done (Reed, 2013), 2) followed by a prediction of the H-point position (Park et al., 2016b), 3) to eventually be able to predict the position for the centre-eye point (Park et al., 2016b). The fore-aft steering wheel position (L6) is a dimension that is set for each vehicle (SAE J3099, 2024). The steering wheel can usually move within an adjustment range, where L6 is defined at the centre of this range. The prediction of the steering wheel location is made by using a model that was initially developed as an ordinal logistic regression to predict the distribution of subjective responses as a function of fore-aft steering wheel position (L6), seat height (H30), and driver stature (Reed, 2013). In the study, subjective responses were measured on a rating scale that varied from 1 to 7. Response 4 was counted as "just right" and in this study inserted into a transformed linear part of the logistic model to be able to predict a preferred fore-aft steering wheel location (L6) (mm) as

$$L6 = \frac{4+0.003441\times\text{Stature}+0.01854\times\text{H30}-0.00004958\times H30^2}{0.021182} \quad (1)$$

If the predicted preferred fore-aft steering wheel location (L6) is outside the adjustment range, then the location is moved to the closest point of the adjustment range area. The predicted preferred fore-aft steering wheel location is subsequently used as the L6 value, instead of the actual L6 value of the vehicle, to predict coordinates for the preferred seat position, i.e. the H-point, with an x-coordinate in relation to the Ball of Foot point (BOF) and a z-coordinate in relation to the ankle heel point (AHP). In the study by Park et al. (2016b), the L6 value was set in different predefined positions, and participants were not allowed to change it. As a result, the prediction models could use different L6 values as predictive variables, where the recommendation is to use the vehicle L6 value in the centre of the steering wheel adjustment range. The motivation for using a predicted steering wheel location instead of the actual vehicle L6 is to statistically determine the L6 position when no predefined value is available, as well as to capture the variability found in real data, thereby reflecting a wider range of outcomes in subsequent predictions. In the third step, coordinates for the centre-eye position are predicted in relation to the H-point position. Following the method by Park et al. (2016b), different posture prediction models for men and women were used (Table 2).

Table 2. Driver Posture-Prediction Models for the Cascade Modelling Approach (Park et al., 2016b).

	Female driver posture-prediction	
Dependent Variable	**Regression Model (mm)**	**RMSE (mm)**
H-point x re PRP	$678 + 0.284 \times S - 494 \times SHS + 2.33 \times BMI - 0.388 \times H30 + 0.426 \times (L6 - 600)$	24.9
H-point z re AHP	$129 - 6.2 \times 10^{-2} \times S + 0.909 \times H30 - 4.65 \times 10^{-2} \times (L6 - 600)$	12.4
Centre-eye x re H-point	$-364 + 7.64 \times 10^{-2} \times S + 540 \times SHS + 0.124 \times (L6 - 600)$	37.7
Centre-eye z re H-point	$-461 + 0.163 \times S + 1445 \times SHS - 15.1 \times BMI + 3.61 \times Age + 9.93 \times 10^{-3} \times S \times BMI - 6.54 \times SHS \times Age$	12.7
	Male driver posture-prediction	
Dependent Variable	**Regression Model (mm)**	**RMSE (mm)**
H-point x re PRP	$-48 + 0.56 \times S + 1.39 \times BMI + 7.53 \times Age - 0.42 \times H + 0.505 \times (L6 - 600) - 3.98 \times 10^{-3} \times S \times Age$	25.3
H-point z re AHP	$123.1 - 5.864 \times 10^{-2} \times S + 0.945 \times H30$	10.6
Centre-eye x re H-point	$-578 + 0.174 \times S + 706 \times SHS - 1.55 \times BMI - 6.48 \times 10^{-2} \times H30$	32.1
Centre-eye z re H-point	$-498 + 0.361 \times Stature + 845 \times SHS + 2.76 \times BMI - 0.175 \times Age$	14.4

Note. S = stature (mm); SHS = sitting height/stature; BMI = body mass index (kg/m^2); H30 = seat height (mm).

Optimization Prediction Method (OPM)

The second method for predicting driving postures involves an optimization approach. In this study, a research version of the DHM tool IPS IMMA was used to represent the optimization prediction method for occupant packaging design (Hanson et al., 2019). The optimization model employed in the software is based on the principles described by Bohlin et al. (2012) and Delfs et al. (2013). This model predicts the driver's posture by solving an optimization problem that seeks to maximize a comfort function, which evaluates ergonomic factors such as joint angles, joint torques, and proximity to obstacles. The optimization is expressed as:

$$\text{maximize } h(x) \qquad \text{while: } g(x) = 0,$$

where h(x) represents the comfort function, prioritizing postures that minimize joint strain and maintain ergonomic alignment, and g(x) refers to kinematic constraints such as body segment alignments (positioning constraints) and biomechanical balance. The driver's optimal joint angles, which are the basis of the optimization in the software, were originally derived from a study by Bergman et al. (2015) and adjusted to fit the biomechanical model of IPS IMMA. In addition, collision avoidance is also included by penalizing postures where body parts approach environmental obstacles. The iterative optimization process refines the posture prediction until it satisfies all the constraints while maximizing the comfort function.

The DHM scene consisted of a group of manikins and the geometries of each of the two vehicle types. The manikin family corresponding to each vehicle represents drivers who participated in the data collection, i.e. manikins were created with the same anthropometric measurements as the participants (Table 1). Then, a standard DHM simulation setup was followed to make manikins adopt an initial seated driving posture (Perez Luque et al., 2022). Several basic constraints were established to ensure that the manikins could perform the necessary driving tasks. These constraints included positioning the feet on the pedals, placing the hands on the steering wheel, ensuring that the hip joint falls within the H-point envelope, and aligning the torso with the mid-back seat using attachment points in the DHM tool (Figure 2). These attachment points restrict the manikins to only move within a defined two or three-dimensional range (such as a line, a plane, or a box), as illustrated in Figure 2. Furthermore, four additional constraints were implemented to improve the accuracy of posture predictions specific to vehicle environments using the DHM tool. These constraints were necessary to address limitations in the DHM tool's default predictive capabilities and better reflect the ergonomic requirements of automotive settings. Three of these constraints addressed collision avoidance with specific vehicle geometries. They were handled using the collision avoidance functionality of the software and included the head-to-roof distance, knee-to-vehicle geometry distance, and thigh-to-steering wheel distance (Figure 2). The fourth constraint, which ensures a minimum downward vision angle, was implemented in the software with the attachment point functionality (Hanson et al., 2019) (Figure 2). This attachment inclination followed the slope of the hood. Then, the "Driving Car" strategy in the DHM tool was selected to make the manikins adopt the driving posture prediction following the OPM.

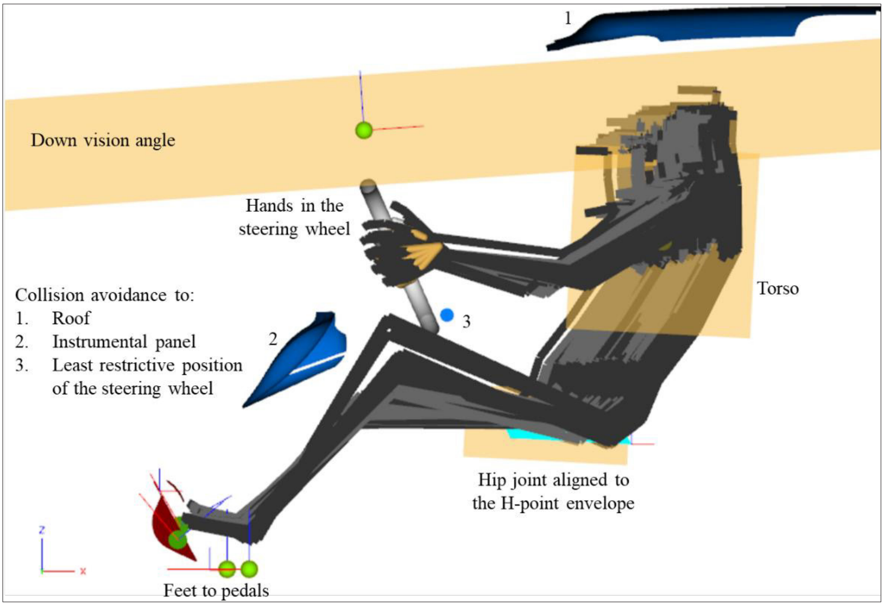


Figure 2. DHM tool scene setup for posture prediction in the case study: manikins seated with basic and additional constraints. Basic constraints ensured necessary driving tasks functionalities (e.g., feet on pedals, hands on wheel), while additional automotive-specific constraints improved realism (e.g., collision avoidance and vision angle).

Results

This section outlines the driving positions obtained for each manikin group in vehicles A and B using the two prediction methods, compared to the real-world collected data. The results show the location of driver key variables: the H-point, eye-point, and steering wheel position.

H-point prediction

Figure 3 illustrates the predicted H-point using SPM (blue points) and OPM (orange points) against the real collected data (green points) for Vehicle A and Vehicle B. Figure 3 also shows results when residual variance (RV), i.e., a degree of randomness, is included in the SPM using the RMSE values given by Park et al. (2016b) together with a stochastic component from a standard normal distribution. Additionally, Figure 3 also illustrates results for the OPM when loosening constraints (LC) are applied, specifically a doubled range of the hip joint attachment in the Z direction.

Vehicle A

In the x-axis (fore-aft direction), there is a noticeable trend where OPM predicts the H-point further back, while SPM predictions tend to be more forward (Figure 3). In this way, OPM predictions could be unable to accurately predict more forward positions, whereas SPM could be failing to cover some of the people who sit further back within the H-point envelope. This tendency of forward prediction by SPM and rearward prediction by OPM is consistent across most data points for Vehicle A, indicating a potential tendency of these methods to position the driver slightly further forward (SPM) or back (OPM) than the real-world measurements. In the z-axis (vertical direction), OPM tends to predict a higher H-point than SPM (Figure 3). SPM's z-axis predictions exhibit a narrower spread in the z-direction, whereas OPM captures a wider range of z-axis positions (Figure 3). However, this increased variability does not necessarily correlate with the actual measurements, indicating a potential overestimation in vertical positioning. This suggests that both methods struggle to accurately capture the variability in vertical seating positions, with OPM overestimating the vertical spread and SPM not fully representing the vertical spread.

Table 3 further details these observations with statistical measures. The average error for the x-axis prediction shows that SPM has a negative bias (-28.4 mm), indicating it generally predicts positions slightly forward compared to the real data. OPM, on the other hand, has a positive bias (27.2 mm), confirming its tendency to predict positions further back. The correlation coefficients for both methods are high (0.8 and 0.9), indicating a strong linear relationship with the actual data. However, the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) show that OPM has slightly better accuracy in the x-axis predictions. For the z-axis, SPM exhibits a lower average error (-2.1 mm), indicating it predicts better the vertical H-point position. OPM shows a slight positive bias (6.7 mm), suggesting that it predicts the H-point slightly higher on average than it actually is. The correlation for SPM is higher (0.7) than for OPM (0.4), indicating better predictive reliability in the vertical dimension. This is verified with both MAE and RMSE values, showing more pronounced errors in OPM predictions.

Vehicle B

For Vehicle B, similar patterns are observed as in Vehicle A, with SPM predicting the H-point further forward and OPM predicting it further back and higher in the z-axis (Figure 3). However, in Vehicle B, the errors are larger for both methods and directions, particularly for SPM in the x-axis predictions, showing a more pronounced forward prediction (-50.3) (Figure 3).

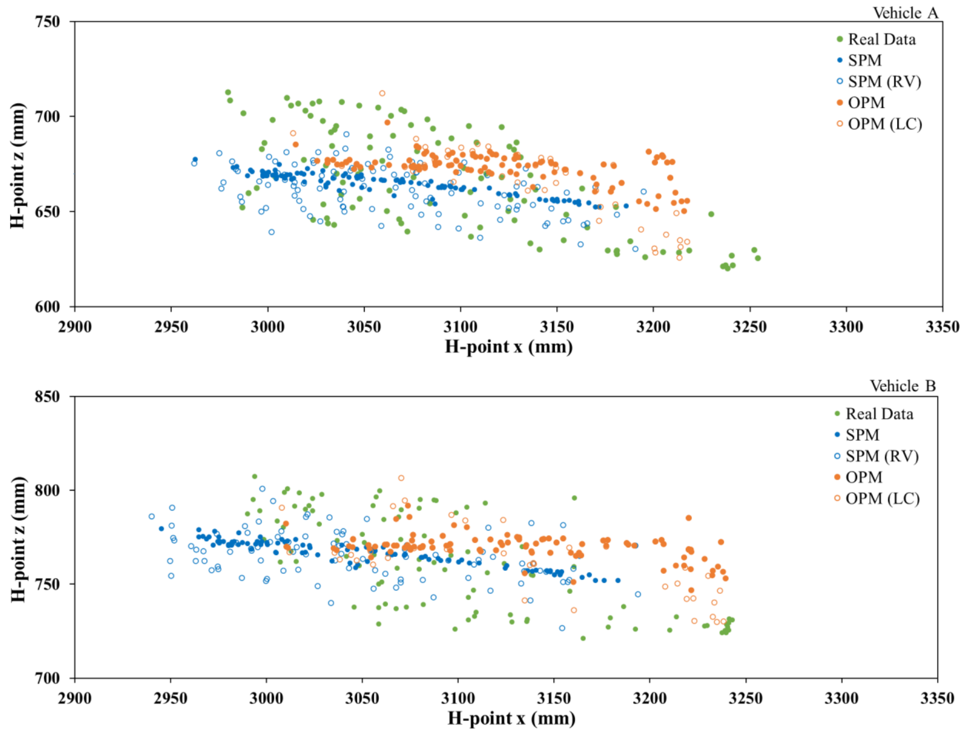


Figure 3. Comparison of H-point prediction by SPM (blue points) and OPM (orange points) against real collected data (green colour) for Vehicle A (top) and Vehicle B (bottom).

Table 3. Comparison of prediction methods for H-point: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		Correlation		MAE		RMSE	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
H-point x (mm)	SPM	-28.4 (40)	-50.3 (33)	0.8	0.9	39.7	53	48.8	59.9
	OPM	27.2 (75)	31.9 (37)	0.9	0.9	38.3	38.4	47.2	48.5
	SPM (RV)	-27.2 (45)	-51.5 (38)	0.8	0.9	42.3	55.8	52.3	63.7
	OPM (LC)	27.2 (39)	31.9 (37)	0.9	0.9	38.3	38.4	47.2	48.5
H-point z (mm)	SPM	-2.1 (23)	6.3 (21)	0.7	0.8	20.4	19.7	23.4	22.3
	OPM	6.7 (26)	9.4 (25)	0.4	0.4	22.6	22.6	26	26.2
	SPM (RV)	-5.1 (24)	4.9 (24)	0.4	0.4	20.9	20.5	24.8	24.5
	OPM (LC)	4.7 (24)	7.1 (24)	0.4	0.4	20.8	21.1	24.8	25.4

Eye-point prediction

Figure 4 and Table 4 compare the predicted eye-point positions using SPM and OPM against the collected data for Vehicles A and B, respectively.

Vehicle A

SPM predicts a more forward eye-point position compared to the real data, with an average error of -60.3 mm, while on the other hand, OPM predicts a more rearward position with a lower average error of 13 mm (Figure 4, Table 4). However, neither method fully captures the spread of the real data in the x-axis, indicating limitations in both predictive models. The correlation for both methods in the x-axis is 0.7, showing a moderate linear relationship with the actual data. In addition, OPM shows lower error values in predicting the eye point in the x-axis than SPM.

In contrast, the z-axis predictions are relatively similar in performance, although OPM shows slightly lower error values, suggesting a marginally better fit to the real data.

Vehicle B

SPM predicts a more forward position, with an average error of -62.7 mm, while OPM predicts a much more rearward position with a significant error of 107.4 mm, indicating a poor fit for Vehicle B (Figure 4, Table 4).

On the z-axis, SPM achieves a low average error of -1.9 mm, closely matching the real data, while OPM shows a slight overestimation at 6.5 mm. The correlation for SPM and OPM on the z-axis is equal (0.8). However, SPM reflects better predictive accuracy in the vertical dimension observing lower error values.

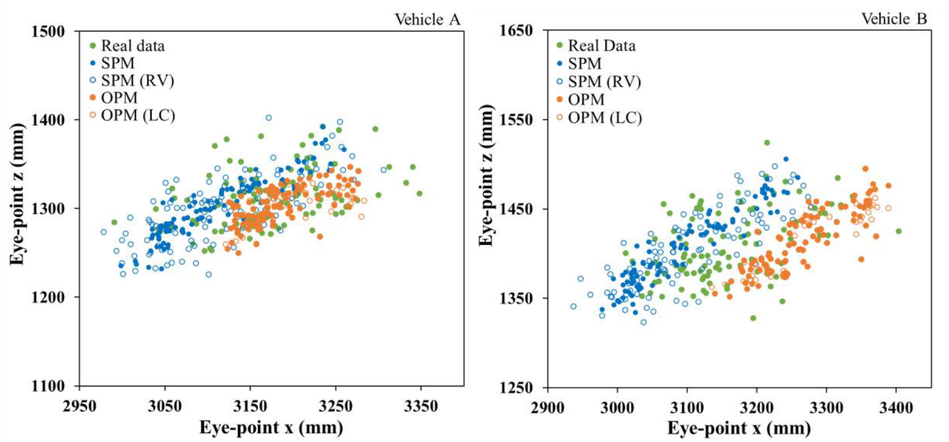


Figure 4. Comparison of eye-point predictions by SPM (blue points) and OPM (orange points) against real collected data (green points) for Vehicle A (left) and Vehicle B (right).

Table 4. Comparison of prediction methods for Eye-point: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		**Correlation**		**MAE**		**RMSE**	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
Eye-point x (mm)	SPM	-60.3 (52)	-62.7 (63)	0.7	0.7	67.8	74.4	79.1	88.9
	OPM	13.0 (72)	107.4 (60)	0.7	0.7	39.9	110	52.3	122.8
	SPM (RV)	-57.4 (61)	-56.2 (74)	0.6	0.6	69	74.4	83.4	92.5
	OPM (LC)	14.3 (51)	107.8 (60)	0.7	0.7	40.2	110.3	52.3	123.2
Eye-point z (mm)	SPM	-10.7 (28)	-1.9 (24)	0.7	0.8	24.5	18.3	29.7	23.8
	OPM	-10.0 (38)	6.5 (26)	0.6	0.8	22.5	19.7	28.5	26.4
	SPM (RV)	-14.3 (32)	-4.6 (27)	0.6	0.8	28.5	21.8	34.8	27.7
	OPM (LC)	-12.0 (29)	4.4 (27)	0.5	0.7	24	19.9	31.2	27.3

Steering wheel position prediction

Table 5 shows the steering wheel position predictions in the x-axis for both Vehicle A and Vehicle B using SPM and OPM, respectively.

Vehicle A

For Vehicle A, OPM closely aligns with the real data, showing an average error of 4.7 mm, while SPM has a more pronounced forward bias with an average error of -18.1 mm (Table 5). OPM demonstrates slightly better predictive accuracy, with a higher correlation (0.6) compared to SPM (0.5) and lower error values for both MAE (15.2 mm) and RMSE (18.1 mm), indicating a better overall fit.

Vehicle B

For Vehicle B, OPM slightly outperforms SPM, with an average error of 8.3 mm compared to SPM's -10.2 mm (Table 5). The correlation for both methods is lower in Vehicle B, with OPM achieving 0.4 and SPM only 0.1, reflecting greater variability in predictive accuracy. OPM also shows lower MAE (18.4 mm) and RMSE (22 mm) than SPM.

Table 5. Comparison of prediction methods for Steering wheel position: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		**Correlation**		**MAE**		**RMSE**	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
Steering wheel position x (mm)	SPM	-18.1 (18)	-10.2 (26)	0.5	0.1	20.5	22	25.8	27.8
	OPM	4.7 (18)	8.3 (20)	0.6	0.4	15.2	18.4	18.1	22

Results summary

The comparison of SPM and OPM revealed that each method demonstrates strengths depending on the specific vehicle configuration and prediction point. Boxplots of the Euclidean distances for SPM and OPM predictions are also included in the Appendix, offering a complementary visualization of the methods' overall prediction accuracy.

Table 6 summarizes the best-performing method in this case study based on all evaluation metrics across both vehicles, highlighting that each method shows better alignment depending on the specific vehicle configuration. While Table 6 presents an overview of the methods' performance, additional insights are available through Bland-Altman plots, which are included in the Appendix providing a more detailed analysis. These plots visually represent the differences between predicted and measured values, helping to identify key trade-offs between precision and adaptability for each method. Boxplots of the Euclidean distances for SPM and OPM predictions are also included in the Appendix, offering a complementary visualization of the methods' overall prediction accuracy.

Table 6. Best performing prediction method for Vehicle A and Vehicle B.

	Best prediction method	
	Vehicle A	**Vehicle B**
H-point x	OPM	OPM
H-point z	SPM	SPM
Eye-point x	OPM	SPM
Eye-point z	OPM	SPM
Steering wheel x	OPM	OPM

Discussion

This case study compares two different methods, SPM and OPM, for predicting seated driving postures and positions within vehicle occupant packaging design. SPM relies on human anthropometrics to predict the location of driver key variables such as the H-point, eye-point, and steering wheel position through a cascade modelling approach, where predictions are made sequentially. However, its base on fixed equations limits adaptability to additional constraints or variations outside the predefined parameters. In contrast, OPM assumes individuals select postures that align their joints as closely as possible to optimal joint angles, maximizing their comfort. This approach is able to adapt to the constraints defined within the DHM scene. However, OPM prediction models may require refinement to represent postural variability across targeted populations better.

Results

The comparison between SPM and OPM for Vehicles A and B reveals differences in how each method performs across the key predicted points: H-point, eye-point, and steering wheel position.

H-point prediction

For the x-coordinate of the H-point, OPM shows better accuracy in the tested scenarios for both vehicles, outperforming SPM. Within the scope of this case, this suggests that OPM is more effective in predicting the fore-aft positioning of the occupant's hip joint for both the low (Vehicle A) and high (Vehicle B) seat configurations. Additionally, the Bland-Altman plots reveal additional insights. For Vehicle A, SPM demonstrates narrower limits of agreement (LOA), indicating better precision, despite having a slight underprediction bias. In contrast, OPM shows wider LOA and more random scatter, which reflects greater adaptability to diverse driver postures but lower precision. A similar trend is observed in Vehicle B, where SPM maintains higher precision but exhibits a stronger underprediction bias, while OPM performs with less systematic bias but greater variability. When considering the z-coordinate (vertical positioning), SPM performs better across both vehicles, which corresponds with the Bland-Altman analysis. This variation in performance between axes highlights the need for further development or complementary consideration of the two methods for H-point predictions.

To further investigate the poor performance of OPM in the z-axis for H-point prediction, the population was divided into three stature groups: short, medium, and tall. This was conducted to determine whether errors increase for specific stature groups, given that stature is a key aspect influencing where individuals sit within the H-point envelope (P. Wolf et al., 2022). Figure 5 shows that the difference error between the measured and predicted increases significantly for OPM as stature increases across both Vehicle A and Vehicle B. While OPM can more accurately predict H-point vertical positions for average-sized occupants, its predictive capability diminishes for individuals at the extremes of the stature range, especially taller drivers. SPM displays similar trends with less pronounced errors, resulting in a more consistent error distribution across all stature groups, making it a more reliable method for predicting H-point z-coordinates in vehicles where accommodating a wide range of driver statures is essential.

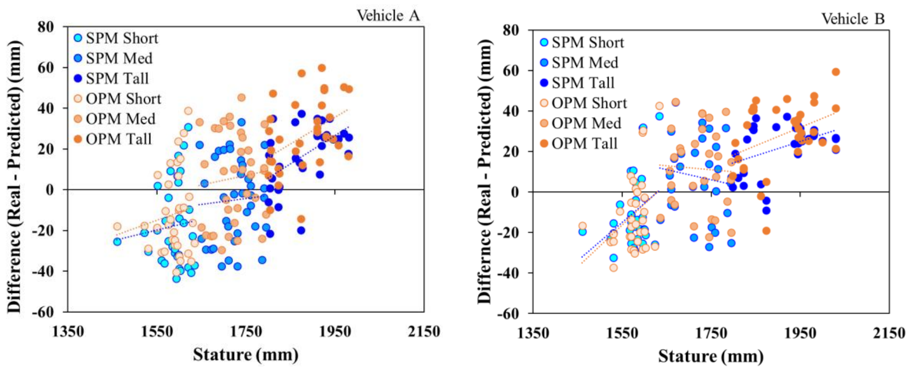


Figure 5. H-point z: Difference errors trend line across stature.

Eye-point prediction

The eye-point prediction results follow another pattern. For Vehicle A, OPM outperforms SPM in both the x- and z-axes. This suggests that OPM is more reliable in predicting eye-point locations for specific vehicle configurations similar to Vehicle A. For Vehicle B, which has a different vehicle configuration with a higher seat height, SPM performs better in both axes. This shift in prediction accuracy suggests that SPM potentially captures better the driver's seating posture and eye-point positioning in vehicle configurations similar to Vehicle B. However, the Bland-Altman plots provide additional context, showing that while OPM performs better according to metrics for Vehicle A, SPM demonstrates narrower LOA and greater precision for both x- and z-coordinates. For Vehicle B, SPM outperforms OPM across all analyses, including Bland-Altman.

Steering wheel position prediction

For the steering wheel x-coordinate, OPM again provides higher accuracy in Vehicle A and Vehicle B, indicating its potential ability to adapt to posture and steering wheel positioning requirements of vehicles with lower and higher seat heights, respectively.

Prediction method by vehicle type

Overall, considering the evaluation metrics, OPM showed better alignment with measured data for Vehicle A, which features a lower seat height in the tested scenarios. Its performance in predicting the H-point, eye-point, and steering wheel position in the x-axis suggests that the current underlying set of optimal angles used in OPM suits better the specific ergonomic configuration of lower-seated vehicles like Vehicle A. Nonetheless, the Bland-Altman analysis highlights that SPM's narrower LOA could result in better precision, even in cases where systematic bias is greater. For Vehicle B, which has a higher seat height, SPM outperforms OPM across most prediction points, particularly in the vertical (z-axis) predictions, although it does not necessarily predict the variability in it. This could indicate that the default set of optimal angles in OPM does not account for the variations in drivers' posture resulting from different vehicle configurations like Vehicle B. Consequently, a different set of optimal angles tailored for higher seat configuration vehicles could improve OPM's prediction performance. Such adjustment could potentially bring OPM's results in Vehicle B closer to the accuracy it achieves in Vehicle A with a lower seat position, suggesting its potential prediction utility across different vehicle types.

Limitations

While the study provides valuable insights, several limitations must be acknowledged. These relate to the number of vehicles analyzed, data collection protocols, and the methods and tools evaluated. First, the study was conducted using two vehicle configurations, which represent contrasting but limited configurations. Given that this is a case study, the results should be interpreted within these specific contexts, and the generalizability of the findings to other vehicle types is limited. Future studies should incorporate a broader range of vehicle configurations to evaluate the methods more comprehensively and enable the generalizability of the results. Additionally, differences in how companies collect or estimate real-world posture data, such as alternatives in pose estimation methods or motion capture technologies, may influence the observed outcomes and their alignment with predictive models. Such methodological variation could result in different patterns and should be taken into account when comparing across industrial contexts.

Second, potential biases may have arisen from the initial seat placement during data collection, as this could have influenced how participants adjusted and selected their seating positions. Although a neutral starting position was adopted based on standard methods from previous research, a recent study has shown that starting conditions can introduce biases (Jung & Park, 2023). However, it is important to note that the methodology for collecting posture data was based on an established methodology used by the company conducting the data collection, where consistency and comparability of results are prioritized, and direct measurement in real vehicles is a key component. Future studies should consider methodologies that reduce these effects to address this, such as conducting multiple trials with varied initial seat positions.

Third, this study was limited to evaluating OPM implemented in a single DHM tool, IPS IMMA. Expanding the analysis to include OPM implemented in other DHM tools could provide a broader understanding of the method's effectiveness and applicability across different DHM tools. Additionally, limitations in the specific IPS IMMA version used, such as the body models and their meshes, could have affected the accuracy of the predictions as they may not realistically represent humans. Enhancing the fidelity of these body models could lead to more accurate and reliable results, particularly in capturing the detailed postural variations of different body sizes.

Prediction methods improvements

While both methods show potential for predicting driver posture in occupant packaging design, their performance varies depending on the specific vehicle configuration. Both methods leave room for improvement, emphasizing the importance of further refinement and validation of predictive models to improve their reliability and applicability. On the one hand, an advantage of SPM is that the prediction of characteristic driving positions is written as closed-form, linear equations (Table 2), which makes them independent of any human model and easier to use. In contrast, OPM must be implemented using a particular kinematic linkage or digital human model. On the other hand, one potential advantage of OPM is that it may provide more generalizable results to new design tasks as its results are a consequence of the digital human model and constraints defined in the virtual environment, whereas SPM is not able to adapt to additional constraints or aspects to the ones considered in their equations. Furthermore, analyzing a specific vehicle configuration can be crucial when choosing prediction methods. This study shows that the set of optimal joint angles for OPM used in this study may yield better prediction results for vehicles with lower seat configurations, while SPM shows better performance for vehicles with higher seat configurations.

While both SPM and OPM present distinct advantages depending on the application, there remain areas where each method could be refined to better capture the complexity of human posture across different vehicle configurations. SPM could be enhanced by adjusting the results using the average error found compared to real world data. Based on the average error of this initial evaluation, adjustments could be made following the same sequence as the cascade modelling approach. Thus, the average error for the steering wheel position would be calculated first, and the adjustment of those predictions would be added before the average error of the H-point position could be calculated next. Lastly, the average error of the centre-eye position could be calculated after adjustments to predictions of the H-point have been added. Different adjustments could be calculated for different vehicle models.

As observed in Figure 5, OPM's error performance particularly increases depending on the stature, suggesting that the optimal angles used in the optimization process could be further refined for specific manikin statures and vehicle configurations. As it currently works, all manikins in a family follow the same optimization values without considering different postures for different body dimensions and postural variability, as unlikely happens in reality (Kyung & Nussbaum, 2009; J. Park et al., 2016b; Brolin et al., 2020; Bubb et al., 2021). Developing a more tailored set of optimal driving angles that consider the unique body dimensions and postural tendencies of different stature groups and for specific types of vehicles could enhance the accuracy of OPM. For example, developing separate sets of optimal angles for distinct manikin sizes or vehicle configurations would allow the optimization process to better reflect the variability in human posture. Such refinements could significantly improve the method's ability to predict driving postures across a diverse population.

Conclusion

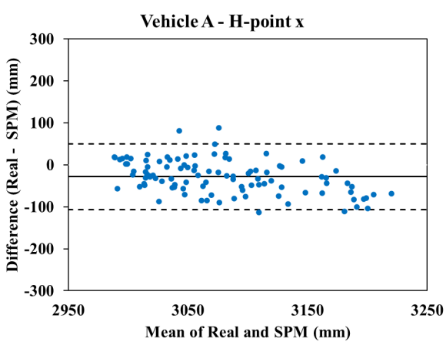
This case study provides an applied comparison of two methods commonly used in the industry for predicting seated postures in occupant packaging design. The SPM and OPM were evaluated using real-world data from two vehicles with contrasting seat configurations. Both methods demonstrate potential viability, but their performance varies depending on the specific tested vehicle configurations and predicted driving position points. SPM, which uses closed-form equations, showed better alignment with measured data for Vehicle B. Its simplicity, ease of use, and computational efficiency make it a practical choice for situations requiring quick predictions without complex simulation setups. In contrast, the evaluation metrics demonstrated that OPM, which uses an optimization approach based on human body models and optimal joint angles, has better alignment for Vehicle A. Yet, the Bland-Alman analysis revealed that SPM, in some cases, has a narrower LOA reflecting more precision despite the systematic bias. OPM has the ability to incorporate a broader range of ergonomic factors, such as joint angles and constraints, providing a comprehensive and flexible approach for occupant packaging design. However, further refinement is required to improve its performance for diverse and specific populations and vehicle configurations.

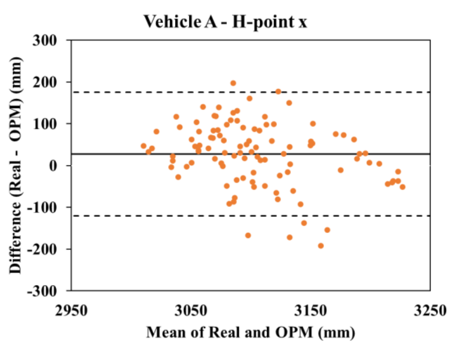
In conclusion, this case study illustrates that both driver posture prediction methods have their strengths and weaknesses, and the specific design context, the vehicle configuration, and the need for precision in certain prediction points should guide the choice between them. SPM is considered ideal for quick predictions and situations where computational simplicity is required, while OPM is considered better suited for more complex occupant packaging designs. Furthermore, these methods should not be limited to drivers but also be extended to the design of seating positions for other occupants in the vehicle, such as passengers. Considering the varying needs and postures of different vehicle occupants is crucial for optimizing comfort and ergonomics across all seating positions. In the context of autonomous vehicles, where driving tasks may no longer remain the primary activity, future research should explore how both SPM and OPM can be adapted for varied occupant positions and activities. Autonomous vehicles offer opportunities for new occupant postures and seating configurations, as users engage in non-driving activities. Thus, incorporating these methods in the design of autonomous vehicle interiors could significantly enhance passenger comfort and safety.

Integrating these methods into the early stages of vehicle occupant packaging design can enhance the overall comfort and safety of vehicle interiors by ensuring that diverse driver populations are better accommodated. Future research should focus on refining both methods, particularly enhancing OPM's adaptability to broader vehicle types, body sizes, and occupant roles (e.g., passengers) and activities (e.g., non-driving tasks), and improving SPM's ability to handle more varied constraints could extend its utility. These refinements will potentially help improve the accuracy of DHM tools and contribute to developing safer and more comfortable vehicle interior designs, particularly as vehicle technologies evolve.

Appendix

This appendix presents Bland-Altman plots for comparing predicted and measured values across key prediction points for SPM and OPM. These plots provide a visual representation of the methods' precision and adaptability, highlighting the differences between predictions and real-world data in the specific design contexts examined in this case study. In each plot, the solid horizontal line represents the mean bias, which is the average difference between the predicted and measured values, indicating whether a method tends to systematically overpredict or underpredict. The dashed horizontal lines represent the limits of agreement (LOA), showing the range within which most differences are expected to fall. Narrow LOA indicate higher precision, while the distribution and pattern of the data points offer insights into each method's adaptability and systematic behaviour.





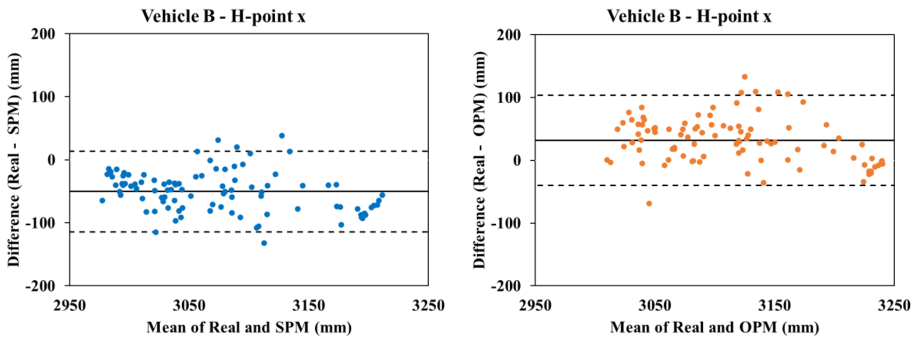


Figure A1. Bland-Altman plots comparing H-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

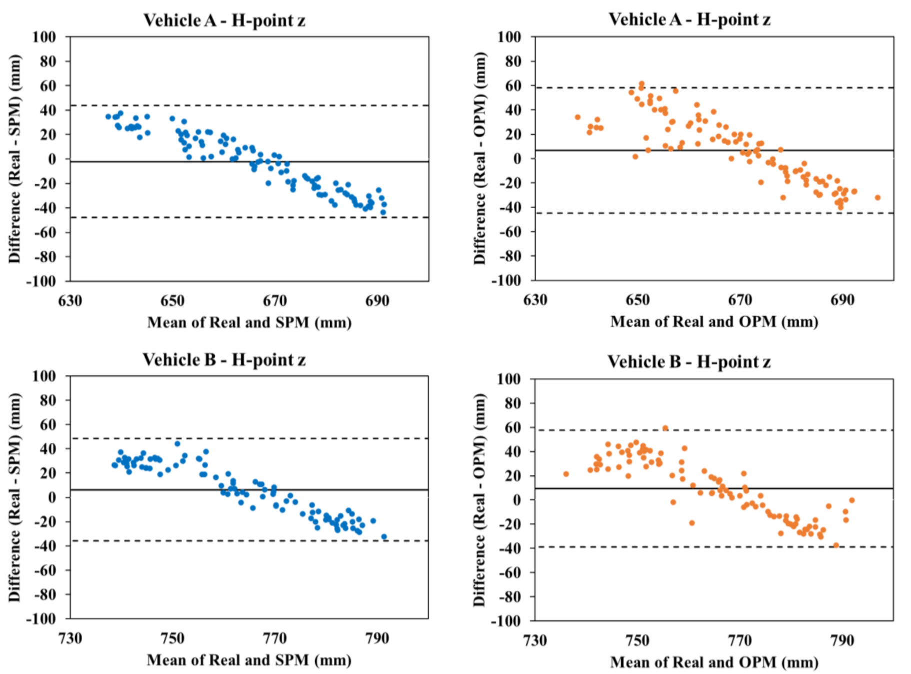


Figure A2. Bland-Altman plots comparing H-point z predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

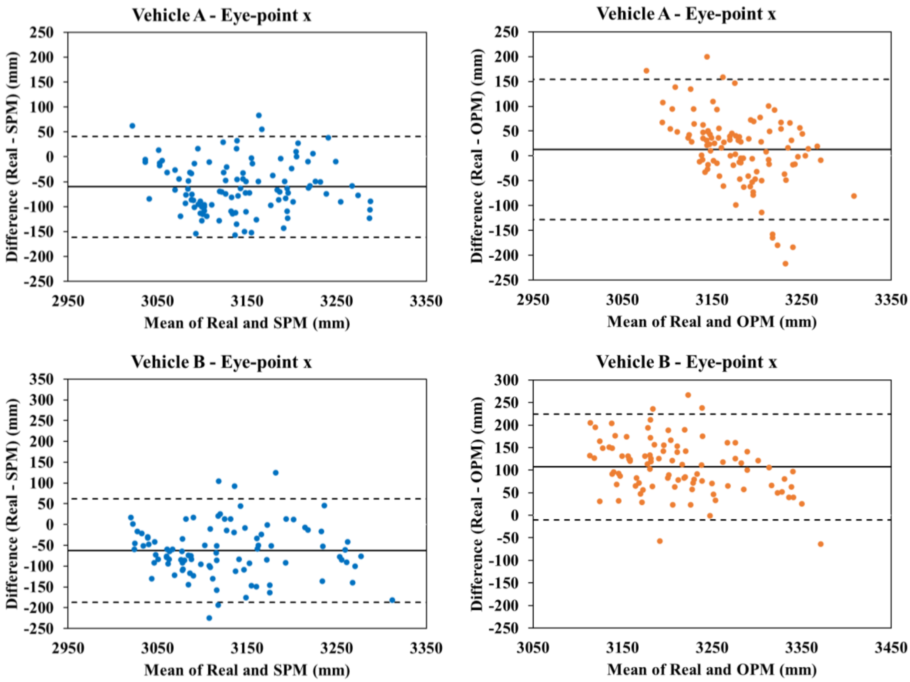
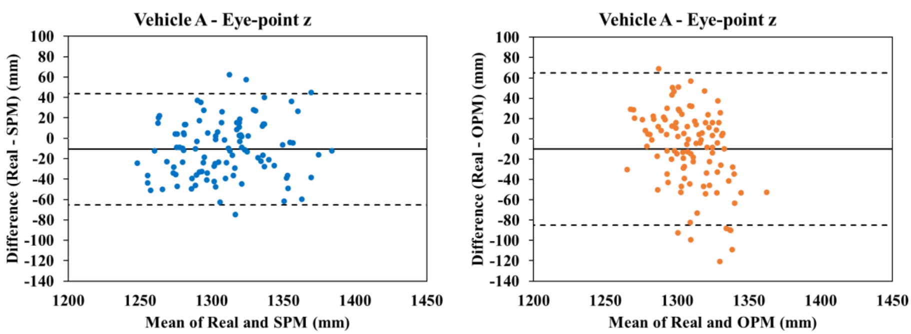


Figure A3. Bland-Altman plots comparing Eye-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).



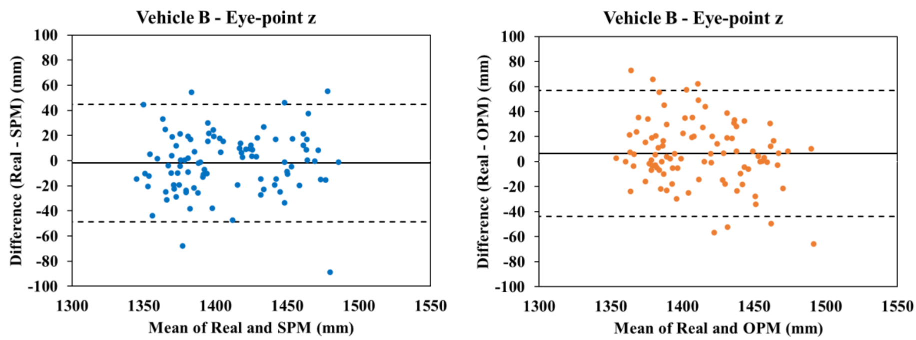


Figure A4. Bland-Altman plots comparing Eye-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

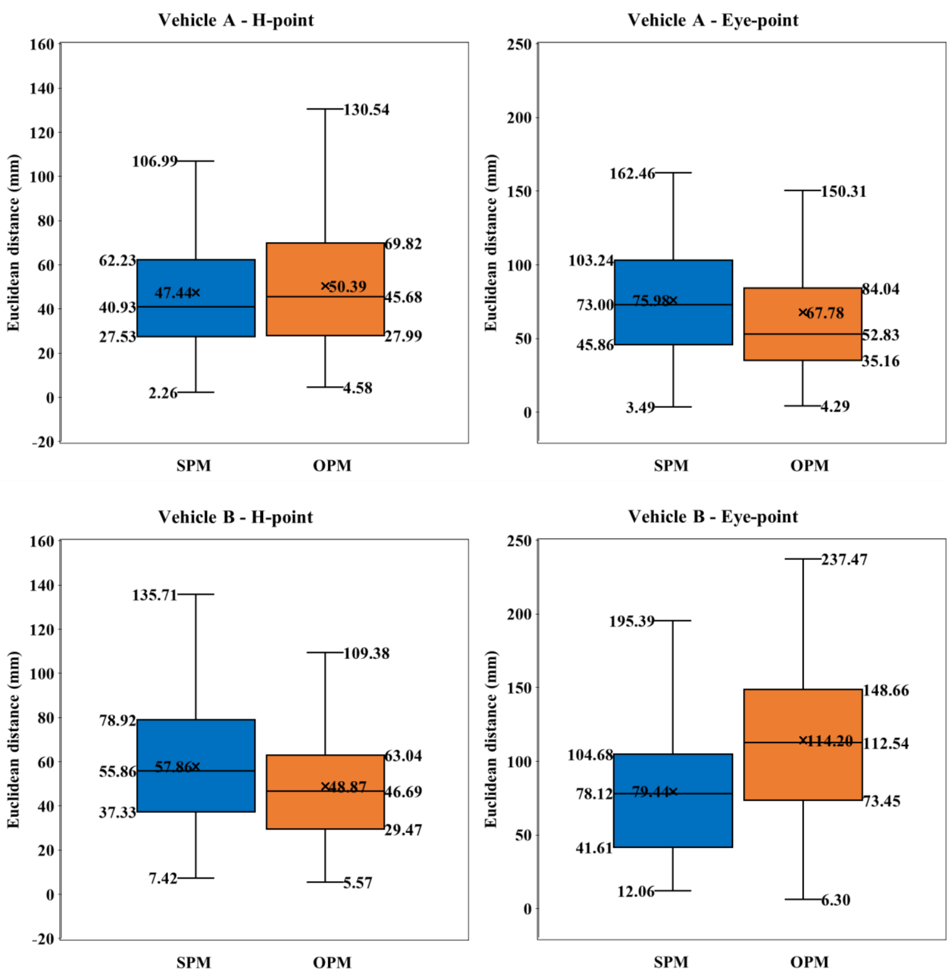


Figure B1. Euclidean distances between predicted and observed values for SPM and OPM across H-point (left) and Eye-point (right) positions in Vehicle A and Vehicle B (bottom).

Acknowledgements

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PAPER E

Improving Occupant Packaging Posture Prediction Through Integration of Simulation and Real Data

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Abstract. This paper presents a process for enhancing driver posture prediction in digital human modelling (DHM) tools by integrating real-world data from driver posture studies. The process leverages key data points such as seat position, steering wheel position, and eye point coordinates to position manikins and extract joint angles for ergonomic analysis. A test study involving 49 drivers was conducted, revealing significant variations in joint angles across different statures. These findings were used to develop stature-based strategies that demonstrated improved predictive accuracy for short and tall stature groups compared to the existing strategy in the DHM tool IPS IMMA. While the results highlight the potential benefits of this approach, limitations such as refined manikin body meshes, seat property considerations, and broader vehicle model validation are recommended. Overall, this method offers a promising solution for addressing incomplete datasets in occupant packaging studies, contributing to the development of more ergonomic and safer vehicle designs.

Keywords: Occupant packaging, digital human modelling, simulation, dataset, process

1 Introduction

Different driver posture models have been developed through data-driven and optimization approaches and implemented in digital human modelling (DHM) tools [1], [2]. Data-driven models predict human body joint angles or positions based on various human and vehicle parameters, while optimization methods aim to minimize discomfort by decreasing deviations from expected neutral or optimal joint angles. Evaluating and validating these models is essential in occupant packaging to be able to ensure ergonomic requirements such as comfort, operating controls, and visibility for a diverse population [3]. However, obtaining comprehensive datasets for occupant packaging validation studies can be challenging in many research and design scenarios. This often results in insufficient and limited data [4], which hinders thorough analysis and enhancement of prediction models in DHM tools. An issue can arise when the prescribed preferred driver angles in the simulation tool are not derived from a similar biomechanical model. Joint chains such as the spine might be simplified in simulations, or the joint angle data can be given only in a two-dimensional sagittal plane. In addition, the validity of laboratory studies for predicting driving posture in real vehicles can, in some cases, be uncertain, and the possibility of doing detailed driving posture studies in real vehicles is limited. However, data of coordinates for the seat position, steering wheel position and driver eye point are relatively easy to measure, even in real vehicles. Still, such data points are not measured directly on the driver, except for the eye-point, which introduces the additional challenge of accurately estimating the mid-hip to h-point offset. This offset measurement, or prediction of it, is often used as input for future posture prediction studies where inverse kinematics posture the digital human models, so-called manikins, towards a preferred driving posture [5].

This paper introduces a process where key data from driver studies in real vehicles, such as coordinates for the seat position, steering wheel position, and driver eye point, is used to position manikins and extract their joint angles. These extracted joint angles can be used to analyze posture and position differences in the population and consequently generate improved posture predictions in less constrained simulation settings for future vehicle ergonomics design. While similar processes have been used in the past [6], the most significant difference lies in its implementation in a DHM tool, allowing the consideration of additional aspects such as anthropometric body meshes or collision avoidance with vehicle geometries.

2 Method

The core idea of this process is to leverage existing real-world data as input for simulations alongside simulations data to fill in missing posture and position data, enriching the dataset and enabling a more comprehensive analysis of occupant packaging scenarios. The process contains three main steps (Figure 1). The first is collecting data on test persons, such as anthropometry and preferred positions of seat (H-point), seat back angle, steering wheel, and eye point. Next, manikins based on each test person's anthropometry are generated. Thirdly, each manikin is positioned in the preferred position of the test person it is representing together with general constraints such as feet on pedals and no collision with surrounding geometry, i.e. roof, dashboard and console. The positioning of the hip and buttock area as well as the back of each manikin are done using a body shape alignment approach [6]. Each manikin's buttock and back surface are aligned with an H-point template positioned in the preferred H-point position of the test person it is representing. From each positioned manikin, body joint angles, positions, and distance to specific vehicle geometries are extracted.

Based on the complete data set containing both anthropometric data and their corresponding preferred body joint angles, statistical analysis can be done to define vehicle or even population-specific optimal joint angles. Lastly, the new optimal joint angles can be used in less constrained simulation settings for future vehicles where the exact position of the seat, steering wheel, and eye point is unknown.

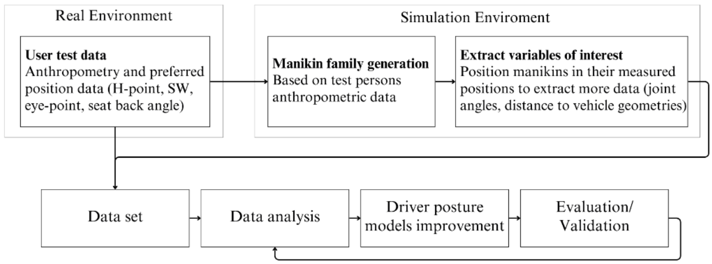

Figure 1. Schematic of the process.

2.1 Evaluation study

A test and evaluation study was conducted to demonstrate the practical application of the proposed method. The data for this study was sourced from an internal experimental study conducted by Volvo Cars in 2017, which involved 49 drivers. This study included an anthropometrically diverse group of 25 men and 24 women. The stature distribution within the test group was intentionally not normally distributed, where 16 participants (all women) were shorter than 1600 mm and 16 participants (all men) were taller than 1937mm (Figure 2). The participants were asked to select their preferred driving posture while driving an SUV model under real road conditions. Data collected during the study included participants' anthropometric measurements and driving position data. The driving position data included several key aspects, such as the seat position (H-point), seat back angle, steering wheel position, and eye point.

The simulation was modeled using the DHM tool IPS IMMA version 2023.R2 [7]. Utilizing the existing functionality in IPS IMMA, a manikin family was generated based on 13 anthropometric measurements from each test participant. Each manikin was then positioned according to the driving posture of the corresponding test participant, and all body joint angles and distance from the manikins to the roof, steering wheel, and instrument panel were recorded.

Following the schematic of the proposed process, the extracted additional variables, such as body joint angles, were analyzed to develop stature-based strategies. This involved determining the optimal joint angle values for driver posture predictions for each of the three groups: short, medium, and tall. Based on these optimal joint angle values per stature group, new strategies were created and evaluated against the "Seated Car" strategy defined in the IPS IMMA software [8].

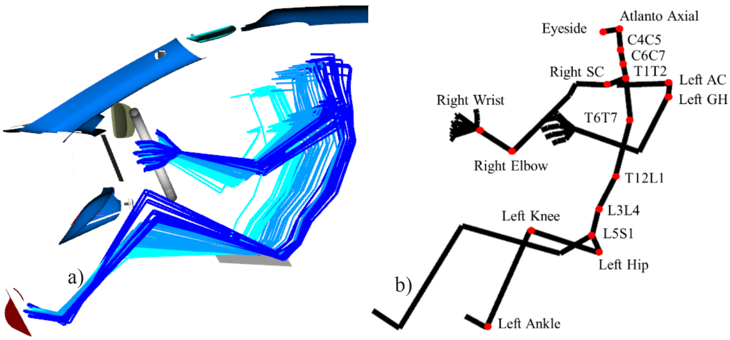

Figure 2. a) Manikin families: short (turquoise), medium (blue), and tall (dark blue); b) IPS IMMA skeleton and joint angle names.

3 Results

After extracting the joint angles, the data was categorized into three groups based on stature to examine variations in driving posture. The groups were defined as follows: short (1460 - 1622 mm), medium (1623 – 1936 mm), and tall (1937 - 2030 mm). Each group contained approximately 1/3 of the data and consisted of 16, 17, and 16 manikins, respectively. The analysis revealed significant differences in preferred joint angles across the three stature groups, with the most pronounced differences observed between the short and tall groups at the C6C7, hip, knee, and wrist joints. When compared to the "Seated Car" strategy of IPS IMMA, the set of predefined values for optimizing the driver posture in the software, the greatest discrepancies were found in the T12L1, hip, and knee joint angles. While differences in the ankle, elbow, and other torso joint angles were noted, they were less substantial compared to the aforementioned joints. These differences are illustrated in Figure 3.

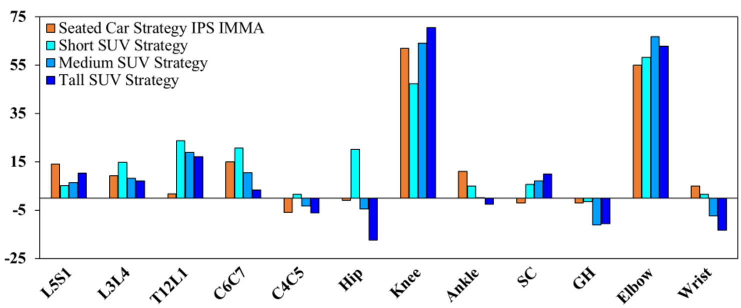

Figure 3. Comparison of simulation strategies angle values: Seated Car default (orange), Short (turquoise), Medium (blue), and Tall (dark blue).

3.1 Improvement of Driver Posture Models

The observed differences in joint angles among the stature groups were used to enhance driver posture prediction capabilities in the software. Three strategies were created based on the average joint angles of the manikins within each stature group. This resulted in a tailored set of preferred values for short, medium, and tall manikins. Then, simulations were done following the default "Seated Car" strategy in IPS IMMA and the developed Stature-based strategies based on the average of the extracted angles. Results from the simulation comparison between the strategies show smaller RMSE values on the stature-based strategies compared to the "Seated Car" in the eye- and h-point in the z-axis for the short and tall stature groups. However, the RMSE values, except for the eye-point x, are lower with the "Seated Car" strategy for the medium stature group, indicating that the software provides good predictions for this specific group. These results are described in Figure 4 and Table 1.

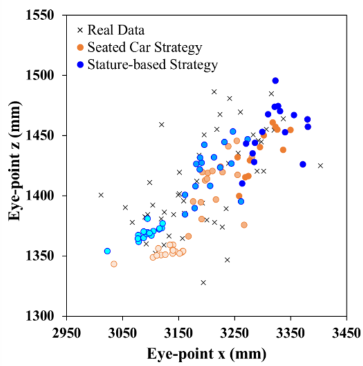

Figure 4. Predicted eye- and h-point positions by strategy and stature group: Seated Car (light orange, mid orange, orange) and Stature-based (turquoise, blue, dark blue) correspond to short, medium, and tall manikins respectively.

Table 1. RMSE (mm) by strategy and stature group.

	H-point x		H-point z		Eye-point x		Eye-point z	
	Seated Car	Stature-based	Seated Car	Stature-based	Seated Car	Stature-based	Seated Car	Stature-based
Short	23.6	21.9	36.4	22.9	55.9	52.2	32.3	22.8
Medium	40.7	44.2	23.8	29.9	44.2	41.5	32.0	36.2
Tall	38.4	42.7	23.4	23.1	93.7	94.6	27.1	22.9

4 Discussion

The proposed integrated approach in this study has the potential to improve the prediction and evaluation of vehicle ergonomics. Leveraging data from previous driver posture studies maximizes the utility of available resources. Data from a constrained environment can be considered valuable for developing strategies even in a less constrained setting due to the original simulation's limited diversity. This approach improves understanding of human-system interaction and predictive accuracy by continuously integrating new data from driver posture studies into the model.

Despite the potential benefits, several limitations exist that need further investigation. While the proposed process enhances predictive accuracy, there is room for improvement in refining manikin body meshes and considering seat properties within the DHM software. Incorporating biomechanical models to calculate seat deflection could further enhance the accuracy of simulations. Additionally, other parameters of how the optimization model runs in the software could be refined to obtain more realistic results. It should also be noted that in this initial study, only data from one SUV vehicle model was analyzed and used to develop the strategies. Validation with more vehicle models would be necessary to ensure the robustness and generalizability of the driver posture models. Addressing these limitations would optimize the outcomes of the proposed process and provide more comprehensive results.

In conclusion, the proposed process presents a promising solution for incomplete datasets in occupant packaging studies and provides researchers with a valuable tool for enhancing analysis and creating specific driver posture prediction models.

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PAPER F

Predicting Human Upper Extremity Reaching Motions: Comparison of Optimization-based Method and Heuristic Method

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Abstract: Predicting human upper extremity reaching motions in 3D space is essential for applications ranging from ergonomics and robotics to rehabilitation and simulation. This study compares two predictive approaches: an optimization-based method (OPM) and a proposed heuristic method (SFM). SFM integrates a steering dynamics model for path planning, an adaptive velocity model, and an inverse kinematics solver. OPM and SFM were validated against motion capture data from ten participants performing four reach tasks, and were evaluated across three motion components: path, velocity, and upper extremity joint configuration. Prediction accuracy and modeled inter-subject variability were assessed using root mean square error (RMSE) and dynamic time warping (DTW). Results show that SFM more accurately predicts spatial path and velocity, while OPM achieves greater precision in joint angle estimation. As input, OPM requires the initial and end postures of the upper extremity as well as the task duration. In contrast, SFM needs the initial posture, initial and target end-effector positions, and initial and estimated peak velocity. The dual-metric evaluation reveals a trade-off between accuracy and behavioral diversity across the two methods, providing valuable insights for selecting predictive models in human motion simulation.

Keywords: Human motion prediction; Optimization; Heuristic; Behavioral dynamics; Trajectory; Path; Velocity; Arm joint angle; Time series; RMSE; Dynamic time warping.

1 Introduction

Predictive human motion methods have long been an interest of researchers from a variety of disciplines (Gang and Wang, 2025). The ability to model and predict human movements without relying on recorded motion data enables advancements in a wide range of applications, including ergonomics, product design, biomechanics, virtual reality, animation, robotics, human-computer interaction, sports, and rehabilitation.

Among the many types of actions that humans perform, reaching motions are one of the most frequently performed types of goal-directed action in daily life (Furuki and Takiyama, 2017). Accurately predicting reaching motions requires capturing both the spatial and temporal dynamics of movement throughout the reaching arm, from the path of the hand (i.e., end-effector) through the angles of each joint in the upper extremity (e.g., wrist, elbow, and shoulder) for the duration of a reach motion. When evaluating observed and predicted reaching motions, three interrelated components of the motion are typically considered: the end-effector path, the end-effector velocity, and the set of upper extremity joint angles configuration, which together define a reaching movement over time (Klein et al., 2022). The end-effector path describes the position of the index fingertip in space as it moves from the starting location to a target location, often represented as a sequence of points in space (Gong et al., 2018). Velocity is a measure of positional or angular change along a path over a period of time and can be measured in terms of a specific point on the body, e.g., the tip of a finger, or a joint, e.g., the elbow (Hamm, 2020). Arm joint configuration defines the specific angular relationship between the segments of the arm (shoulder, elbow, and wrist) that define the limb posture and coordination throughout the movement (Clark et al., 2020). Together, these components characterize the spatial and temporal aspects of a motion, forming what is commonly referred to as a trajectory, a continuous description of how the body, in this case an upper extremity, moves through space and time (Rudenko et al., 2020). Models that predict reaching motions should accurately predict at least one of these dimensions of the reaching trajectory and ideally should predict all three.

A range of modelling approaches have been developed to predict path, velocity, and posture (joint angles), both in isolation and together (Farahani et al., 2015; Kim et al., 2016; Wolf et al., 2020). These methods fall into the following groups: data-driven, optimization-based, and rule-based or heuristic approaches. Data-driven methods can involve using previously recorded motion data directly, either in its original form or rescaled. This approach serves as an alternative to more advanced prediction techniques that rely less directly on prior observations. Data-driven methods can also include statistical models, reinforcement learning, and artificial neural networks, which identify and/or learn patterns from recorded motion data, and then predict human motions that exhibit those patterns given limited initial conditions and task data (Wang et al., 2019; Porzio and Coraggio, 2025). Data-driven methods can excel when prediction contexts are similar to the training or observational datasets, but quickly diverge for novel prediction contexts (Cui et al., 2020; Xia et al., 2022). In contrast, optimization-based methods use a knowledge-based approach to identify a set of cost functions that represent features of human motion that are assumed to be controlled or prioritized in human motion. Cost functions may include muscle activity, energy consumption, comfort, dynamic effort, or joint displacements (Xiang et al., 2010; Yang et al., 2011; Zaman et al., 2021). Selected cost functions are integrated with optimization algorithms that aim to minimize or maximize the outputs of the cost functions for a given context, ideally resulting in an optimal motion prediction. Optimization-based methods typically perform better in novel prediction contexts than data-driven methods, as they are not as dependent on the specifics of previous observations. Lastly, heuristic methods also take a knowledge-driven approach, but instead of searching in a space of possible solutions based on cost functions, heuristic solvers apply a set of rules following specific patterns or regularities grounded in observations and theories of human motion to find a solution. Heuristic algorithms generally build on geometric operations involving points, distances, and angles in an iterative fashion, offering solutions that are more computationally efficient than data-driven or optimization-based approaches (Braun et al., 2021; Cao et al., 2023). Heuristic methods can include a behavioral dynamics approach to motion prediction, which predicts motion paths and velocity based on mathematical models derived from observations of human motion (Fajen and Warren, 2003). Thus, behavioral dynamics models are mathematical models based on observations and theories of human motion, like optimization-based approaches. Unlike optimization-based approaches, which search a space of possible trajectories to minimize a cost function, behavioral dynamics models generate a single prediction given a set of initial conditions and model parameters. Like optimization-based approaches, heuristic approaches have the potential to perform well in novel prediction contexts with the added potential of greater flexibility and computational efficiency.

For human motion prediction, the dominant prediction methods are data-driven and optimization-based methods. The choice between these methods typically relies on the type of data available and the specific context or goals of the prediction task. For these approaches, path, velocity, and joint angle predictions are typically integrated into a single prediction model, though some models predict velocity independent of path and joint angle predictions. Heuristic planners are relatively underexplored within fields investigating human motion, and there are currently no empirically validated heuristic-based approaches to human motion prediction that integrate path, velocity, and joint angle predictions. To this end, we would like to explore heuristic planning methods to compare with the optimization-based method to determine their trade-offs. Our proposed heuristic prediction approach integrates an empirically validated behavioral dynamics model for path prediction, a lowpass filter-based approach to velocity prediction, and a heuristic inverse kinematics solver that has been widely deployed in computer graphics but not validated for accuracy in human motion prediction. For the path, our model includes a modified version of the steering dynamics model (SDM) introduced by Fajen and Warren (2003, 2007), and later adapted by Lamb and colleagues to predict locomotion paths for a 2D reaching task (2017, 2019). While the 2D version of the SDM has demonstrated a good fit to a variety of human behaviors in 2D, we will implement a 3D variant to predict naturalistic reaching paths. For the velocity prediction component, the velocity model builds on early research into the relationship between muscle activation and reaching motions, where it was observed that physiological properties of muscles acted like a lowpass filter on observed velocity profiles (Gottlieb et al., 1989). Finally, we propose that the Forward and Backward Inverse Kinematics (FABRIK) solver can provide a solid basis for joint angle predictions given the velocity and path prediction of the lowpass velocity model and the SDM. The FABRIK solver has been widely deployed in computer graphics contexts, primarily for video game animations, but has also been deployed as a robot controller and demonstrates potential as an accurate method for accurate biomechanical predictions (Aristidou et al., 2016; Aristidou and Lasenby, 2011; Lamb et al., 2022; Moreno and Alcántara, 2022; Santos et al., 2021). Our combined heuristic model, named SFM (short for Steering-FABRIK Method), is a first-of-its-kind heuristic prediction model with the aim of providing an integrated prediction of path, velocity, and joint angle configuration over an entire reaching motion.

In this article, we will compare an initial formulation of the SFM to an established optimization-based modeling (OPM) approach. Understanding the differences between motion prediction models is essential for selecting the right approach in a given context. Different methods entail trade-offs in accuracy, precision, generalizability, and computational cost. In this article, we will primarily focus on the relative potential of the SFM and OPM as regards accuracy. We will consider both how accurately the models predict specific individual motions and how well they reflect the patterns of variability in human motion. By comparing these models, researchers can make more informed choices tailored to their specific application, whether in robotics, ergonomics, rehabilitation, or animation.

The aim of the study is to compare two different motion prediction methods for modeling upper extremity reaching in 3D space. The OPM approach applied in this study uses cost functions related to all three components, i.e., end-effector path, end-effector velocity, and upper extremity joint configuration. The second is our proposed SFM heuristic approach. For the comparison, prediction output from OPM and SFM are validated against motion capture experimental data, evaluated across three motion components: path, velocity, and upper extremity joint configuration throughout the upper extremity reaching motion.

2 Method

To compare the proposed SFM approach with an existing OPM approach, we collected a unique motion capture experimental dataset consisting of four unconstrained upper extremity reaching motions. Because this study is performed in the context of a project focused on methods for vehicle cabin design, we selected motions related to a vehicle cabin from the perspective of the driver.

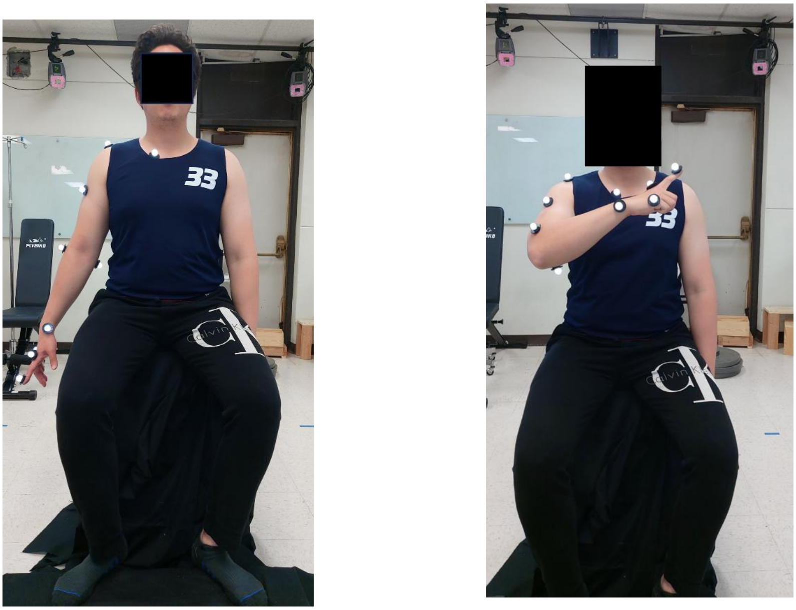
2.1 Experimental collected data

A total of 10 (5 male and 5 female) participants (age: 27.5 $\pm$ 4 years) volunteered for the experiment. All of them were physically and mentally sound. They were right-handed and had no restrictions on their right arm during testing. The experiment was conducted at Texas Tech University (TTU), following the protocol approved by the Institutional Review Board at TTU. All participants signed the informed consent form and performed 4 tasks, 3 trials for each task.

A motion capture system with 8 cameras (Motion Analysis Corporation, CA) was used to collect the right upper extremity motion data (60 Hz). Ten retroreflective markers were placed on the subject's right half of the upper body, as shown in Figure 1. The markers were attached to the bony landmarks following the protocol (Cloutier et al., 2011).

All participants were instructed to "touch" four target points separately, assuming they were sitting in their driver seat with the initial posture as the arms straight down (Figure 1b): the rearview mirror (Task 1), seat belt (Task 2), gear shifter (Task 3), and passenger seat (Task 4). Each participant was instructed to perform the abovementioned tasks one by one in the following order: 1) have a relaxed posture sitting on a chair, 2) reach the target point from the initial posture in 1), and 3) stop for 3 seconds at the target location and then return to the initial posture described in 1).

(a)



(b) (c)

Figure 1. Lab experiments: (a) Motion capture environment with an 8-camera system, (b) initial posture with both arms straight down, and (c) final posture of Task 2 (front view).

The Generator of Body Data (GEBOD) was used to create the subjects' anthropometric data (arm segment lengths) listed in Table 1 based on the subject's gender, stature, and weight (Cheng et al., 1994). Markers were labeled and their position curves were smoothed in Cortex (Motion Analysis Corporation, Rohnert Park, CA, USA). After the post-processing, the data were converted into a C3D file and became the input for Visual 3D (C-Motion Inc., Germantown, MD, USA). In Visual 3D, motion capture experimental data were utilized to build a model (upper arm, forearm, and hand), and all joint angles (the shoulder, elbow, and wrist joint) were calculated.

Table 1. Anthropometric data of the right upper extremity.

Subject	Gender	Height (m)	Weight (kg)	Upper arm length (m)	Forearm length (m)	Hand length (m)
1	Male	1.87	110	0.318	0.266	0.207
2	Male	1.74	75	0.301	0.258	0.188
3	Male	1.89	90	0.324	0.278	0.201
4	Male	1.85	106	0.314	0.264	0.199
5	Male	1.88	77	0.326	0.282	0.199
6	Female	1.65	47	0.270	0.222	0.185
7	Female	1.48	46	0.230	0.200	0.170
8	Female	1.60	53	0.254	0.214	0.181
9	Female	1.58	48	0.252	0.212	0.179
10	Female	1.70	65	0.270	0.224	0.191

Skeletal model

The right upper extremity model is constructed with seven degrees-of-freedom (DOFs). Figure 2 shows Denavit-Hartenberg (DH) representations (Denavit and Hartenberg, 1955), and the corresponding DH parameters are shown in Table 2. The first three DOFs (q_1, q_2, q_3) represent the shoulder flexion, abduction, and rotation, respectively. The fourth DOF (q_4) corresponds to the elbow flexion. The final three DOFs (q_5, q_6, q_7) are the wrist flexion, deviation, and rotation, respectively. In the kinematic chain, l_1 corresponds to the upper arm length, l_2 to the forearm length, and l_3 to the hand length, which is reflected in Table 2 and Figure 2. The definitions of the DH parameters (θ, d, a, α) in Table 2 follow the original convention presented by Denavit and Hartenberg (1955).

Table 2. DH table of the right upper extremity model.

Segment	DOF	θ (rad)	d (m)	a (m)	α (rad)
Shoulder	shoulder_z (q_1)	0	0	0	$\pi/2$
	shoulder_y (q_2)	$\pi/2$	0	0	$\pi/2$
	shoulder_x (q_3)	$\pi/2$	Upper arm length (l_1)	0	$\pi/2$
Elbow	elbow_z (q_4)	$\pi/2$	0	Forearm length (l_2)	0
Wrist	wrist_z (q_5)	0	0	0	$\pi/2$
	wrist_y (q_6)	$\pi/2$	0	0	$\pi/2$
	wrist_x (q_7)	0	Hand length (l_3)	0	0

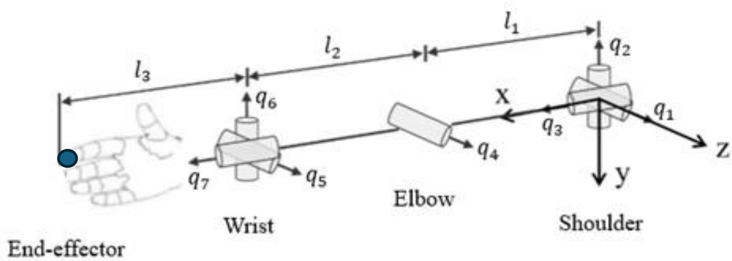


Figure 2. 3D right upper extremity model.

2.2 Optimization-based prediction method (OPM)

The right upper extremity motion is formulated as a nonlinear programming problem, shown in Equations 1-3 for the OPM, and the results of the optimization-based method are the end-effector path, velocity, and joint angle profiles and instantaneous upper extremity configuration. The optimization formulation was solved through the commercial software MATLAB (The MathWorks Inc., Natick, MA). A local optimization solver, FMINCON, was selected to solve the nonlinear problem, and sequential quadratic programming was applied.

Optimization formulation

The control points ($\boldsymbol{P}_c$) of cubic B-spline curves for joint angles are design variables. The cost function is to minimize the joint displacements from the neutral position and maximize the end-effector velocity, shown as follows:

$$J = \int_0^T \sum_{i=1}^{7} \frac{w_i (q_i(\boldsymbol{P}_c, t) - q_i^N)^2}{\left(\left\|\dot{\boldsymbol{P}}_{end-effector}(t)\right\|_2^2 + \alpha\right)} dt \qquad (1)$$

where *T* is the total task time that is from the subject's experimental data, q_i^N is the *i*th neutral joint angles, and $\dot{\boldsymbol{P}}_{end-effector}$ is a velocity vector of the end-effector, and α is a positive constant that avoids the denominator to be zero, and 100 is used in this study. The neutral joint angles are defined by our previous research (Yang et al., 2004), representing a relatively comfortable posture. w_i is the *i*th objective function weight coefficient, where $w_1, w_2, w_5 \sim w_7$ are 0.4, and w_3 and w_4 are 0.6, respectively. Note that in the cost function, the reason to maximize the end-effector velocity is to prevent the zig-zag motion of the end-effector because the upper extremity model is a redundant system.

Constraints

The right upper extremity reaching motion is subject to time-dependent and time-independent constraints. For the time-dependent constraint, joint angle limits are applied to avoid hyperextension and hyperflexion. The ith joint angle limits are defined in Eq. (2).

$$q_i^L \leq q_i(\boldsymbol{c}, t) \leq q_i^U \tag{2}$$

where q_i^L and q_i^U represent the ith lower and upper joint angle limits, respectively. The joint angle limits are listed in Table 3.

Table 3. Joint angle limits.

DOF	Lower limit (deg)	Upper limit (deg)
shoulder_z (q_1)	-180	60
shoulder_y (q_2)	-120	60
shoulder_x (q_3)	-90	45
elbow_z (q_4)	-130	10
wrist_z (q_5)	-45	45
wrist_y (q_6)	-45	45
wrist_x (q_7)	-45	45

The time-independent constraints are the initial and final postures. The right upper extremity joint angles constrain the experimental data at the initial and final task, shown as:

$$q_i(\boldsymbol{c}, t) = q_i^E(t), \quad t = 0, T \tag{3}$$

where q_i^E is the ith joint angle from experiments.

2.3 Heuristic prediction method (SFM)

The SFM heuristic method integrates three interconnected sub-models to simulate and predict human reaching motions in 3D space (Table 4 and

). First, the SDM predicts the heading of the end-effector as well as aligns the end-effector orientation with the goal hand pose orientation. Second, a velocity model predicts the distance the end effector travels along the predicted heading during each time step of the prediction. Finally, an arm joint angle configuration that places the end-effector in the newly predicted position and orientation while respecting joint constraints is predicted using the FABRIK inverse kinematics solver. The combined use of these three models enables the generation of full arm reaching motions in 3D space.

Table 5 outlines in pseudocode this integrated approach (SFM).

Table 4. Components of the SFM (Heuristic method).

Motion Component	SFM Sub-Models Used	Function
Path	Steering Dynamics Model (Fajen and Warren, 2003)	Computes the spatial trajectory of the end-effector using the input velocity
Velocity	Adaptive lowpass velocity model (Gottlieb et al., 1989)	Defines the speed profile of the end-effector based on its distance to the target
Upper extremity configuration	FABRIK (Aristidou and Lasenby, 2011)	Determines the joint angles needed to achieve the desired end-effector path

Table 5. Input parameters (left) and pseudocode for the SFM algorithm (right), outlining the iterative prediction of hand path, velocity and arm joint configuration at each timestep.

Input:	**Psuedocode:**
Initial joint angles, Initial end-effector position, Goal position, Initial velocity, estimated peak velocity.	$n = 1$ *Initialize* M with current state of arm, m_n.

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Constraints:
Joint Angle Limits.

Parameters:
g_{pos} = goal position.
h_{pos} = end-effector position.
h_{ort} = end-effector bone orientation.
h_{fwd} = forward vector of end-effector heading.
h_{vel} = end-effector velocity.
h_{dist} = distance (meters) between h_{pos} and g_{pos}.
h_{err} = angular error between h_{fwd} and vector from h_{pos} to g_{pos}.
$\{q_1, q_2, \ldots . q_7\}$ = Set of joint angles defining the arm configuration.

$m = \{g_{pos}, h_{pos}, h_{ort}, h_{fwd}, h_{vel}, h_{dist}, h_{err}, \ldots$ $\{q_1, q_2, \ldots . q_7\}\}$

$M = \{m_1, m_2, \ldots, m_n\}$ where n is the length of the final predicted motion trajectory.

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While (h_{dist} > 0.01)
$n = n + 1$

Estimate change in h_{fwd} (SDM).
Estimate change in h_{ort} (SDM).
Estimate change in h_{vel} (low-pass filter velocity model).
Predict next h_{pos} and h_{ort} based on estimated changes in h_{fwd}, h_{ort}, and h_{vel}.
Estimate new $\{q_1, q_2, \ldots . q_7\}$ given current $\{q_1, q_2, \ldots . q_7\}$ and predicted h_{pos} and h_{ort} (FABRIK).

Calculate new h_{dist} and h_{err}.

Set m_n to updated state estimates.

Add m_n to list of M.

End

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Steering dynamics model (SDM)

The SDM, originally developed in cognitive science by Fajen and Warren (2003), is a behavioral dynamics model that predicts human steering behavior toward goals while navigating through space. The SDM predicts the change in heading of a system, e.g., the end effector of the arm, as it approaches a goal location. Thus, the SDM is a second-order differential equation based on a damped-mass spring model that describes changes in heading direction, ϕ, over time (Equation 4). The angular acceleration, $\ddot{\phi}$, is influenced by the damping, b, stiffness controlling the strength of the goal attraction, k_g, the angular heading to the goal, ψ_g, from the current position, and a set of decay parameters, c_1 and c_2, that modulated the effect of goal distance, d_g, on turning.

$$\ddot{\phi} = -b\dot{\phi} - k_g\,(\phi - \psi_g)\left(e^{-c_1 d_g} + c_2\right) \tag{4}$$

A detailed description of the original model, its derivation, and validation can be found in Fajen and Warren (2003). The parameters used were selected through manual testing across a range of values to ensure plausible steering trajectories for the reaching tasks.

While the SDM was originally formulated in 2D, for the current comparison, we extended the model to 3D space using quaternion algebra, enabling smooth orientation transitions in 3D space. The original SDM formulation assumes a single plane of rotation and where the predicted headings are defined relative to the plane normal centered on the predicted system's position. The generalized 3D SDM makes the angle-axis representation of the system's heading explicit so that the axis of rotation can change throughout the path. To select a single axis of rotation at each timestep, the 3D SDM uses a quaternion representation of the hand's current motion heading to define ϕ relative to an axis of rotation, v_{hr}. The axis of rotation v_{hr}, was defined by treating the hand's forward motion as the vector, v_{hf}, defined by normalized difference between the hand's current position and its previous position. An arbitrary orthogonal vector, v_{ho}, was defined as $v_{ho} = v_{hf} \times v_{wu}$where $v_{wu} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ representing a global vertical vector, thus $v_{hr} = v_{hf} \times v_{ho}$. Thus, the orthonormal basis of the current motion heading can be defined as $[v_{hf}, v_{ho}, v_{hr}]$ from which the quaternion q_h can be derived. A similar approach to the goal heading starts with the vector, v_{gf}, defined by the normalized difference between the hand's position and the goal position. The orthogonal vector is defined as $v_{go} = v_{gf} \times v_{wu}$ and the rotational vector $v_{gr} = v_{gf} \times v_{go}$, from which a quaternion to the goal, q_g, can be derived. The rotation from q_h to q_g is then defined as $q_{hg} = q_g g_h^{-1}$ and enforce the shortest rotation path (Equation 5).

$$q_{hg}\begin{cases} -q_{hg}, & if\ (q_g \cdot q_h) < 0 \\ q_{hg}, & otherwise \end{cases} \tag{5}$$

Standard quaternion algebra allows for the derivation of the desired angular rotation, $(\phi - \psi_g)$, from q_{hg}, as well as the relevant axis of rotation, v_{hgr}, used to rotate from one heading to the other.

At each timestep of the prediction, a plane of rotation, defined by the plain normal vector, v_n, and the end-effector location, h_{pos}, was identified and the goal heading was calculated relative to that plane. Continuous updating of the plane of rotation allows the 3D SDM to accommodate changes in task and environment, including obstacles. The model output is a predicted change in heading that can be used to derive a positional change once the velocity is predicted.

Adaptive lowpass velocity model

One approach to velocity prediction is to assume a total time to arrive at the reach target, typically based on distance and difficulty of the reach task, combined with some form of Fitts Law (Fitts, 1954). However, timing estimates can be highly variable and can be affected by active movement, suggesting that while time estimates may play a role in human motion generation, they may be too noisy to play the strong role assumed in these types of velocity prediction models. We propose a novel approach to velocity prediction that does not assume accurate time estimates and instead builds on observed neuromuscular properties combined with an estimate of braking ability grounded in Ecological Psychology (Gibson, 1979; Gottlieb et al., 1989; Harrison et al., 2016). Our proposed model leverages the observation that muscle activations during reaching can be modeled as a lowpass filter applied to an increasing or decreasing activation of motoneuron pools (Gottlieb et al., 1989). Thus, the velocity prediction model is a simple damped-mass spring model that stands in for the detailed muscle dynamics of the arm system (Equation 6).

$$\ddot{v} = -b_v\dot{v} - k_v\left(v - (v_{peak} * s)\right) \tag{6}$$

Where v is velocity, b_v is a frictional damping force, and k_v is a stiffness term. The peak velocity is estimated prior to prediction using a linear regression model based on the distance to the goal, which was the most predictive variable among a wide set of motion and anthropometric parameters. This variable showed a strong correlation with peak velocity (R=0.70), and its use enables prediction without requiring subject-specific measurement. The resulting regression model is (Equation 7):

$$v_{peak} = 1.1241 * d_g + 25.788 \tag{7}$$

The current level of activation $\{s \in \mathbb{R}\ |0 \le s \le 1\}$ modulates the maximum value of $\ddot{v}$ and depends on the system's distance to the target and current ability to stop, where the ability to stop is estimated based on the current velocity and the settling time of the system described by (Equation 6) assuming $s = 0$, a transient response of 5%, and that the system is critically damped. The resulting settling time estimate is used to determine the rough minimum stopping distance, b_d, given the current state of the system. The ratio, described in Equation 8,

$$b_a = \frac{b_d}{d_g} \tag{8}$$

thus characterizes the ability of the agent to stop, b_a, before reaching the target if the activation signal is set to 0, where $b_a < 1$ indicates that the system will overshoot the target resulting in a collision and $b_a \ge 1$ indicates that the system is able to stop at or before reaching the target. Thus, given the current activation level, s_{cur}, a new value, s, can be estimated using an adaptive low-pass filter formulation (Equation 9),

$$s = \begin{cases} s_{cur} - \dfrac{s_{cur}}{d_g/c_3}, & if\ b_a < 2 \\ s_{cur} + \dfrac{s_{cur}}{d_g/c_3}, & otherwise \end{cases} \tag{9}$$

where c_3 is a constant value modulating the effect of distance on the filtered s value. The value of $b_a < 2$ was selected because the low-pass filter formulation does not allow for an instantaneous reduction of s to 0. In this way the velocity model predicts the change in velocity of the hand given only the state of the reaching system and an expected maximum velocity. The neuromuscular properties of the system are represented by a simple damped-mass spring system and a low-pass filter combined with a mathematical description of the system's stopping ability to stop, i.e., its stopping affordance. When combined with the predicted changes in heading and velocity are taken together, one can predict the change in hand position from the current point in the reaching trajectory. The predicted hand position can then be passed to the FABRIK solver to predict remaining arm joint configuration.

FABRIK solver

To determine the right arm joint angle configuration that follows the predicted end-effector path generated by the SDM and velocity models, the FABRIK (Forward and Backward Reaching Inverse Kinematics) model is employed. FABRIK is a heuristic iterative algorithm that solves the inverse kinematics problem by adjusting joint positions based on simple geometric rules (Aristidou and Lasenby, 2011). Each joint is updated by projecting it onto a line defined by its neighbor and preserving segment lengths, making it computationally efficient and free from the singularities that affect Jacobian-based methods.

The algorithm operates in two stages per iteration: a forward reaching phase where the end-effector is placed at the target position and joint positions are recalculated backward along the chain, followed by a backward reaching phase where the base joint is reset to its original position and joint positions are updated outward toward the end-effector (Figure 3). This cycle continues until the end-effector reaches the target within a predefined tolerance. FABRIK also supports joint constraints, and in this implementation, the same joint angle limits as those used in the optimization-based method (OPM) were applied to ensure a fair comparison (Table 3).

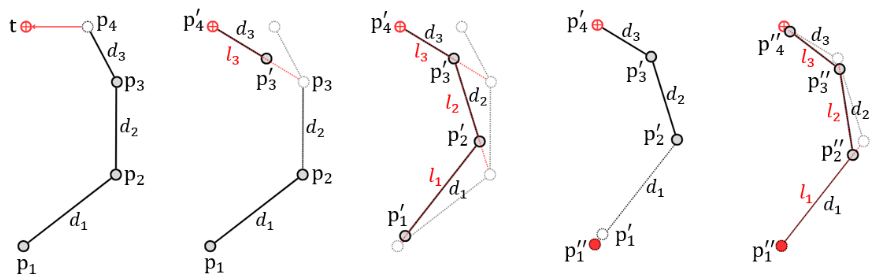


Figure 3. Illustration of a full iteration cycle of the FABRIK algorithm for a 4-joint chain.

2.4 Data analysis

Accuracy evaluation

The accuracy of the two prediction methods, OPM and SFM, was evaluated against experimental motion capture data. The analysis focused on three motion components: end-effector (hand) path, end-effector velocity profile, and arm joint configuration over time. For each of these, the corresponding time series data were preprocessed, normalized, and quantitatively assessed using two metrics: root mean square error (RMSE) and dynamic time warping (DTW).

RMSE is a widely used metric for evaluating the accuracy of prediction models. It quantifies the average magnitude of the error between predicted and observed values at each time point. However, RMSE treats motion profiles as purely numerical sequences and does not account for differences in the geometric shape of the movement. This can lead to misleading conclusions when comparing complex trajectories or when trajectories differ in spatial or temporal scale and alignment. To address these limitations, we applied the DTW method as a complementary evaluation metric. Unlike RMSE, DTW compares sequences by non-linearly aligning them in time, allowing for shifts, delays, or differences in duration (Senin, 2008). This makes DTW particularly suitable for analyzing 3D human motion, where similarities in trajectory shape can be obscured by differences in spatial or temporal scale and alignment (Cleasby et al., 2019). To illustrate the differences, Figure 4 shows a conceptual example contrasting RMSE (Euclidean distance) and DTW alignment.

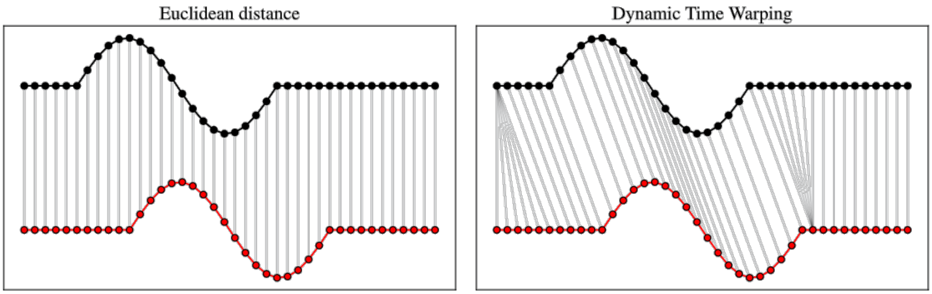


Figure 4. Visual comparison between RMSE (left) and DTW (right) for evaluating time series similarity. RMSE compares values at the same time steps, while DTW allows for flexible temporal alignment. Adapted from (Tavenard, 2021).

For the current comparison study, DTW provides a cost matrix representing the distance between each point in a predicted sequence and each point in the observed trajectory. The cost matrix then represents all possible alignments between the two sequences. Given this matrix, one can compute an optimal warping path, consisting of the best alignment between the two datasets, by finding the path through the cost matrix that has the lowest summed distance value. The resulting, i.e., the minimal summed distance, DTW value represents the minimal cumulative cost required to align the sequences (Senin, 2008). A baseline DTW value was derived by applying DTW task-wise to observed data. Thus, each trajectory in a task was individually compared to every other trajectory produced for that task. The average and range of these DTW values for each task provides an expected similarity range for the prediction data. We reason that DTW values within this range can be considered as similar to a specific observed data as any other human trajectory for that task would be. This is a minimal success criterion providing the upper bounds for what might be considered a successful prediction. In the ideal case, the DTW value for a successful prediction of any specific observed trajectory would approach 0.

To ensure that DTW reflects shape similarity rather than magnitude differences, each sequence was centered by subtracting its mean and scaling it by normalization using the Euclidean (L2) norm (Latash, 2010). This normalization step removes global offset and amplitude variation, aligning sequences to emphasize their temporal and geometric structure.

Evaluation of modeled inter-subject variability

In addition to evaluating the accuracy of each method against experimental data, a complementary analysis was performed to assess the extent to which each method preserved natural inter-individual variability in movement. The goal of this analysis was to determine whether the predictions generated by each method could reflect the behavioral differences observed across different subjects.

This was achieved by comparing motion sequences between all subjects within each dataset (experimental, OPM, and SFM). Pairwise RMSE and DTW values were calculated for every subject-to-subject combination within each group (e.g., real-to-real, OPM-to-OPM, and SFM-to-SFM). These within-group comparisons provided a quantitative measure of the inter-subject variability generated by each method.

Rather than comparing predictions to experimental data, this analysis focused on evaluating the variation across subjects within each dataset. The resulting distributions of variability scores establish a baseline for interpreting each method's ability to simulate human motion diversity.

Preprocessing and normalization

Before calculating RMSE and DTW, all motion sequences underwent the same preprocessing. First, the 3D position of the end-effector (right hand tip) was computed for each frame for both observed and simulated data. From these positions, absolute velocity curves were derived. In the SFM predictions, the velocity occasionally became asymptotic near the target. To prevent this low-motion "tail" from skewing temporal alignment, the sequences were truncated when velocity dropped below 5% of the peak value (Sethi et al., 2017). Because the proposed SFM method does not rely on an initial estimate of movement duration, its predicted trajectory times often differed from those of the OPM and the observed data. To enable comparison, all data were resampled onto a percentage-normalized time axis, ranging from 0% to 100% of task duration. This allowed consistent application of both RMSE and DTW across subjects and methods, regardless of execution speed or original recording length. Two trials were removed as outliers because their peak velocity was more than 2 standard deviations above the average velocity for their respective tasks.

At this point, RMSE was computed at three levels: per frame, per subject, and per task. As described above, prior to DTW calculation, each sequence was normalized by subtracting its mean value and scaling to unit L2 norm. This allowed DTW to focus on the shape of the motion profiles rather than absolute differences in magnitude or offset.

Each motion components (path, velocity, and arm configuration) were processed and analyzed using the same processing and analysis pipeline, scaled for the component's dimensionality. The path was represented as a 3D multivariate time series (x, y, z), capturing the Cartesian trajectory of the end-effector. Velocity consisted of a univariate time series representing the speed magnitude of end-effector movement over time. Arm configuration was modeled as a 7D multivariate time series composed of seven joint angles, one for each degree of freedom in the right upper extremity model (see Figure 2). Table 6 summarizes the data structure for each motion component, clarifying their dimensionality and how they are represented for the analysis.

RMSE was computed for each motion component by comparing their time series on a per-dimension basis (e.g., velocity, each path axis, or each joint angle). In contrast, DTW was applied to each multivariate time series as a whole, providing a similarity score characterizing the similarity of the respective n-dimensional shapes.

Table 6. Data structure of the analyzed motion components.

Motion Component	Data Type	Description
Path	3D multivariate time series	Cartesian position (x, y, z) of the end-effector
Velocity	1D univariate time series	Magnitude of end-effector velocity over time
Arm joint configuration	7D multivariate time series	Joint angles for the 7-DOF right upper extremity model

3 Results

Results are reported individually for each motion component.

3.1 End-effector path

Figure 5 displays the end-effector paths of all ten subjects in Task 4, comparing observed paths to predictions from the OPM and SFM methods. Each colored line represents a different subject, and the same color-subject pair is used across all visualization methods. In this task, observed paths exhibit a range of spatial patterns, reflecting diverse individual strategies. OPM predictions generally follow a more arc-like path, deviating in some cases from the observed path direction and curvatures. SFM predictions appear slightly more adapted to individual variations in path direction and curvature for several subjects.

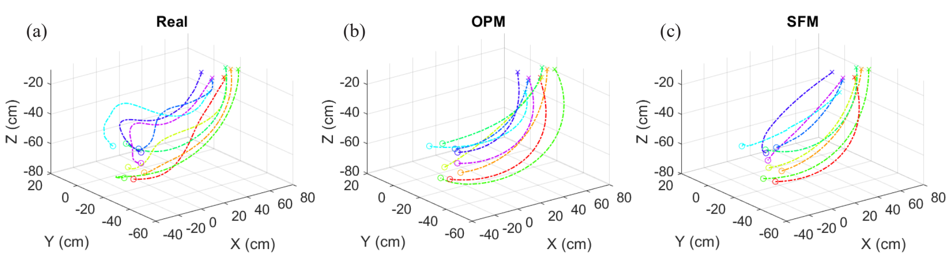


Figure 5. End-effector path of all subjects for Task 4: (a) Experimental data, (b) OPM predictions, and (c) SFM predictions. Each color represents a different subject, consistent across all plots. Circles denote starting positions; crosses indicate end points.

Table 7 presents the quantitative evaluation of spatial path prediction accuracy using RMSE and DTW metrics across all four tasks. RMSE measures the overall spatial error between the predicted and observed end-effector paths, while DTW reflects differences in the shape of the paths. The comparison of RMSE values shows that SFM outperforms OPM with lower RMSE values in Tasks 1 (23.4 for SFM vs. 31.3 for OPM) and 4 (21.1 for SFM vs. 25 for OPM) and shows nearly equal performance in Task 3. Only in Task 2, OPM achieves a slightly better RMSE. DTW values reveal a similar trend. SFM matches or exceeds OPM with closer to observed shape similarity in Tasks 1 and 4. Both methods perform equally in Task 3, and OPM slightly outperforms in Task 2.

Table 7. Summary of Path Prediction Results: RMSE and DTW values across all tasks. RMSE in cm. Bold values indicate better performance.

		Task 1		Task 2		Task 3		Task 4	
		OPM	SDM	OPM	SDM	OPM	SDM	OPM	SDM
RMSE	Mean (SD)	31.3 (8.9)	**23.4 (5.3)**	**19.1 (7.2)**	22.3 (8.3)	**14.3 (6.1)**	**14.5 (5)**	25 (6)	**21.1 (8.2)**
	Std error	2.8	1.7	2.4	2.8	1.9	1.6	2.0	2.7
	Best Pred	Sub 8	Sub 8	Sub 7	Sub 9	Sub 8	Sub 1	Sub 1	Sub 5
	Worse Pred	Sub 5	Sub 4	Sub 3	Sub 4	Sub 10	Sub 6	Sub 2	Sub 6
DTW	Mean (SD)	2.4 (0.9)	**2 (0.8)**	**1 (0.4)**	1.2 (0.4)	**1.3 (0.6)**	**1.3 (0.4)**	1.8 (0.5)	**1.5 (0.8)**
	Std error	0.3	0.3	0.1	0.1	0.2	0.1	0.2	0.3
	Best Pred	Sub 6	Sub 6	Sub 7	Sub 1	Sub 8	Sub 1	Sub 5	Sub 5
	Worse Pred	Sub 4	Sub 7	Sub 3	Sub 4	Sub 10	Sub 6	Sub 8	Sub 9

Best and worst path predictions

Figure 6 shows the best and worst path planning predictions for each method based on the DTW metric. The top row presents the best-matching predictions. In both cases, the predicted trajectories closely follow the observed paths. The bottom row shows the least accurate predictions for each method, where predictions show visible differences in the trajectory's initial direction, spatial curvature, and endpoint location, as can be seen for the SFM prediction.

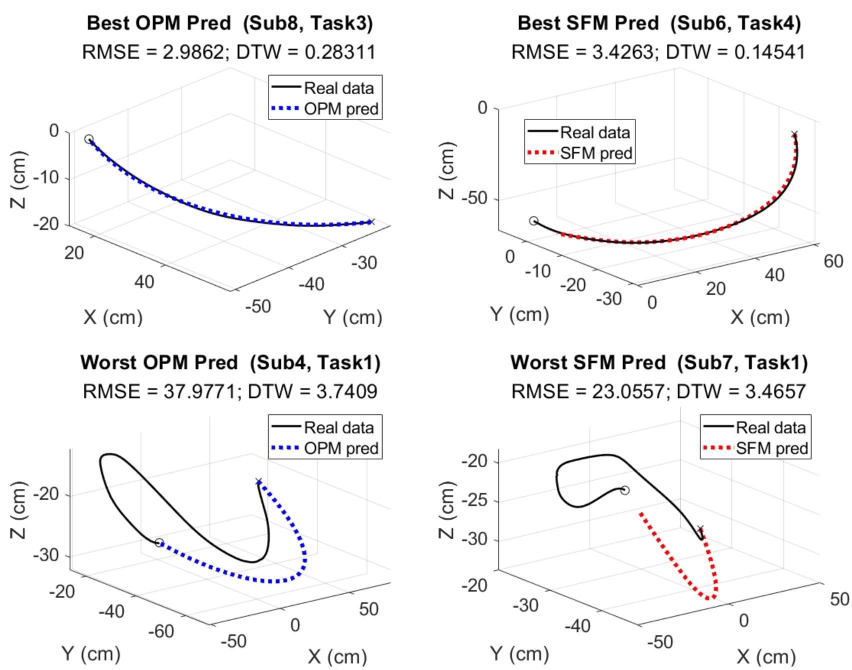


Figure 6. Examples of best (top) and worst (bottom) path predictions for the OPM (left) and SFM (right) methods, based on the DTW evaluation metric.

Modeled inter-subject path variability

Table 8 reports the modeled path planning inter-subject variability for the observed and predicted data using RMSE and DTW metrics. The observed motions exhibit the highest variability across all tasks in magnitude and shape. Among the two methods, SFM consistently shows higher mean and standard deviation values than OPM across both metrics, meaning its variability is greater and ultimately more similar to the observed variability in magnitude and path shape. This indicates that SFM better captures the natural differences in path planning across individuals.

Table 8. Modeled inter-subject variability for path predictions. RMSE and DTW reported as Mean ± SD (Min-Max). RMSE in cm. Bold indicates the method with an average closer to observed values.

	Task 1		Task 2		Task 3		Task 4	
	RMSE	DTW	RMSE	DTW	RMSE	DTW	RMSE	DTW
Real	31.8 ± 9.9 (11.5–53.9)	2.3 ± 0.6 (0.5–3.7)	25.9 ± 7.2 (16.1–45.6)	1.2 ± 0.5 (0.4–2.4)	28.1 ± 8.5 (11.8–44.8)	2.0 ± 0.6 (0.6–3.1)	28.2 ± 9.2 (13.3–54.8)	2.0 ± 0.7 (0.4–3.0)
OPM	22.0 ± 8.4 (5.4–39.0)	1.5 ± 0.7 (0.2–2.6)	23.8 ± 8.7 (8.2–48.9)	**0.9 ± 0.5 (0.2–1.9)**	20.9 ± 7.3 (8.1–34.9)	1.1 ± 0.8 (0.1–2.5)	**24.7 ± 9.7 (7.9–48.5)**	1.1 ± 0.5 (0.1–2.0)
SFM	**25.9 ± 8.9 (8.7–46.4)**	**1.6 ± 0.6 (0.3–2.5)**	**25.9 ± 8.2 (7.0–39.2)**	**0.9 ± 0.4 (0.2–1.7)**	**21.3 ± 7.9 (5.7–36.6)**	**1.3 ± 0.6 (0.2–2.5)**	19.6 ± 6.1 (7.1–32.0)	**1.2 ± 0.7 (0.1–2.2)**

3.2 End-effector velocity

Figure 7 shows the absolute velocity curves for all ten subjects performing Task 1. The recorded curves typically display a rapid acceleration phase followed by a smooth deceleration as the hand reaches the target position. Individual deceleration phases regularly exhibited one or more short periods of acceleration prior to reaching the target (see Figure 7a). Occasionally, a similar moment of deceleration occurred during the initial acceleration phase. OPM predictions produce smooth and symmetric bell-shaped curves, generally underestimating the peak velocities (see Figure 7b). On the other hand, SFM predictions typically exhibit a closer visual match to the shape and magnitude of the observed data's velocity curves, especially during deceleration, though SFM tended to reach the target earlier than was observed in the recorded data (see Figure 7c).

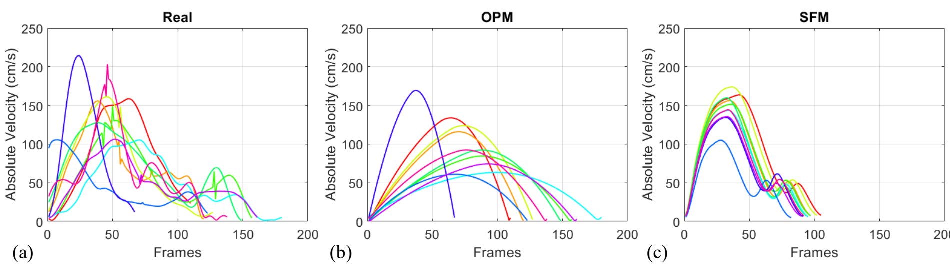


Figure 7. End-effector velocity profiles of all subjects for Task 1: (a) Experimental data), (b) OPM predictions, and (c) SFM predictions. Each color represents a different subject, and the same color-subject pairing is used across the prediction plots.

Table 9 summarizes the velocity prediction accuracy across all four tasks using two evaluation metrics: RMSE, measuring the overall amplitude errors, and DTW, measuring the profile shape similarity. A task-by-task comparison reveals that, according to the RMSE evaluation metric, SFM outperforms OPM in Task 1 (with an RMSE value of 34 compared to 39.9) and Task 4 (29 vs. 31.2). On the other hand, OPM achieves lower RMSE in Task 2 (30.9 vs. 47.1) and Task 3 (24.7 vs. 26.5), indicating a better match of overall (average) velocity magnitude and task duration.

When looking at DTW values, SFM generally shows better or equivalent performance compared to OPM. Specifically, in Task 1 and Task 4, SFM clearly outperforms OPM with lower DTW values (1.6 vs. 2.6 and 1.4 vs.

1.9, respectively). Task 3 shows identical DTW values for both methods (2.3), and in Task 2, OPM has a slightly lower DTW (1.1 vs. 1.2), indicating a very close match in curve shape between the two methods for that task.

Table 9. Summary of Velocity Prediction Results: RMSE and DTW values across all tasks. RMSE in cm/s. Bold values indicate better performance.

		Task 1		Task 2		Task 3		Task 4	
		OPM	SFM	OPM	SFM	OPM	SFM	OPM	SFM
RMSE	Mean (SD)	39.9 (11.4)	**34 (9.3)**	**30.9 (14.9)**	47.1 (15.9)	**24.7 (8.4)**	26.5 (9.3)	31.2 (15.9)	**29 (10.1)**
	Std error	3.6	3.0	5.0	5.3	2.7	3.0	5.3	3.4
	Best Pred	Sub 1	Sub 2	Sub 2	Sub 5	Sub 4	Sub 7	Sub 7	Sub 5
	Worse Pred	Sub 8	Sub 1	Sub 9	Sub 3	Sub 3	Sub 6	Sub 2	Sub 1
DTW	Mean (SD)	2.6 (2.1)	**1.6 (1.2)**	**1.1 (0.5)**	1.2 (0.4)	**2.3 (1.2)**	**2.3 (0.8)**	1.9 (0.8)	**1.4 (0.5)**
	Std error	0.7	0.4	0.2	0.1	0.4	0.3	0.3	0.2
	Best Pred	Sub 1	Sub 1	Sub 3	Sub 2	Sub 2	Sub 1	Sub 5	Sub 4
	Worse Pred	Sub 7	Sub 7	Sub 9	Sub 7	Sub 9	Sub 8	Sub 2	Sub 6

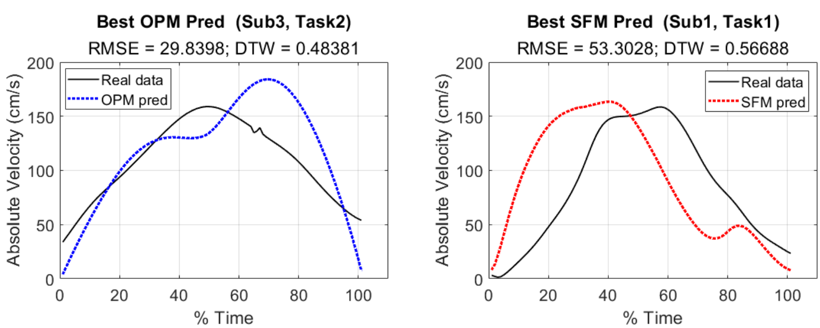
Table 10 presents the average peak velocities across subjects for each task. The observed data exhibits different peak velocities depending on the task, with Task 2 showing the highest peak on average (152.8 cm/s). Compared to the observed data, OPM underestimates the peak velocity in all tasks, with an average deviation of 31.6 cm/s, and reaching a maximum underestimation of 47.7 cm/s in Task 1. In contrast, SFM matches the experimental data more closely, with an average deviation of 2.8 cm/s.

Table 10. Average Peak Velocity values per method across all tasks in cm/s. Bold indicates the method with an average peak velocity closer to observed values.

	Task 1	**Task 2**	**Task 3**	**Task 4**
Average Peak Vel Observed (SD)	148.7 (38.9)	152.8 (30.4)	82.9 (24.4)	111.1 (29.5)
Average Peak Vel OPM (SD)	101.0 (34.4)	132.1 (34.6)	57.2 (24.4)	78.6 (23.5)
Average Peak Vel SFM (SD)	**146.7 (19.2)**	**140.6 (14.1)**	**79.7 (13.3)**	**117.3 (13.6)**

Best and worst velocity predictions

To further illustrate the velocity prediction performance, Figure 8 shows each method's best and worst velocity predictions, comparing the observed and predicted profiles for the corresponding subjects and tasks. The top row displays the best cases based on the DTW metric, where both OPM and SFM predicted curves follow the general shape well. The bottom row shows the worst cases, where both methods' predictions deviate from the curve shape of the observed data.



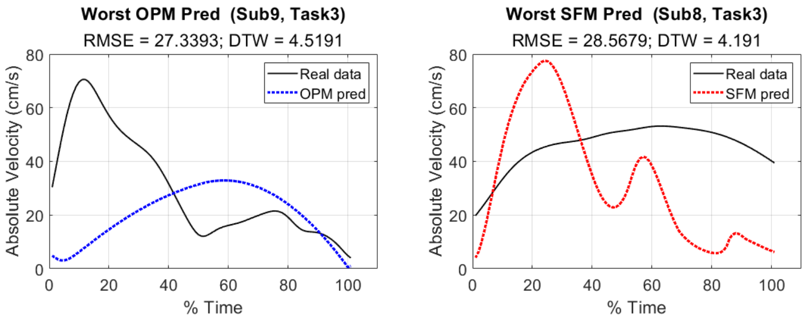


Figure 8. Examples of best (top) and worst (bottom) velocity predictions for the OPM (left) and SFM (right) methods, based on the DTW evaluation metric.

Modeled inter-subject velocity variability

Table 11 shows how observed velocity profiles exhibit substantially higher variability in both magnitude (RMSE) and shape (DTW) across all tasks. In contrast, both OPM and SFM consistently produce lower variability, suggesting more uniform predictions across subjects. OPM generally reproduces a higher level of variability than SFM, especially in terms of RMSE values, suggesting that its predictions better reflect the magnitude differences in how individual participants perform the tasks. SFM, on the other hand, consistently shows higher variability in DTW, reflecting greater shape variability across predictions.

Taken together, these results suggest that neither model fully reproduces the variability observed in human motion velocities. The range of variability produced by OPM is more like the observed inter-subject variability measured by RMSE and the range of variability produced by SFM is more like the observed inter-subject variability measured by DTW.

Table 11. Modeled inter-subject variability in velocity profiles for each task. Reported as Mean ± SD (Min-Max). RMSE in cm/s. Bold indicates the method with an average closer to observed values.

	Task 1		Task 2		Task 3		Task 4	
	RMSE	DTW	RMSE	DTW	RMSE	DTW	RMSE	DTW
Real	43.5 ± 18.2 (16.5–95.1)	2.4 ± 2.0 (0.4–9.1)	46.4 ± 15.4 (12.9–76.9)	1.0 ± 0.5 (0.3–2.5)	32.4 ± 10.2 (10.9–57.7)	2.7 ± 1.5 (0.6–7.0)	36.2 ± 14.0 (15.2–74.9)	2.0 ± 0.8 (0.4–4.3)
OPM	**27.8 ± 18.8** (2.5–73.9)	0.2 ± 0.1 (0.2–0.4)	**30.8 ± 17.5** (4.0–69.1)	0.3 ± 0.1 (0.1–0.5)	16.2 ± 9.0 (0.7–37.2)	0.4 ± 0.2 (0.0–1.1)	**19.8 ± 11.0** (3.5–45.0)	0.3 ± 0.1 (0.2–0.7)
SFM	16.0 ± 10.4 (2.8–45.8)	**0.5 ± 0.3** (0.1–1.1)	21.0 ± 9.9 (4.9–45.6)	**0.8 ± 0.3** (0.3–1.6)	**17.9 ± 8.2** (3.1–41.6)	**1.4 ± 0.7** (0.3–3.2)	15.4 ± 7.3 (1.9–29.9)	**0.8 ± 0.4** (0.2–1.6)

3.3 Upper extremity joint configuration

This section evaluates the performance of the two prediction methods, OPM and SFM, in predicting the seven degrees of freedom (DOFs) of the right arm: shoulder (3 DOFs), elbow (1 DOF), and wrist (3 DOFs). Figure 9 shows the distribution of joint angles for Task 2 across all subjects, comparing experimental data with predictions from OPM and SFM. Each boxplot presents the interquartile range (IQR), median, and distribution of joint angles, allowing for a visual comparison of predicted and observed motion ranges for each DOF for Task 2.

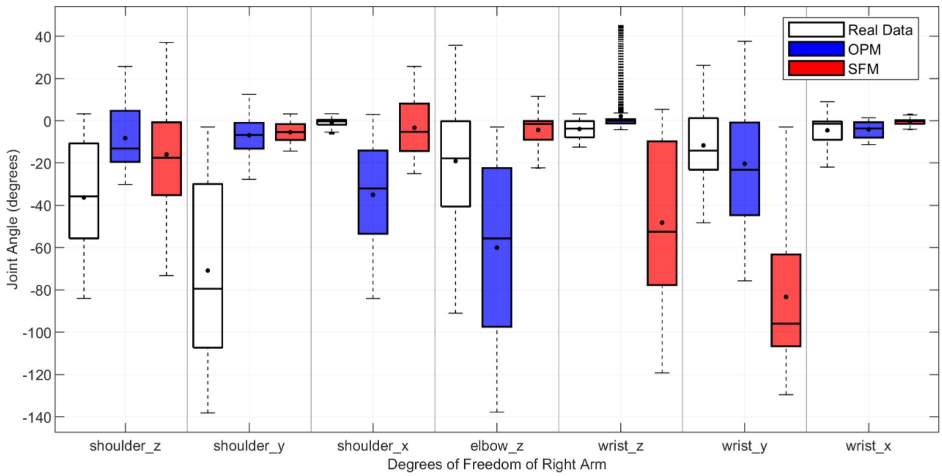


Figure 9. Distribution of joint angle values for Task 2 across the seven DOFs of the right upper extremity, encompassing the shoulder, elbow, and wrist. For each DOF, three boxplots are shown representing the observed data (white), the OPM (blue), and the SFM (red). Each box displays the IQR with the black horizontal line indicating the median, and black circles denoting the mean joint angle across all subjects. Whiskers extend to 1.5x the IQR, and outliers are shown as black ticks.

The observed joint angles for the shoulder_z, shoulder_y, and wrist_y exhibit broad distributions, indicating substantial variability in how participants approached the task using those joints. In contrast, shoulder_x, and wrist_z exhibit narrower distributions, indicating more narrow use across subjects.

OPM predictions (blue) align well with the experimental data in joints like wrist_y and wrist_x but deviate in others, such as shoulder_y, and elbow_z, overestimating or underestimating respectively, and shifting the distribution range. SFM predictions (red) show closer alignment in shoulder_x, and wrist_x but underestimate variability in wrist_y and overestimates in elbow_z. Thus, neither prediction method fully captures the general range of joint angles used during the reaching motions in the observed data.

Table 12 presents the RMSE evaluation per joint angle across all four tasks. The elbow_z joint consistently yields the highest RMSE values across all tasks for both methods, particularly in Tasks 1 and 4, with scores reaching up to 39.4° for OPM and 42.5° for SFM, indicating greater difficulty in accurately predicting the elbow joint during the reaching motion. In contrast, the wrist_y and wrist_x joints consistently show the lowest RMSE values, with OPM achieving values as low as 1.4° and SFM as low as 1.0° across tasks.

Although SFM performs better in specific joints for certain tasks, OPM yields a lower average RMSE across all joints and tasks (10.8° vs 12.8°), suggesting a slightly better prediction of the magnitude of joint configurations.

Table 13 summarizes DTW scores for the full 7-DOF joint angles trajectories. Across all four tasks, OPM consistently achieves lower DTW values than SFM, with the largest differences observed in Task 2 (3.5° for OPM vs. 4.4° for SFM) and Task 4 (5.9° for OPM vs. 6.7° for SFM), indicating that OPM more accurately captures the overall shape of joint coordination patterns.

Table 12. RMSE evaluation per joint (shoulder, elbow, and wrist) across all tasks, comparing OPM and SFM. RMSE in cm/s. Bold values indicate better performance.

	OPM							SFM						
Task 1	s_z	s_y	s_x	e_z	w_z	w_y	w_x	s_z	s_y	s_x	e_z	w_z	w_y	w_x
Average of RMSE	**14.6**	**10.9**	**13.3**	39.4	**6.8**	**3.3**	**1.6**	18.1	14.4	22.6	**33.7**	7.6	4.0	2.1
SD of RMSE	6.7	3.3	6.7	16.5	3.9	2.0	1.4	7.7	6.9	4.2	14.0	3.4	2.4	2.2
Std error of RMSE	2.1	1.1	2.1	5.2	1.2	0.6	0.4	2.4	2.2	1.3	4.4	1.1	0.7	0.7

	OPM							SFM						
Task 2	s_z	s_y	s_x	e_z	w_z	w_y	w_x	s_z	s_y	s_x	e_z	w_z	w_y	w_x
Average of RMSE	**6.2**	**7.7**	10.6	**16.1**	**5.3**	**2.6**	3.9	18.2	16.6	**7.8**	22.2	5.6	**2.6**	**1.0**
SD of RMSE	2.8	3.4	11.7	6.8	3.9	1.0	8.3	8.4	8.0	3.9	8.7	3.7	0.9	0.9
Std error of RMSE	0.9	1.1	3.9	2.3	1.3	0.3	2.8	2.8	2.7	1.3	2.9	1.2	0.3	0.3

	OPM							SFM						
Task 3	s_z	s_y	s_x	e_z	w_z	w_y	w_x	s_z	s_y	s_x	e_z	w_z	w_y	w_x
Average of RMSE	7.0	**2.6**	**6.1**	20.0	**6.5**	**3.8**	**1.4**	**6.5**	5.3	7.2	**15.7**	7.3	4.1	1.6
SD of RMSE	4.9	2.1	3.9	11.3	6.3	2.6	1.2	3.2	3.2	4.3	8.3	7.1	2.7	1.3
Std error of RMSE	1.6	0.6	1.2	3.6	2.0	0.8	0.4	1.0	1.0	1.4	2.6	2.2	0.8	0.4

	OPM							SFM						
Task 4	s_z	s_y	s_x	e_z	w_z	w_y	w_x	s_z	s_y	s_x	e_z	w_z	w_y	w_x
Average of RMSE	**24.4**	**10.2**	**28.1**	**35.0**	**8.8**	**3.9**	**3.5**	25.5	11.7	36.5	42.5	9.2	4.8	3.6
SD of RMSE	12.3	4.0	14.6	16.8	6.6	2.1	3.5	12.5	7.8	12.1	20.4	6.0	2.4	3.3
Std error of RMSE	4.1	1.3	4.9	5.6	2.2	0.7	1.2	4.2	2.6	4.0	6.8	2.0	0.8	1.1

Table 13. DTW values across all joint angles for each task. Bold values indicate better performance.

	Task 1		Task 2		Task 3		Task 4	
	OPM	SFM	OPM	SFM	OPM	SFM	OPM	SFM
Average of DTW	**5.0**	5.2	**3.5**	4.4	**5.2**	5.4	**5.9**	6.7
SD of DTW	1.5	1.3	1.0	1.4	0.8	1.0	1.0	1.5
Std error of DTW	0.5	0.4	0.3	0.5	0.3	0.3	0.3	0.5

Modeled inter-subject joint configuration variability

Table 14 presents the RMSE-based inter-subject variability for each joint angle across all tasks. Again, the observed data show the highest variability in nearly all joints and tasks, reflecting differences in individual movement strategies. Between the two models, OPM generally preserves more inter-subject variability than SFM, particularly in the shoulder and elbow joints. However, it occasionally also exhibits greater variability relative to the observed data.

Table 15 presents DTW measured inter-subject variability for each joint angle across all tasks. The variability of the observed data is the highest across all tasks, and SFM captures closer to the average and range exhibited by the observed data.

Table 14. Inter-subject variability for arm joint angle configuration. RMSE reported as Mean ± SD (Min-Max). Bold indicates the method with an average closer to observed values.

	RMSE - Task 1						
Method	shoulder_z	shoulder_y	shoulder_x	elbow	wrist_z	wrist_y	wrist_x
Real	24.6 ± 12.3 (7.1–53.0)	13.2 ± 9.2 (2.7–32.2)	34.9 ± 17.0 (6.1–72.0)	37.1 ± 19.8 (10.4–87.4)	16.0 ± 8.2 (2.3–37.4)	5.5 ± 2.3 (1.7–12.0)	5.4 ± 4.8 (1.0–17.6)
OPM	9.1 ± 4.5 (2.1–20.0)	**11.2 ± 8.6** (1.4–28.4)	**24.4 ± 12.5** (1.5–47.1)	**18.5 ± 11.0** (3.7–43.2)	**12.5 ± 7.8** (0.6–34.0)	**4.5 ± 2.0** (0.8–9.7)	**3.9 ± 3.9** (0.3–13.7)
SFM	**12.1 ± 5.3** (3.9–26.2)	15.5 ± 8.4 (1.5–35.0)	19.2 ± 11.1 (1.0–45.8)	18.2 ± 7.8 (5.3–37.1)	11.2 ± 7.7 (0.5–33.6)	3.9 ± 2.0 (0.3–9.3)	2.8 ± 2.4 (0.3–9.7)

	RMSE - Task 2						
Method	shoulder_z	shoulder_y	shoulder_x	elbow	wrist_z	wrist_y	wrist_x
Real	14.0 ± 6.2 (3.2–27.1)	7.8 ± 2.6 (1.9–13.7)	20.0 ± 10.2 (2.7–35.7)	17.1 ± 9.7 (2.8–46.9)	10.2 ± 4.9 (2.9–25.8)	5.9 ± 2.6 (1.9–12.3)	2.7 ± 1.4 (0.4–6.3)
OPM	**9.9 ± 5.5** (0.5–17.8)	**6.9 ± 3.3** (1.7–15.8)	28.7 ± 18.3 (2.3–69.2)	7.1 ± 4.4 (0.5–16.4)	**9.7 ± 6.1** (0.4–25.9)	**5.3 ± 2.9** (1.1–10.6)	7.9 ± 10.7 (0.3–29.5)
SFM	20.6 ± 11.2 (1.9–43.5)	20.4 ± 11.0 (4.5–47.3)	**23.1 ± 11.4** (3.7–45.3)	**16.2 ± 10.2** (4.2–40.4)	9.3 ± 6.2 (0.3–25.1)	4.9 ± 2.8 (1.1–10.3)	**2.0 ± 1.4** (0.1–5.6)

Method	RMSE - Task 3 shoulder$_z$	shoulder$_y$	shoulder$_x$	elbow	wrist$_z$	wrist$_y$	wrist$_x$
Real	15.7 ± 7.2 (4.2–37.3)	7.1 ± 3.3 (1.5–14.9)	35.4 ± 19.7 (5.0–78.8)	24.2 ± 15.2 (6.7–58.4)	12.2 ± 5.2 (2.8–21.6)	4.8 ± 2.0 (1.9–8.8)	4.1 ± 2.5 (0.5–11.7)
OPM	**13.0 ± 8.3** (2.8–39.4)	5.9 ± 3.4 (0.7–14.9)	**31.7 ± 17.7** (4.7–69.8)	**19.1 ± 11.2** (2.7–46.9)	10.3 ± 5.2 (1.9–23.2)	**3.6 ± 1.6** (0.7–7.0)	**3.5 ± 2.0** (0.3–8.7)
SFM	10.1 ± 4.9 (1.6–21.5)	**7.3 ± 4.2** (1.5–18.6)	28.9 ± 16.8 (4.2–70.3)	13.7 ± 6.9 (3.2–31.0)	**10.5 ± 6.1** (1.3–25.9)	2.7 ± 1.3 (0.3–6.1)	3.3 ± 2.0 (0.2–7.3)

Method	RMSE - Task 4 shoulder$_z$	shoulder$_y$	shoulder$_x$	elbow	wrist$_z$	wrist$_y$	wrist$_x$
Real	39.9 ± 19.7 (6.1–85.4)	14.5 ± 8.5 (3.4–34.5)	54.8 ± 27.0 (9.7–117.6)	33.3 ± 17.0 (8.3–76.2)	16.3 ± 8.7 (2.9–35.0)	7.8 ± 3.4 (1.8–13.9)	6.6 ± 4.3 (0.8–17.0)
OPM	**38.0 ± 22.5** (1.9–89.0)	5.4 ± 2.7 (0.9–11.6)	42.9 ± 18.4 (12.3–76.0)	8.3 ± 4.0 (1.2–18.9)	10.1 ± 7.8 (0.3–34.0)	**5.7 ± 2.6** (0.9–11.4)	2.2 ± 1.4 (0.3–5.8)
SFM	34.5 ± 17.3 (3.0–66.2)	**7.8 ± 3.8** (1.6–15.5)	**46.7 ± 24.3** (12.1–99.4)	**12.3 ± 5.1** (3.6–23.7)	**10.2 ± 8.2** (0.2–35.2)	4.3 ± 1.9 (0.8–8.0)	**2.3 ± 1.5** (0.4–5.9)

Table 15. Inter-subject variability for upper extremity joint angle configuration. DTW reported as Mean ± SD (Min-Max). Bold indicates the method with an average closer to observed values.

DTW	Task 1	Task 2	Task 3	Task 4
Real	5.7 ± 1.3 (2.9–8.6)	4.2 ± 1.2 (1.5–6.1)	6.4 ± 1.1 (4.4–9.9)	7.1 ± 1.4 (3.9–10.0)
OPM	2.9 ± 2.0 (0.1–8.0)	2.3 ± 1.1 (0.0–4.9)	3.5 ± 1.7 (0.1–8.0)	3.7 ± 2.4 (0.3–10.2)
SFM	**5.0 ± 1.9** (1.6–9.9)	**3.9 ± 1.3** (0.8–6.4)	**4.2 ± 1.6** (1.1–8.9)	**4.8 ± 1.4** (2.5–9.3)

4 Discussion

We compared two distinct methods for predicting human reaching motions in 3D space. The comparison focused on three key components of reaching behavior: path, velocity, and arm joint configuration. By evaluating predictions across these motion components, the study aimed to assess both the accuracy and the ability of each method to capture inter-individual variability.

Results

The evaluation of prediction accuracy and inter-subject variability across the three motion components revealed distinct performance patterns for the two methods (Table 16).

For the path, SFM consistently outperformed OPM in both accuracy and variability. It achieved lower RMSE and DTW values in most tasks and produced subject-to-subject differences in spatial paths that more closely matched those observed in the experimental data. These results indicate that SFM better preserves both the magnitude and shape of individual spatial strategies.

For the velocity, SFM achieved higher accuracy overall. It produced lower DTW values across most tasks and replicated observed peak velocities more closely than OPM. However, OPM better captured inter-subject differences in overall velocity levels, as indicated by RMSE-based inter-subject variability. In contrast, SFM showed lower DTW-based variability, suggesting it better captured the differences in the shape of individual velocity profiles.

For the arm joint configuration, OPM yielded higher prediction accuracy, with lower RMSE and DTW values across tasks. It also better reproduced variability in joint angle magnitude across individuals, while SFM captured more shape variability in the coordination patterns. This suggests that OPM provides more precise joint estimates, whereas SFM reflects more variation in how subjects perform the same task.

These results reflect a general trade-off between accuracy and diversity representation. Methods that achieve high prediction accuracy often produce more uniform outputs, while those that preserve inter-subject variability may sacrifice precision. Incorporating both RMSE and DTW in the evaluation provides a more complete understanding of these dynamics across motion components.

To synthesize the findings across the three evaluated components: path, velocity, and arm configuration, Table 16 provides a summary of the best-performing method for each evaluated aspect, considering both predictive accuracy and the ability to replicate behavioral variability observed in human data. Accuracy reflects performance based on RMSE and DTW metrics, while behavioral variability refers to how well each method preserves the diversity of individual movement patterns.

Table 16. Summary of the best-performing prediction method for each movement aspect, based on accuracy (RMSE and DTW) and behavioral variability across subjects.

	Accuracy		Modeled inter-subject variability	
	Magnitude (RMSE)	Shape (DTW)	Magnitude (RMSE)	Shape (DTW)
Path	SFM	SFM	SFM	SFM
Velocity	SFM	SFM	OPM	SFM
Upper extremity joint configuration	OPM	OPM	OPM	SFM

Evaluation metrics: Complementary roles of RMSE and DTW

In this study, RMSE and DTW were utilized together to assess the accuracy and modeled variability of motion predictions across subjects for different components of reaching behavior. RMSE is widely used for quantifying point-wise differences between predicted and observed values. However, RMSE evaluates each time step independently and does not consider the overall structure or timing of the motion, which can lead to misleading conclusions when sequences exhibit differences in spatial or temporal scale and alignment. To address this limitation, DTW was introduced as a complementary metric to compare time-dependent sequences by non-linearly aligning them in time. DTW provides a single value that reflects the best alignment between two multivariate sequences, offering a more holistic view of motion similarity.

DTW has been widely used in fields such as gait recognition (Boulgouris et al., 2004), activity classification (Seto et al., 2015), and trajectory clustering (Zhang et al., 2022), where its ability to align time-series data non-linearly makes it particularly suitable for shape-based comparisons. However, despite its demonstrated strengths, DTW has rarely been adopted as a quantitative evaluation metric for predictive modeling in multivariate human motion contexts. In this study, we extend DTW's use to evaluate how well model-generated multivariate trajectories (path, velocity, joint configuration) replicate collected experimental motions.

Prior work in other domains has demonstrated DTW's robustness across both simulated and real-world data. For example, Cleasby et al. (2019) found DTW and Fréchet distance to be the most responsive metrics when comparing animal trajectories, outperforming commonly used measures for trajectory similarity. Likewise, Laperre et al. (2020) illustrated that DTW can outperform RMSE in evaluating predictions with temporal shifts in the context of geomagnetic storm forecasting, where DTW more effectively captured structural and temporal mismatches between predicted and actual data. Similarly, in time-series classification and motion gesture studies, DTW has shown superior performance due to its sensitivity to sequence shape and flexibility to temporal misalignment (Jiang et al., 2017; Yu and Xiong, 2019; Gloumakov et al., 2020).

Our findings show that RMSE and DTW often lead to different conclusions about model performance depending on what aspect of the motion is being assessed. For instance, as seen in Tables 6 and 9, the best- and worst-predicted subjects per task often differ depending on whether RMSE or DTW is used. This divergence reflects the distinct components of motion each metric captures, highlighting that relying on a single metric can bias or mislead conclusions about critical aspects of model behavior and performance. Understanding not just how many errors exist, but also what kind of errors are measured and each model is prone to, is essential for meaningful evaluations.

Taken together, our results suggest that combining RMSE and DTW enables a more complete understanding of model behavior by capturing both the local and structural accuracy, as well as the variability, of motion predictions.

By applying DTW to full multivariate outputs, this study highlights the metric's potential as a rigorous evaluation metric for multivariate, time-dependent human motion prediction models.

Differences between prediction methods

The differences in performance between OPM and SFM can be attributed to how each method operates and the type of assumptions embedded in their design. OPM formulates reaching motion as a constrained optimization problem. It uses cubic B-spline curves for joint angles and optimizes a multi-objective cost function combining joint displacement minimization and end-effector velocity maximization. This produces smooth, globally coordinated joint trajectories, resulting in higher accuracy for arm configuration prediction. However, OPM tends to produce uniform trajectories across subjects, which may limit its ability to capture the diversity observed in human movement. In contrast, the SFM combines three rule-based models: an adaptive velocity profile, a dynamic path planner (SDM), and an inverse kinematics solver (FABRIK). These components are applied sequentially but are also interconnected: the velocity profile influences the path planner, which then informs the joint configuration solver, which may adjust the path if the path target is not within the arm's reaching space. This modular and reactive structure allows SFM to better replicate variation in velocity and spatial paths between individuals, resulting in higher shape similarity and modeled variability, particularly for path and velocity.

To better illustrate the differences between the two methods, Table 17 provides a side-by-side overview of key characteristics, including input requirements, solution strategy, and performance trade-offs. The models also differ in the nature and specificity of their inputs, OPM emphasizes temporal constraints and initial joint configuration, while SFM relies more on behavioral cues such as peak velocity and heading direction. These differences influence not only computational demand and implementation complexity, but also the adaptability and generalizability of each method across different use cases. Ultimately, the choice between OPM and SFM depends on the intended application. Based on our results, tasks that require biomechanical precision, full-body coordination, or high-fidelity posture simulation may benefit more from OPM, while the flexibility and efficiency of SFM may better serve tasks emphasizing behavioral variability, real-time interaction, or adaptive simulation.

Table 17. Overview of prediction methods and their key characteristics.

	OPM	SFM
Input	Initial spatial position; Goal spatial position; Total task time; Initial arm posture;	Initial spatial position; Goal spatial position; Initial velocity; Estimated peak velocity; Initial arm posture
Output	7-DOF joint angle trajectories (path, velocity, arm joint configuration)	7-DOF joint angle trajectories (path, velocity, arm joint configuration)
How solution is found	Search into the full solution space to find the optimal solution	Applies rule-based logic to generate a feasible solution sequentially
Pros	+ Allows optimization of multiple objectives	+ Modular and adaptable + Low computational cost
Cons	– Computationally expensive – Requires complete initial data	– May not always reach the target – Dependent on parameter tuning

Limitations

While the study provides valuable insights, several limitations must be acknowledged. First, the parameters of the SDM and velocity models within SFM were manually selected based on qualitative performance. No formal optimization or sensitivity analysis was conducted, which may affect the generalizability and robustness of the results. Future work will focus on methods for consistent and grounded parameterization of the SDM and velocity models with a focus on identifying methods that ensure the best possible predictions for these methods.

Second, the experimental tasks were performed without physically defined start or end points. Instead, participants were instructed verbally to perform movements as if seated in a vehicle. This open-ended task design may have led to inconsistencies in how participants interpreted and executed the movements, particularly regarding initial posture and endpoint location, which could have contributed to larger prediction errors in certain tasks and made some components of the motion more difficult to model accurately.

Third, the study involved a limited sample of ten participants performing four distinct reaching tasks. While this was sufficient to support the methodological comparison and identify key differences between the models, the scope of the dataset constrains broader generalization. Future studies incorporating a larger and more diverse participant pool and a wider range of reaching scenarios could provide deeper insight into model performance across contexts.

Model improvements and future work

The comparative evaluation of OPM and SFM highlights clear areas where each method can be refined to improve overall prediction quality. For OPM, refining the cost function, particularly to better capture motion shape, could enhance shape similarity and improve the model's ability to reflect inter-subject differences. This adjustment may lead to more realistic trajectory predictions while maintaining joint accuracy. For SFM, improvements may be achieved by systematically tuning the SDM parameters to reduce spatial magnitude errors and improve alignment with observed joint configurations. In addition, incorporating more rigorous definition of joint or posture-related constraints within the FABRIK module could improve the plausibility and accuracy of predicted arm configurations.

In addition to refining each model, future work could expand by including a broader set of reaching tasks. Tasks involving obstacle avoidance or continuous movement within constrained environments may offer further insights into model behavior and expose limitations that were not evident in the current set of tasks. Lastly, the findings of this study emphasize the importance of using multiple evaluation metrics when assessing predictive performance in human motion modeling. Further work to gain a clearer understanding of what each metric captures and what it may fail to represent is essential for improving both model evaluation and future model development, particularly when aiming to reproduce the complexity and richness of human movement.

5 Conclusion

This study presents a novel prediction model, the SFM, and an initial evaluation of its performance relative to an establish prediction model, the OPM. While the results were mixed, each model demonstrated distinct strengths and weaknesses.

The use of DTW, alongside RMSE, enables a more comprehensive and nuanced evaluation of the models, revealing differences in both how closely the predicted motions match the observed data and how well they preserve the temporal structure and coordination patterns of natural human movement. This dual-metric approach proves value in identifying trade-offs between accuracy and the ability to replicate inter-subject variation, an aspect often overlooked in conventional evaluations, which typically solely evaluates accuracy using RMSE. A critical contribution of this current work is the demonstration of the additional insights provided by including DTW as a prediction evaluation metric.

Results show that SFM performs better in predicting end-effector velocity and spatial paths, particularly in capturing the diversity of movement strategies across individuals. OPM, in contrast, achieves higher accuracy in joint angle configuration, producing smoother trajectories. The distinct model structures, input requirements, and computational characteristics explain these differences and highlight the importance of aligning model selection with application-specific priorities. Ultimately, the choice between OPM and SFM depends on the intended data available or use case. Overall, this initial evaluation indicates that the SFM may be a promising new prediction method for human motion, but further development is required.

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ESTELA PÉREZ LUQUE

Estela Pérez Luque holds a BSc in Mechanical Engineering from the University of Málaga in Spain and a MSc in Intelligent Automation from the University of Skövde. Her research interests include digital human modelling, simulation, and prediction of human posture and motion in product and production development processes.

In this dissertation, Estela investigates how the functionality and accuracy of digital human modelling tools can be improved to support decision-making when addressing ergonomics in vehicle occupant packaging design. Her work identifies current industrial practices, proposes improved simulation methods, and introduces posture and motion prediction models that support more efficient and reliable ergonomics assessments. Together, these contributions enable a more proactive, systematic, and human-centred approach to designing for human diversity in product development.

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